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Post-Quantum TLS 1.3 Handshake from CPA-Secure KEMs with Tighter Reductions

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Abstract

TLS 1.3 is at the heart of secure modern internet communications. With the rise of quantum attacks, post-quantum TLS 1.3, built on post-quantum key encapsulation mechanisms (KEMs), has naturally become a major research focus. At Eurocrypt 2022, Huguenin-Dumittan and Vaudenay demonstrated that KEMs secure against chosen-plaintext attacks (CPA) are sufficient to construct a secure TLS 1.3 handshake in the random oracle model (ROM), but their security reduction incurs an $\mathcal{O}(q^6)$ loss, where q is the number of random oracle queries. Improving their security bounds was left as an open problem. To address this problem, Zhou et al. took the first step at Asiacrypt 2024, improving the loss factor to $\mathcal{O}(q^2)$ in the ROM and $\mathcal{O}(q^4)$ in the quantum ROM (QROM) for OW-CPA secure KEMs, and to $\mathcal{O}(q)$ (ROM) and $\mathcal{O}(q^2)$ (QROM) for IND-CPA secure KEMs.

In this work, we advance the state-of-the-art by providing tighter security reductions for TLS 1.3 handshake based on CPA-secure KEMs. We introduce a new security notion, *IND-ICCA-IMAC*, and show that with a slight ciphertext expansion, the reduction losses can be significantly improved to $\mathcal{O}(q)$ (ROM) and $\mathcal{O}(q^2)$ (QROM) for OW-CPA secure KEMs, and to only $\mathcal{O}(1)$ in both models for IND-CPA secure KEMs. Moreover, we prove that without additional modifications such as ciphertext expansion, the loss of $\mathcal{O}(q)$ (ROM) and $\mathcal{O}(q^2)$ (QROM) is unavoidable. Finally, we analyze the security of TLS 1.3 from CPA-secure KEMs in the hybrid key exchange setting, and provide experimental evidence that ciphertext expansion is a practical trade-off for mitigating reduction losses.

Keywords: Post-quantum TLS 1.3 · tight reduction · QROM.

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1 Introduction

As quantum computing advances, the foundations of classical public-key cryptosystems face an unprecedented security threat [2]. Cryptosystems based on the hardness of factorization of large numbers and on the discrete logarithm problem, such as RSA [72] and (elliptic curve) Diffie-Hellman ((EC)DH) key exchange [33, 61], are particularly vulnerable to quantum attacks [74]. The TLS 1.3 protocol [71], a cornerstone of modern secure internet communications, relies on the (EC)DH key exchange mechanism, making it inherently insecure against quantum adversaries. This vulnerability has driven significant research into post-quantum TLS 1.3 capable of withstanding quantum threats [52, 73, 81].

Post-Quantum TLS 1.3 from KEMs. With the ongoing post-quantum cryptography standardization process [63] and the proposal of post-quantum key encapsulation mechanisms (KEMs) [11, 25, 50, 62], KEMs have emerged as a promising replacement for (EC)DH key exchange in TLS 1.3 [58, 75]. Previously, many researchers believed that replacing the (EC)DH key exchange with KEMs would require CCA security (i.e., security against chosen-ciphertext attacks) [38, 67]. However, a breakthrough study by Huguenin-Dumittan and Vaudenay [52] showed that CPA-secure KEMs (i.e., secure against chosen-plaintext attacks) are sufficient to ensure security in the random oracle model (ROM) [6].

They introduced a special intermediate security notion called IND-1CCA-MAC for KEMs, and proved that any OW-CPA secure KEM (i.e., one-way against chosen-plaintext attacks) implies the IND-1CCA-MAC security in the ROM, and then used the IND-1CCA-MAC security to prove TLS 1.3 security in the multi-stage model [38]. However, their reduction from IND-1CCA-MAC security to OW-CPA security of the KEM incurs a significant loss factor of $\mathcal{O}(q^6)$, where q is the number of random oracle queries made by the adversary, and, further, their analysis did not extend to quantum ROM (QROM) [13], a critical model for post-quantum security assessments. They left the improvement of the ROM bounds and the proof for the QROM security as open problems.

State of the Art. Motivated by the above challenges, recent work by Zhou et al. [81] at Asiacrypt 2024 has made good advances by tightening the reduction loss in the ROM and providing the first security proof in the QROM. They introduced a weaker security notion termed IND-1CCA-MAC*, which they showed is sufficient to establish the security of the TLS 1.3 handshake, while maintaining the same tightness as the proof based on IND-1CCA-MAC. In particular, they proved that the reductions from IND-1CCA-MAC* security to the OW-CPA security of the KEM incur only an $\mathcal{O}(q^2)$ loss in the ROM and an $\mathcal{O}(q^4)$ loss in the QROM, and that when the KEM is IND-CPA secure, the loss factor can be further reduced to $\mathcal{O}(q)$ and $\mathcal{O}(q^2)$ in the ROM and the QROM, respectively.

Motivating Question. While Zhou et al. [81] have made very good progress on the open problems of Huguenin-Dumittan and Vaudenay [52], there remains room to further reduce the loss factors in their reductions. In practice, improving the reduction tightness is highly desirable from the efficiency consideration, as it eliminates the need to increase security parameters to compensate for the loss factor [47, 66]. This point is especially important in practical protocols such as TLS 1.3. The state of the art, in turn, naturally leads to the following question:

Is it possible to further improve the reduction loss factor for post-quantum TLS 1.3 from CPA-secure KEMs (in both the ROM and the QROM)?

1.1 Our Contributions

In this work, we provide an affirmative answer to the aforementioned question. We introduce a new intermediate security notion, *IND-1CCA-1MAC*, which suffices to prove TLS 1.3 handshake

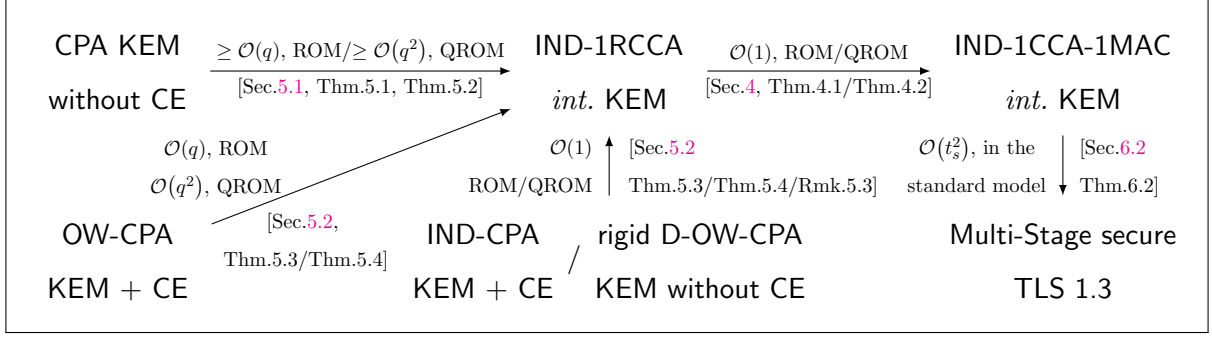


Figure 1: Result overview of our work, where *int.* KEM represents an *intermediate* KEM used in our security proofs, CE stands for ciphertext expansion, q is the number of random oracle queries, and t_s indicates the bounded number of sessions.

Table 1: Reduction tightness from the IND-1CCA-MAC*/IND-1CCA-1MAC secure KEM to the CPA-secure KEM. ϵ_R denotes the advantage of the reduction algorithm R w.r.t. the CPA security of the underlying KEM, $\epsilon_{\mathcal{A}}/\epsilon'_{\mathcal{A}}$ represents the advantage of the adversary $\mathcal{A}$ w.r.t. the IND-1CCA-MAC*/IND-1CCA-1MAC security of the obtained KEM, and q is the number of the random oracle queries made by $\mathcal{A}$.

Underlying KEM	Tightness	Model
OW-CPA [52, Eurocrypt'22]	$\epsilon_R \approx \mathcal{O}(1/q^6)\epsilon_{\mathcal{A}}^{\dagger}$	ROM
OW-CPA [81, Asiacrypt'24]	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}$	ROM
OW-CPA (Our work) [†]	$\epsilon_R \approx \mathcal{O}(1/q)\epsilon'_{\mathcal{A}}$	ROM
IND-CPA [81, Asiacrypt'24]	$\epsilon_R \approx \mathcal{O}(1/q)\epsilon_{\mathcal{A}}$	ROM
IND-CPA (Our work) [†]	$\epsilon_R \approx \epsilon'_{\mathcal{A}}$	ROM
OW-CPA [52, Eurocrypt'22]	-	QROM
OW-CPA [81, Asiacrypt'24]	$\epsilon_R \approx \mathcal{O}(1/q^4)\epsilon_{\mathcal{A}}^2$	QROM
OW-CPA (Our work) [†]	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}'^2$	QROM
IND-CPA [81, Asiacrypt'24]	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}^2$	QROM
IND-CPA (Our work) [†]	$\epsilon_R \approx \epsilon_{\mathcal{A}}'^2$	QROM
rigid D-OW-CPA [81, Asiacrypt'24]	-	QROM
rigid D-OW-CPA (Our work)	$\epsilon_R \approx \epsilon_{\mathcal{A}}'^2$	QROM

[†] The ciphertext expansion is required in the Handshake Secret **HS** negotiation.

[‡] The reduction provided by [52] is actually from IND-1CCA-MAC, but as pointed by [81], it implies the security of IND-1CCA-MAC*.

security with the same reduction tightness as IND-1CCA-MAC [52] and IND-1CCA-MAC* [81], but enables us to construct tighter reductions to the CPA security of the underlying KEM. Fig. 1 depicts our result overview, and the reduction tightness is presented in Table 1.

Specifically, our contributions are as follows:

1. We first propose an intermediate security notion, *IND-1CCA-1MAC*, and show that the

IND-1RCCA secure KEM¹ tightly implies IND-1CCA-1MAC security in both the ROM and the QROM. This can greatly simplify the security analysis, as we only need to focus on the IND-1RCCA security.

2. We prove that, if the underlying CPA-secure KEM has malleability, then any reduction from the IND-1RCCA security to the CPA security inherently incur a loss factor of $\mathcal{O}(q)$ (or $\mathcal{O}(q^2)$) in the ROM (or QROM) in general. This can also clarify why the reductions presented in [52, 81] are non-tight.
3. We show that the reduction from the IND-1RCCA security to the CPA security can be improved by incorporating a slight tag in the ciphertext (called ciphertext expansion hereafter). Particularly, if the KEM is IND-CPA secure, the reduction has a loss factor of only $\mathcal{O}(1)$ in both models.
4. We prove that TLS 1.3 based on IND-1CCA-1MAC KEMs is secure in the multi-stage model [38], with the same reduction tightness as those from IND-1CCA-MAC and IND-1CCA-MAC*, respectively.

Putting all results together, we achieve a post-quantum TLS 1.3 handshake from CPA-secure KEMs with tighter reductions. We examine our work in the context of the hybrid key agreement [75]. Experimental results confirm that ciphertext expansion is a viable trade-off for mitigating reduction losses.

Remark 1.1. We found that if the underlying KEM exhibits a specific property (termed rigid DKEM or rigid D-OW-CPA KEM in [81]), the loss factor remains $\mathcal{O}(1)$ in both the ROM and the QROM without requiring ciphertext expansion, further improving the corresponding result² in [81].

1.2 Related Work

Research on TLS protocol security has long been a major focus in academia [10, 16–20, 53, 57]. Since the introduction of the TLS 1.3 draft specifications, its security has been rigorously analyzed alongside the standardization process. These efforts included a critical examination of the handshake mechanism [40, 48, 59, 60] and in-depth analysis of the record layer’s cryptographic guarantees [44, 68]. Notably, some studies employed novel frameworks beyond conventional reduction-based approaches [9, 27, 28].

With the formal adoption of TLS 1.3 as an internet standard [71], subsequent research has systematically addressed the finalized architecture of the protocol. For instance, privacy features of the handshake procedure were thoroughly investigated by Arfaoui et al. in [4]. Diemert and Jager [32], along with Davis and Günther [30], refined computational security proofs for the full 1-RTT handshake by adapting techniques from Cohn-Gordon et al. [26], resulting in tighter reductions. Other researchers have proposed improvements or modifications to TLS 1.3, such as enhancing the forward secrecy of 0-RTT handshakes [5, 29, 31, 43], designing and analyzing post-quantum TLS 1.3 [21, 22, 69, 70], performing practical deployment of post-quantum TLS 1.3 [77, 78], defining hybrid key exchange in TLS 1.3 [58, 75], or proposing a KEM-based alternative handshake to support post-quantum deployment [24, 45, 73].

¹Informally, IND-1RCCA is a relaxation of IND-1CCA (i.e., IND-CCA but the adversary is allowed only one decapsulation oracle query), with the modification that decapsulation of a ciphertext $c \neq c^*$ resulting in one of the challenge keys $\{k_0, k_1\}$ returns a dedicated symbol `test` instead of the actual key. See Definition 3.1 for details.

²The QROM proof was absent in [81] and is first provided here.

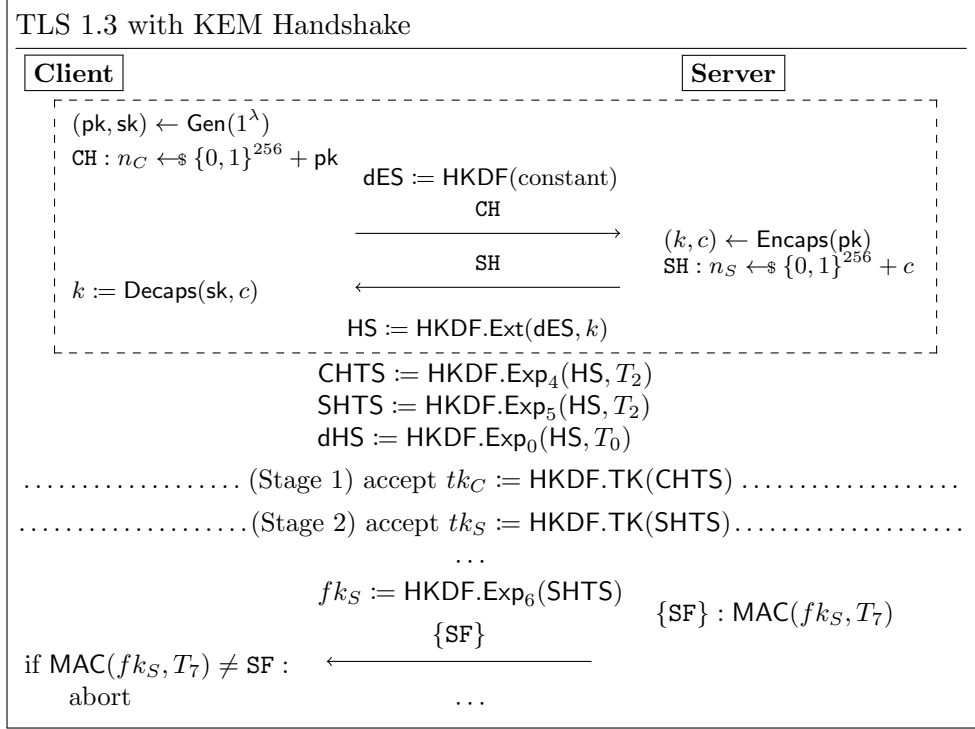


Figure 2: TLS 1.3 handshake with KEM [52]. The notation aligns with the conventions established in [52]. $\{\dots\}$ is the message encrypted with tk_S , and T_i is the hash of the transcript up to the i -th message. For brevity, **ClientHello** and **ClientKeyShare** are combined as **CH**, and **ServerHello** and **ServerKeyShare** as **SH**. Only proof-relevant steps are shown. Keys in the remaining stages (3-6, omitted) are derived from **dHS**.

In the security analysis of the TLS 1.3 handshake protocol, Dowling et al. [38] formalized a multi-stage key exchange model to capture various properties associated with each stage's key. This work builds on previous analyses of TLS 1.3 drafts [35–37, 39, 42], extending them to the finalized version of TLS 1.3. It demonstrated the security of the three handshake modes in the standard model under the PRF-ODH assumption [18]. Huguenin-Dumittan and Vaudenay [52] were the first to prove, within this multi-stage model, that when DH key exchange in TLS 1.3 is replaced by a KEM, CPA security is sufficient to ensure the security of the handshake. They also noted that the CDH assumption is sufficient to prove the security of the original TLS 1.3. Zhou et al. [81] refined the conclusions of Huguenin-Dumittan and Vaudenay, optimizing the bounds of security reductions in the ROM and extending their conclusions to the QROM.

2 Technique Overview

To demonstrate that a CPA-secure KEM is sufficient for TLS 1.3 handshake, Huguenin-Dumittan and Vaudenay [52] represented the relevant part of the TLS 1.3 handshake with KEM in Fig. 2. This representation simplifies the notation that does not affect the proof and omits steps irrelevant to the proof. In Fig. 2, the client computes $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$ and sends pk as its share (instead of g^x) to the server. Upon receiving pk , the server computes $(k, c) \leftarrow \text{Encaps}(pk)$, and sends c as its share (instead of g^y) to the client. Then, the client computes $k := \text{Decaps}(sk, c)$. Finally, the shared secret is set to k (instead of g^{xy}).

The IND-1CCA-MAC/IND-1CCA-MAC* Game for $\mathcal{A}$:
1 : $(pk, sk) \leftarrow \text{Gen}(1^\lambda), (k^*, c^*) \leftarrow \text{Encaps}(pk), HS^* := G(k^*), n^* \leftarrow \$\{0, 1\}^n$ 2 : $(CHTS_0, SHTS_0, dHS_0) := (H_1(HS^*, H_T(c^*, n^*)), H_2(HS^*, H_T(c^*, n^*)), H_3(HS^*))$ 3 : $(CHTS_1, SHTS_1, dHS_1) \leftarrow \$\{0, 1\}^{3n}, b \leftarrow \$\{0, 1\}, cnt_0, cnt_1 := 0$ 4 : $\hat{b} \leftarrow \mathcal{A}^{O_{\text{Dec}}, O_{\text{MAC}}^{\text{Dec}}}(\text{pk}, c^*, n^*, (CHTS_b, SHTS_b, dHS_b))$ 5 : return $[b = \hat{b}]$ Decapsulation Oracle $O_{\text{Dec}}((c, n) \neq (c^*, n^*))$:
6 : if $cnt_0 \neq 0$ then : return $\perp$ 7 : $cnt_0 := cnt_0 + 1, k' := \text{Decaps}(sk, c)$ 8 : if $k' = \perp$ then : return $\perp$ 9 : $HS' := G(k'), CHTS := H_1(HS', H_T(c, n)), SHTS := H_2(HS', H_T(c, n))$ 10 : $tk_C := H_D(CHTS), tk_S := H_D(SHTS)$ 11 : return (tk_C, tk_S) MAC Oracle $O_{\text{MAC}}^{\text{Dec}}((c, n) \neq (c^*, n^*), \text{tag}, \text{txt})$:
12 : if $cnt_1 \neq 0$ then : return $\perp$ 13 : if $O_{\text{MAC}}^{\text{Dec}}(c, n, \text{tag}, \text{txt})$ is not invoked after $O_{\text{Dec}}(c, n)$ or (c, n) in the two oracles are distinct then : return $\perp$ // IND-1CCA-MAC* 14 : $cnt_1 := cnt_1 + 1, k' := \text{Decaps}(sk, c)$ 15 : if $k' = \perp$ then : return $\perp$ 16 : $HS' := G(k'), SHTS := H_2(HS', H_T(c, n)), fk_S := H_4(SHTS)$ 17 : if $\text{Vrf}(fk_S, \text{tag}, \text{txt}) = \text{accept}$ then : return HS' 18 : return $\perp$

Figure 3: IND-1CCA-MAC(*) game for the adversary $\mathcal{A}$ (adapted from [52, 81]).

2.1 Previous Techniques and Results

Foundational Work by Huguenin-Dumittan and Vaudenay [52]. In the study by Huguenin-Dumittan and Vaudenay [52], an intermediate security notion called IND-1CCA-MAC was introduced, as shown in Fig. 3. Secrets such as $(CHTS_0, SHTS_0, dHS_0)$ in Fig. 3 are derived in a manner consistent with their counterparts in Fig. 2, where $G(\cdot)$ corresponds to $\text{HKDF.Ext}(\text{dES}, \cdot)$, $H_i(\cdot, H_T(c, n))$ (for $i = 1, 2$) corresponds to $\text{HKDF.Exp}_j(\cdot, T_2)$ (for $j = 4, 5$), $H_3(\cdot)$ corresponds to $\text{HKDF.Exp}_0(\cdot, T_0)$, $H_4(\cdot)$ corresponds to $\text{HKDF.Exp}_6(\cdot)$, and $H_D(\cdot)$ corresponds to $\text{HKDF.TK}(\cdot)$.

Simply speaking, the IND-1CCA-MAC notion is used to simulate the honest protocol session for the adversary $\mathcal{A}$ in the multi-stage security model [38] (cf. Appendix J). Specifically, when a server session label_S is chosen as the test session, the reduction algorithm must simulate its honest client counterpart label_C to complete the handshake and respond to $\mathcal{A}$. When SH sent by label_S does not match the one received by label_C , this indicates potential tampering of SH by $\mathcal{A}$. In this case, the reduction can use the decapsulation oracle O_{Dec} to obtain the corresponding traffic secrets (tk_C, tk_S) , allowing it to continue simulating label_C until the **ServerFinished** message **SF** is received. The reduction can then decrypt it with tk_S and verify it by the MAC oracle $O_{\text{MAC}}^{\text{Dec}}$. If the verification is successful, the reduction can use the response HS' to derive all subsequent secrets and complete the handshake. The overall security reduction for the TLS 1.3 handshake is bounded by $\mathcal{O}(t_s^2 \epsilon_A)$, where t_s is the maximum number of sessions and ϵ_A is

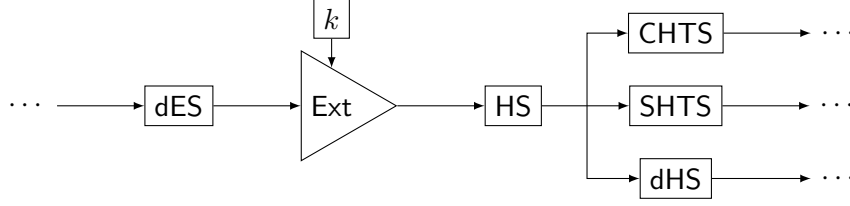


Figure 4: The TLS 1.3 key schedule, simplified from [38, Fig. 2]. Arrows indicate the direction of key derivation, where `Ext` denotes the `HKDF.Ext` function in Fig. 2.

$\mathcal{A}$'s advantage in breaking the IND-1CCA-MAC security of the KEM.

Subsequent Improvement by Zhou et al. [81] Note that in the security proof of TLS 1.3 handshake from IND-1CCA-MAC, the MAC oracle is only queried after the decapsulation oracle query, and both query inputs target the same $\text{SH} = (c, n_S)$. Zhou et al. discovered that the IND-1CCA-MAC security notion can be further simplified in two aspects.

The first restricts $\mathcal{A}$ to query the MAC oracle only after the decapsulation oracle has been queried. The second requires (c, n) in both oracle queries must be consistent. Zhou et al. refer to this weakened security notion as IND-1CCA-MAC*, as shown in Fig. 3. The security of TLS 1.3 handshake proven from IND-1CCA-MAC* maintains the same reduction tightness as that proven from IND-1CCA-MAC. Due to these restrictions, when analyzing the IND-1CCA-MAC* security of the underlying KEM, the values generated in the decapsulation oracle can be directly used to simulate the MAC oracle, thus reducing the ROM reduction loss. In the QROM, they employed O2H lemmas [3, 12] and the refined optional-query measure-and-reprogram technique [54] to address the random oracle reprogramming encountered in the proof, thus achieving the QROM proof.

2.2 Our Main Technique: IND-1CCA-1MAC Security

Note that in the TLS 1.3 key schedule, as shown in Fig. 4, the shared secret k does not directly participate in subsequent key derivation but derives the Handshake Secret (HS) through the `HKDF.Ext` algorithm, which is then used to derive subsequent keys. Therefore, we can regard the part in the dashed box of Fig. 2 as an *intermediate* KEM obtained during the handshake, where HS is the “key” generated by this *intermediate* KEM, and the secrets such as CHTS are derived from this key. For this new defined KEM, we provide a new intermediate security notion, called IND-1CCA-1MAC, similar to IND-1CCA-MAC*, as shown in Fig. 5. It should be noted that our defined IND-1CCA-1MAC security pertains to the *intermediate* KEM illustrated in the dashed box in Fig. 2, while the IND-1CCA-MAC* security is related to the underlying KEM used to replace the (EC)DH key exchange. The benefits of our notion are twofold.

1. Our IND-1CCA-1MAC and IND-1CCA-MAC/IND-1CCA-MAC* both aim to demonstrate that, with a single decapsulation oracle query and a single MAC oracle query, the three secrets (CHTS, SHTS, dHS) are indistinguishable from random secrets for any PPT adversary. We can prove that our IND-1CCA-1MAC is sufficient to prove the security of TLS 1.3, with a security loss equivalent to that proved by IND-1CCA-MAC/IND-1CCA-MAC*.
2. We can prove that any IND-1RCCA secure KEM is also IND-1CCA-1MAC secure, with a tight reduction. This implies that regardless of the security of the KEM used to replace the (EC)DH key exchange, as long as the *intermediate* KEM is IND-1RCCA secure, the security of TLS 1.3 handshake can be guaranteed. Furthermore, the overall tightness of

The IND-1CCA-1MAC Game for $\mathcal{A}$:		
1 :	$(pk, sk) \leftarrow \text{Gen}(1^\lambda), (k^*, c^*) \leftarrow \text{Encaps}(pk),$	$\text{HS}^* := k^*, n^* \xleftarrow{\$} \{0, 1\}^n$
2 – 5 :	...	
Decapsulation Oracle $O_{\text{Dec}}((c, n) \neq (c^*, n^*))$:		
6 – 8 :	...	
9 :	$\text{HS}' := k', \text{CHTS} := H_1(\text{HS}', H_T(c, n)), \text{SHTS} := H_2(\text{HS}', H_T(c, n))$	
10 – 11 :	...	
MAC Oracle $O_{\text{MAC}}^{\text{Dec}}((c, n) \neq (c^*, n^*), \text{tag}, \text{txt})$:		
12 – 15 :	...	
16 :	$\text{HS}' := k', \text{SHTS} := H_2(\text{HS}', H_T(c, n)), fk_S := H_4(\text{SHTS})$	
17 – 18 :	...	

Figure 5: The IND-1CCA-1MAC game for adversary $\mathcal{A}$. Differences from the IND-1CCA-1MAC* game in Fig. 3 are highlighted, with all other procedures remaining identical.

$\text{Gen}_{\text{tag}}(1^\lambda)$	$\text{Encaps}_{\text{tag}}(pk)$	$\text{Decaps}_{\text{tag}}(sk, c)$
1 : $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$	1 : $(k', c) \leftarrow \text{Encaps}(pk)$	1 : $k' := \text{Decaps}(sk, c)$
2 : return (pk, sk)	2 : $k := G(k')$	2 : if $k' = \perp$ then : return $\perp$
	3 : return (k, c)	3 : return $k := G(k')$

Figure 6: $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G]$.

the reduction depends primarily on the reduction loss from the IND-1RCCA security of the *intermediate* KEM to the security of the underlying KEM. This facilitates further exploration of the TLS 1.3 handshake with tighter reductions.

CONCLUSION 1: IND-1RCCA $\Rightarrow$ IND-1CCA-1MAC is tight (Section 4). We show that an IND-1RCCA secure KEM tightly implies IND-1CCA-1MAC security in both the ROM and the QROM, as shown in Theorems 4.1 and 4.2. The proof uses the decapsulation oracle from the IND-1RCCA game to simulate the IND-1CCA-1MAC decapsulation oracle, then gradually replaces secrets (starting with HS^*) with random values. The QROM proof follows a similar approach but uses O2H lemmas (Lemmas 3.1 and 3.2) instead of the difference lemma and applies random oracle merging techniques [55, 76] to ensure correct application of O2H.

Since an IND-1RCCA secure KEM tightly implies IND-1CCA-1MAC security, and the reduction from TLS 1.3 handshake security to IND-1CCA-1MAC security exhibits the same tightness as that to the IND-1CCA-MAC/IND-1CCA-MAC* security, it follows that as long as the tightness of the reduction from the IND-1RCCA security of the *intermediate* KEM to the CPA security of the underlying KEM is optimal, the overall reduction tightness from TLS 1.3 handshake security to the underlying CPA-secure KEM will also be optimal.

CONCLUSION 2: IND-1RCCA $\Rightarrow$ CPA has unavoidable loss (Section 5.1). We can see that the *intermediate* KEM shown in the dashed box of Fig. 2 can be viewed as being constructed from the underlying KEM via a transformation illustrated in Fig. 6, denoted as T_{tag} ³. Inspired by Jiang et al. [54, Section 5], we can prove that the reduction from the IND-1RCCA security of

³The notation T_{tag} is used to distinguish it from T_{tag} , which will be introduced later.

$\text{Gen}_{\text{tag}}(1^\lambda)$	$\text{Encaps}_{\text{tag}}(\text{pk})$	$\text{Decaps}_{\text{tag}}(\text{sk}, (c, \text{tag}))$
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	1 : $(k', c) \leftarrow \text{Encaps}(\text{pk})$	1 : $k' := \text{Decaps}(\text{sk}, c)$
2 : return (pk, sk)	2 : $k := G(k')$	2 : if $k' = \perp$ or $H'(k') \neq \text{tag}$
	3 : $\text{tag} := H'(k')$	then : return $\perp$
	4 : return $(k, (c, \text{tag}))$	3 : return $k := G(k')$

Figure 7: $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G, H']$. Differences from T_{tag} (Fig. 6) are highlighted.

KEM_{tag} to the CPA security of the underlying KEM is inherently non-tight. Informally, consider an adversary $\mathcal{B}$ against the IND-1RCCA security of KEM_{tag} , who receives a challenge ciphertext c^* (from $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk}^*)$) and need to distinguish $k_0 := G(k^*)$ from random k_1 . If $\mathcal{B}$ can generate a ciphertext c' from c^* such that $f(k^*) = \text{Decaps}(\text{sk}, c') \notin \{k^*, \perp\}$ (this property is referred to as malleability), then querying c' to the decapsulation oracle allows $\mathcal{B}$ to obtain $k := G(f(k^*)) \notin \{\perp, \text{test}\}$ with high probability. Then, $\mathcal{B}$ can query the random oracle G to find a value k^* satisfying $G(f(k^*)) = k$, and use $G(k^*)$ to distinguish k_0 from k_1 .

Based on this, we can show that $\mathcal{B}$'s distinguishing advantage is at least $\mathcal{O}(q \cdot 2^{-\lambda})$ in the ROM and $\mathcal{O}(q^2 \cdot 2^{-\lambda})$ in the QROM, where q is the number of random oracle queries and λ is the security parameter. For a λ -bit secure KEM, any PPT adversary has advantage at most $\mathcal{O}(2^{-\lambda})$. Thus, the simulation of the decapsulation oracle for KEM_{tag} inevitably introduces loss factors of $\mathcal{O}(q)$ (ROM) and $\mathcal{O}(q^2)$ (QROM), ruling out a tight reduction.

CONCLUSION 3: Ciphertext expansion tightens IND-1RCCA $\Rightarrow$ CPA (Section 5.2). In the above attack, $\mathcal{B}$ exploits malleability in the CPA-secure KEM to produce a *valid* ciphertext that can be correctly decapsulated, granting $\mathcal{B}$ extra advantage via the IND-1RCCA decapsulation oracle. To counteract this, we append a slight verifiable tag to the ciphertext, thereby reducing the chance that $\mathcal{B}$ can fabricate a valid ciphertext without knowing the decapsulated value, thus blocking this advantage.

Concretely, we augment the ciphertext with a $\text{tag} := H'(k)$ via a new hash H' (i.e. ciphertext expansion) and verify that $H'(k)$ matches the tag during decapsulation. This defines a new transformation T_{tag} as shown in Fig. 7. In the ROM, if an adversary queries a valid (c, tag) , it must have queried H' at $k' := \text{Decaps}(\text{sk}, c)$, allowing the reduction to extract k' from the query log with $H'(k') = \text{tag}$ and simulate decapsulation with only $\mathcal{O}(1)$ loss. In the QROM, this is achieved using the extractable RO-simulator technique by Don et al. [34].

Combined with Conclusion 1, this yields reductions from IND-1CCA-1MAC security to the CPA security of the underlying KEM, enabling a post-quantum TLS 1.3 handshake with tighter reductions (Section 6) as shown in Fig. 8.

A special case without requiring ciphertext expansion (Remark 5.3 in Sec. 5.2). If the underlying KEM satisfies a specific property, namely, each pair $(k, c) \leftarrow \text{Encaps}(\text{pk})$ satisfies a deterministic bijection $g(k) = c$, we can show that the reduction from the IND-1RCCA security of the *intermediate* KEM to the OW-CPA security of the underlying KEM can be further optimized.

From the attack perspective in Conclusion 2, the decapsulation oracle query provides a deterministic oracle $G(f(\cdot)) = k$, aiding the adversary $\mathcal{B}$ in finding k^* by checking $G(f(k^*)) = k$. However, if (k^*, c^*) satisfies $g(k^*) = c^*$ for a deterministic bijection g , the KEM itself offers a deterministic oracle $g(\cdot) = c^*$ to recover k^* , making the decapsulation oracle query redundant and the impossibility result in Conclusion 2 inapplicable.

As shown in [81], such KEMs can be built from OW-CPA secure rigid deterministic public-key encryptions (such as those in Classic McEliece [25] and NTRU [65]), called rigid DKEM or

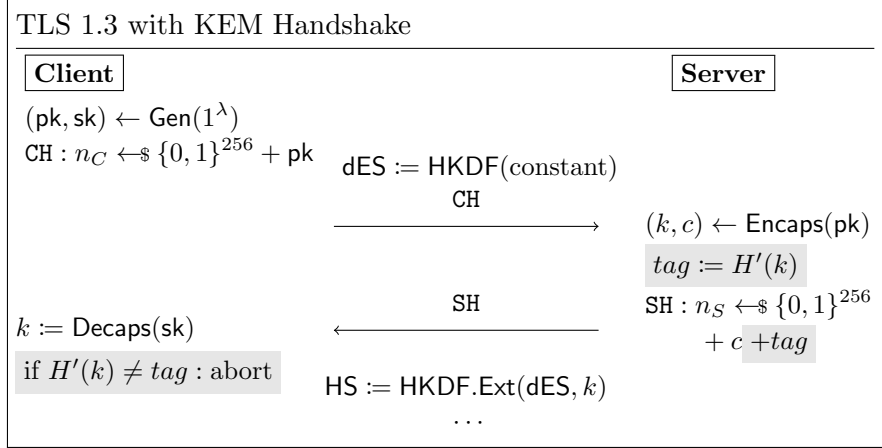


Figure 8: TLS 1.3 handshake with ciphertext expansion. Omitted process is as in Fig. 2.

rigid D-OW-CPA KEM in [81]. We prove that the reduction for these KEMs achieves a loss factor of $\mathcal{O}(1)$ without requiring ciphertext expansion in both models, a result that may support standardization of schemes like Classic McEliece [25] and NTRU [65].

On the practical efficiency impact. Given that a substantial fraction of global TLS 1.3 traffic currently uses post-quantum algorithms (e.g., in approximately 36–45% of HTTPS requests processed by Cloudflare⁴) with the vast majority employing hybrid methods [46, 75], we further demonstrate the security of our approach in a hybrid key agreement setting (Section 7.1). To quantify the performance benefits of reducing the loss factor through ciphertext expansion, we conducted experiments based on OpenSSL⁵ and liboqs⁶. Our results confirm that, for the standard OW-CPA/IND-CPA secure KEMs, using ciphertext expansion to compensate for the loss factor is more economical than increasing security parameters (Section 7.2).

3 Preliminaries

This section introduces the basic notation conventions, a security definition, and several key lemmas. Formal definitions of the (quantum) random oracle model and relevant cryptographic primitives are available in Appendices A and B, respectively.

3.1 Notation

Let $H : \mathcal{X} \rightarrow \mathcal{Y}$ denote the function H with domain $\mathcal{X}$ and codomain $\mathcal{Y}$, and Ω_H the set of such functions. For a set $\mathcal{S}$, let $|\mathcal{S}|$ be its cardinality, and $s \leftarrow \mathcal{S}$ mean s is chosen uniformly from $\mathcal{S}$. For a probabilistic (or deterministic) algorithm A with input x , $y \leftarrow A(x)$ (or $y := A(x)$) is indicated that y is the output of $A(x)$. An oracle algorithm with classical (or quantum) access to the oracle H is written A^H (or $A^{(H)}$). The indicator $[x = y]$ is 1 if $x = y$ and 0 otherwise. Let λ be the security parameter, and PPT abbreviate *probabilistic polynomial time*.

3.2 IND-1(R)CCA Security of KEM

Definition 3.1 (IND-1(R)CCA Security of KEM). The IND-1CCA (resp. IND-1RCCA) security of $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$ defined over $(\mathcal{K}, \mathcal{C})$ is defined via a game played between

⁴<https://radar.cloudflare.com/adoption-and-usage#post-quantum-encryption-adoption>

⁵<https://github.com/openssl/openssl/>

⁶<https://github.com/open-quantum-safe/liboqs>

IND-1(R)CCA Game:	Decapsulation Oracle $O_{\text{Dec}}(c \neq c^*)$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	1 : if $\text{cnt} \neq 0$ then : return $\perp$
2 : $(k_0, c^*) \leftarrow \text{Encaps}(\text{pk})$	2 : $\text{cnt} := \text{cnt} + 1$, $k := \text{Decaps}(\text{sk}, c)$
3 : $k_1 \leftarrow \mathcal{K}$, $b \leftarrow \mathcal{S}\{0, 1\}$	3 : if $k = \perp$ then : return $\perp$
4 : $\text{cnt} := 0$	4 : if $k \in \{k_0, k_1\}$ then
5 : $\hat{b} \leftarrow \mathcal{A}^{O_{\text{Dec}}}(\text{pk}, c^*, k_b)$	return test // IND-1RCCA
6 : return $[b = \hat{b}]$	5 : return k

Figure 9: Security games for IND-1(R)CCA security of KEM = (Gen, Encaps, Decaps).

a challenger and an adversary $\mathcal{A}$ as shown in Fig. 9. The adversary's advantage is defined as $\text{Adv}_{\text{KEM}}^{\text{IND-1CCA}}(\mathcal{A}) := |\Pr[b = \hat{b}] - 1/2|$ (resp. $\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{A}) := |\Pr[b = \hat{b}] - 1/2|$). If this advantage is negligible for any PPT adversary, then KEM is said to be IND-1CCA (resp. IND-1RCCA) secure. The keys k_0, k_1 and ciphertext c generated by the challenger are referred to as the challenge keys and challenge ciphertext, respectively.

3.3 The One-Way to Hiding Lemma

Lemma 3.1 (Regular One-Way to Hiding [3, Theorem 3], adapted). *Let $S \subseteq \mathcal{X}$ be a random subset. Let $G, H : \mathcal{X} \rightarrow \mathcal{Y}$ be random functions such that $\forall x \notin S, G(x) = H(x)$. Let z be a random value, and suppose (S, G, H, z) have an arbitrary joint distribution. Let A be a quantum oracle algorithm making q random oracle queries (not necessarily unitary). Define $B^{|H\rangle}$ as an oracle algorithm which, on input z , proceeds as follows: it randomly samples $i \leftarrow \mathcal{S}\{1, \dots, q\}$, runs $A^{|H\rangle}(z)$ until (just before) the i -th query, measures all query input registers in the computational basis, and outputs the set T of measurement outcomes. Then*

$$|\Pr[1 \leftarrow A^{|G\rangle}(z)] - \Pr[1 \leftarrow A^{|H\rangle}(z)]| \leq 2q\sqrt{\Pr[S \cap T \neq \emptyset : T \leftarrow B^{|H\rangle}(z)]}.$$

Lemma 3.2 (Double-Sided One-Way to Hiding [12, Lemma 5], adapted). *Let $G, H : \mathcal{X} \rightarrow \mathcal{Y}$ be random functions such that $\forall x \neq x^* \in \mathcal{X}, G(x) = H(x)$, and let z be a random value, where (G, H, x^*, z) may have an arbitrary joint distribution. Let $A^{|H\rangle}(z)$ be an oracle algorithm with quantum access to the random oracle H (not necessarily unitary). Then, there exists a double-sided oracle algorithm $B^{|G\rangle, |H\rangle}(z)$ with quantum access to both G and H , such that*

$$|\Pr[1 \leftarrow A^{|G\rangle}(z)] - \Pr[1 \leftarrow A^{|H\rangle}(z)]| \leq 2\sqrt{\Pr[x^* = \hat{x} : \hat{x} \leftarrow B^{|G\rangle, |H\rangle}(z)]}.$$

Lemma 3.3 (Search in Double-Sided Oracle [54, Lemma 2.3], adapted). *In Lemma 3.2, if $\mathcal{X} = \mathcal{X}_1 \times \mathcal{X}_2$, $x^* = (x_1, x_2) \in \mathcal{X}_1 \times \mathcal{X}_2$, and $x_1 \in \mathcal{X}_1$ is uniformly random and independent of G (or H) and z , then*

$$\Pr[x^* = \hat{x} : \hat{x} \leftarrow B^{|G\rangle, |H\rangle}(z)] \leq q_G^2/|\mathcal{X}_1| \text{ (or } q_H^2/|\mathcal{X}_1|),$$

where q_G (or q_H) is the bounded number of the G (or H) random oracle queries made by $A^{|G\rangle}(z)$ (or $A^{|H\rangle}(z)$).

3.4 Difference Lemma

Lemma 3.4 (Difference Lemma [14, Theorem 4.7]). *Let W_1, W_2 and Z denote the events defined over some probability space, and let $\bar{Z}$ be the complement of Z . If $\Pr[W_1 \wedge \bar{Z}] = \Pr[W_2 \wedge \bar{Z}]$, we have $|\Pr[W_2] - \Pr[W_1]| \leq \Pr[Z]$.*

4 IND-1RCCA Secure KEM Tightly Implies IND-1CCA-1MAC Security

This section will discuss one of our core conclusions: if a KEM is IND-1RCCA secure, then when combined with an EUF-0T secure MAC and a collision-resistant hash function H_T , modeling other key-derivation functions as random oracles, this KEM is IND-1CCA-1MAC secure (as defined in Definition 4.1).

Definition 4.1 (IND-1CCA-1MAC Security of KEM). We define the IND-1CCA-1MAC security of a KEM $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$ defined over $(\mathcal{K}_{\text{kem}}, \mathcal{C}_{\text{kem}})$ through an attack game, as shown in Fig. 5, between a challenger and an adversary $\mathcal{A}$, where H_1, H_2, H_3, H_4 , and H_D are key-derivation functions, H_T is a hash function, and $\text{MAC} = (\text{Sig}, \text{Vrf})$ is a MAC system defined over $(\mathcal{K}_{\text{mac}}, \mathcal{M}_{\text{mac}}, \mathcal{T}_{\text{mac}})$. We define the advantage of $\mathcal{A}$ with respect to KEM as $\text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) := |\Pr[b = \hat{b}] - 1/2|$. If this advantage is negligible for all PPT adversaries, we say KEM is IND-1CCA-1MAC secure.

4.1 Security of the KEM in the ROM

The security of the KEM in the ROM is captured in Theorem 4.1. Note that H_T is not required to be modeled as random oracle.

Theorem 4.1 (Security of the KEM in the ROM). *Let $\{H_i\}_{i \in \{1,2,3,4,D\}}$ be the key-derivation functions and H_T be the hash function as in Fig. 5. Assume $\{H_i\}_{i \in \{1,2,3,4,D\}}$ are modeled as random oracles, and H_T is collision resistant. For any PPT adversary $\mathcal{A}$ that breaks the IND-1CCA-1MAC security of KEM, there exists a PPT adversary $\mathcal{B}$ that breaks the IND-1RCCA security of KEM, a PPT adversary $\mathcal{C}$ that breaks the collision resistance of H_T , and a PPT adversary $\mathcal{D}$ that breaks the EUF-0T security of MAC, such that*

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) \leq & 2\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{B}) + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) \\ & + (2(q_1 + q_2 + q_3) + 1)/|\mathcal{K}_{\text{kem}}| + (q_4 + q_D)/2^n, \end{aligned}$$

where q_i ($i = 1, 2, 3, 4, D$) is a bound on the number of H_i random oracle queries made by $\mathcal{A}$, and $\mathcal{K}_{\text{kem}}$ is the key space of KEM.

Proof Sketch. We first replace HS^* (including HS' obtained in the decapsulation oracle $O_{\text{Dec}}(c, n)$ when $\text{Decaps}(\text{sk}, c) = k^*$) with a random value. By simulating O_{Dec} using the decapsulation oracle from the IND-1RCCA game, we reduce $\mathcal{A}$'s advantage in detecting this change to the IND-1RCCA security of KEM. Next, we replace $(\text{CHTS}_0, \text{SHTS}_0, \text{dHS}_0)$ and $(\text{CHTS}, \text{SHTS})$ from O_{Dec} (when $\text{Decaps}(\text{sk}, c) = k^*$) with random values. By the difference lemma (Lemma 3.4), $\mathcal{A}$ cannot notice this unless it queries H_1, H_2, H_3 on an input involving the randomized HS^* . Since $(\text{CHTS}, \text{SHTS})$ are already random, another application of the difference lemma shows that replacing tk_C, tk_S, fk_S in O_{Dec} and $O_{\text{MAC}}^{\text{Dec}}$ (for $\text{Decaps}(\text{sk}, c) = k^*$) with random values goes undetected unless $\mathcal{A}$ queries H_D and H_4 on those random values. Moreover, as fk_S in $O_{\text{MAC}}^{\text{Dec}}$ is already random, if MAC is EUF-0T secure, we can let $O_{\text{MAC}}^{\text{Dec}}$ return $\perp$ without affecting $\mathcal{A}$'s behavior. At this point, both $(\text{CHTS}_0, \text{SHTS}_0, \text{dHS}_0)$ and $(\text{CHTS}_1, \text{SHTS}_1, \text{dHS}_1)$ are uniformly random and independent from $\mathcal{A}$'s view, so $\mathcal{A}$'s distinguishing advantage is at most $1/2$. The full proof is given in Appendix C. $\square$

4.2 Security of the KEM in the QROM

Theorem 4.2 (Security of the KEM in the QROM). *Let $\{H_i\}_{i \in \{1,2,3,4,D\}}$ be the key-derivation functions and H_T be the hash function as in Fig. 5. Assume $\{H_i\}_{i \in \{1,2,3,4,D\}}$ are modeled as*

quantum-accessible random oracles, and H_T is collision resistant. For any adversary $\mathcal{A}$ that attacks the IND-1CCA-1MAC security of KEM with quantum access to the random oracles $\{H_i\}_{i \in \{1,2,3,4,D\}}$, there exists an adversary $\mathcal{B}$ that breaks the IND-1RCCA security of KEM, an adversary $\mathcal{C}$ that breaks the collision resistance of H_T , and an adversary $\mathcal{D}$ that breaks the EUF-0T security of MAC, such that

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 2\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{B}) + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) \\ &\quad + (4(q_1 + q_2 + q_3) + 1)/\sqrt{|\mathcal{K}_{\text{kem}}|} + 2(q_4 + q_D)/2^{n/2}, \end{aligned}$$

where q_i is the bounded number of the queries made by $\mathcal{A}$ to the H_i random oracles for $i \in \{1, 2, 3, 4, D\}$, and $\mathcal{K}_{\text{kem}}$ is the key space of KEM.

Proof Sketch. The proof strategy for Theorem 4.2 is similar to that of Theorem 4.1, with the main distinction being that the difference lemma (Lemma 3.4) utilized in the proof of Theorem 4.1 is replaced by the regular O2H lemma (Lemma 3.1) in the proof of Theorem 4.2. Moreover, special handling of the random oracle is necessary to apply the regular O2H lemma. The full proof can be found in Appendix D. $\square$

5 Reduction from IND-1RCCA Security

This section starts our analysis of the IND-1RCCA security for the *intermediate* KEM. Specifically, we first analyze the IND-1RCCA security of $\text{KEM}_{t_{\text{tag}}} := T_{t_{\text{tag}}}[\text{KEM}, G]$ (Fig. 6). We prove that any reduction from its security to the CPA security of the underlying KEM necessarily incurs a loss factor of $\mathcal{O}(q)$ in the ROM and $\mathcal{O}(q^2)$ in the QROM. We then analyze the IND-1RCCA security of our proposed $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G, H']$ (Fig. 7), demonstrating that it achieves a tighter loss factor compared to $T_{t_{\text{tag}}}$.

5.1 Tightness of the Reduction for $\text{KEM}_{t_{\text{tag}}}$

The IND-1RCCA security of $\text{KEM}_{t_{\text{tag}}} := T_{t_{\text{tag}}}[\text{KEM}, G]$ is relatively straightforward. Note that our main focus lies in accessing the tightness of the reduction for $\text{KEM}_{t_{\text{tag}}}$, the specific details of the reduction are not our primary concern. For the sake of completeness, we provide the full proof of the IND-1RCCA security for $\text{KEM}_{t_{\text{tag}}}$ in Appendix E, and the reduction result is presented in Theorem 5.1.

Theorem 5.1 (IND-1RCCA Security of $\text{KEM}_{t_{\text{tag}}}$). *Assume the key-derivation function G depicted in Fig. 6 is modeled as a random oracle. For any PPT adversary $\mathcal{A}$ attacking the IND-1RCCA security of $\text{KEM}_{t_{\text{tag}}} := T_{t_{\text{tag}}}[\text{KEM}, G]$, there exist a PPT adversary $\mathcal{B}$ attacking the OW-CPA security of KEM and a PPT adversary $\mathcal{B}'$ attacking the IND-CPA security of KEM s.t.*

$$\begin{aligned} \text{Adv}_{\text{KEM}_{t_{\text{tag}}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq q_G(q_G + 3)\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}), \\ \text{Adv}_{\text{KEM}_{t_{\text{tag}}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 2(q_G + 3)\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + q_G(q_G + 3)/|\mathcal{K}_{\text{kem}}|, \end{aligned}$$

where q_G is the bounded number of the random oracle G queries made by $\mathcal{A}$, and $\mathcal{K}_{\text{kem}}$ is the key space of the underlying KEM.

When the random oracle G is quantum-accessible, we have

$$\begin{aligned} \text{Adv}_{\text{KEM}_{t_{\text{tag}}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 10(q_G + 1)^2 \sqrt{\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + 1/|G|}, \\ \text{Adv}_{\text{KEM}_{t_{\text{tag}}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 10(q_G + 1) \sqrt{2\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + (q_G + 1)^2/|\mathcal{K}_{\text{kem}}| + 1/|G|}, \end{aligned}$$

where $|G|$ is the output space size of G .

We now turn to the central focus of this section: we prove that the reduction in Theorem 5.1 has an unavoidable loss factor of $\mathcal{O}(q)$ in the ROM and $\mathcal{O}(q^2)$ in the QROM in general.

Theorem 5.2. *Let $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$ be a KEM with the malleability property. There exists an adversary $\mathcal{B}$ against the IND-1RCCA security of $\text{KEM}_{\text{tag}} = T_{\text{tag}}[\text{KEM}, G]$ in the ROM (resp. QROM) with advantage*

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{B}) \geq q/(7.39|\mathcal{K}_{\text{kem}}|) \quad \left(\text{resp. } \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{B}) \geq q^2/|\mathcal{K}_{\text{kem}}| \right),$$

where q is the number of G -queries made by $\mathcal{B}$ with $\sqrt{\frac{1}{|G|} + \frac{1}{|\mathcal{K}_{\text{kem}}|}} \leq \sin\left(\frac{\pi}{6q+3}\right)$ and $q \leq |G| - 2$, and $|G| \geq 3$ is the output space size of the random oracle G .

Theorem 5.2. Let (pk^*, c^*, k_b) be the IND-1RCCA challenge for $\text{KEM}_{\text{tag}} = T_{\text{tag}}[\text{KEM}, G]$, where $(\text{pk}^*, \text{sk}^*) \leftarrow \text{Gen}(1^\lambda)$, $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk}^*)$, $k_0 := G(k^*)$, $k_1 \leftarrow \$ \text{Range}(G)$, and $b \leftarrow \$ \{0, 1\}$. WLOG, we assume $k_0 \neq k_1$. Since KEM is malleable, there exists an algorithm $\bar{B}$ that takes (pk^*, c^*) as input and outputs (f, c') such that: (1) $f(k^*) = \text{Decaps}(\text{sk}^*, c') \notin \{k^*, \perp\}$, and (2) $f(\tilde{k}) \neq \text{Decaps}(\text{sk}^*, c')$ for all $\tilde{k} \in \mathcal{K}_{\text{kem}} \setminus \{k^*\}$.

Define an indicator function $\mathbb{I}_{f,G}^k : \mathcal{K}_{\text{kem}} \rightarrow \{0, 1\}$ as

$$\mathbb{I}_{f,G}^k(k) = \begin{cases} 1 & \text{if } G(f(k)) = k, \\ 0 & \text{otherwise.} \end{cases}$$

We first consider the case in the ROM. Let $\mathcal{B}^{G, O_{\text{Dec}}}(\text{pk}^*, c^*, k_b)$ be a ROM adversary that proceeds as follows:

1. Run $\bar{B}(\text{pk}^*, c^*)$ to obtain (f, c') .
2. Query O_{Dec} with c' to obtain k . If $k \in \{\perp, \text{test}\}$, output $\perp$.
3. Randomly sample q different $\bar{k}_1, \dots, \bar{k}_q \leftarrow \$ \mathcal{K}_{\text{kem}}$, and for each $i \in \{1, \dots, q\}$, compute $\mathbb{I}_{f,G}^k(\bar{k}_i)$ by querying G .
4. If there exists a $\bar{k}_i$ such that $\mathbb{I}_{f,G}^k(\bar{k}_i) = 1$, output $1 - [G(\bar{k}_i) = k_b]$. Otherwise, output $\perp$.

Since $f(k^*) \neq \perp$, $k = O_{\text{Dec}}(c') \neq \perp$. The case $k = \text{test}$ implies $G(f(k^*)) \neq k^* \in \{k_0, k_1\}$, which occurs with probability $\frac{2}{|G|}$. Hence, the probability that $k \notin \{\perp, \text{test}\}$ is $1 - \frac{2}{|G|}$. Under the condition $k \notin \{\perp, \text{test}\}$, we have $\mathbb{I}_{f,G}^k(k^*) = 1$ with probability 1, and for any $\tilde{k} \in \mathcal{K}_{\text{kem}} \setminus \{k^*\}$, $\mathbb{I}_{f,G}^k(\tilde{k}) = 1$ with probability $\frac{1}{|G|}$. Note also that $\Pr[k^* \in \{\bar{k}_1, \dots, \bar{k}_q\}] = \frac{q}{|\mathcal{K}_{\text{kem}}|}$. Therefore, we can obtain

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{B}) &\geq (1 - \frac{2}{|G|}) \left(\frac{q}{|\mathcal{K}_{\text{kem}}|} \right) (1 - \frac{1}{|G|})^{q-1} \geq (1 - \frac{2}{|G|})^q \left(\frac{q}{|\mathcal{K}_{\text{kem}}|} \right) \stackrel{(a)}{\geq} \\ &(1 - \frac{2}{|G|})^{|G|-2} \left(\frac{q}{|\mathcal{K}_{\text{kem}}|} \right) = \left((1 - \frac{2}{|G|})^{\frac{|G|}{2}-1} \right)^2 \left(\frac{q}{|\mathcal{K}_{\text{kem}}|} \right) \stackrel{(b)}{\geq} \left(\frac{1}{e} \right)^2 \left(\frac{q}{|\mathcal{K}_{\text{kem}}|} \right) \geq \frac{q}{7.39|\mathcal{K}_{\text{kem}}|}, \end{aligned}$$

where (a) is derived from $q \leq |G| - 2$ and (b) uses the inequality $(1 - 1/x)^{x-1} \geq 1/e$ which holds for all $x > 1$.

Next, we consider the case in the QROM. Let $\mathcal{B}^{(G), O_{\text{Dec}}}$ be a QROM adversary that proceeds as follows:

1. Run $\bar{B}(\text{pk}^*, c^*)$ to obtain (f, c') .

2. Query O_{Dec} with c' to obtain k . If $k \in \{\perp, \text{test}\}$, output $\perp$.
3. Use Grover's algorithm [41] to search k^* . It applies q times Grover iterations, and each Grover iteration uses the oracle $O : |k\rangle \mapsto (-1)^{\mathbb{I}_{f,G}^k(k)} |k\rangle$.
4. After Grover's algorithm outputs $\bar{k}$, output $1 - [G(\bar{k}) = k_b]$ if $\mathbb{I}_{f,G}^k(\bar{k}) = 1$, and output $\perp$ otherwise.

As in the ROM case, the probability that $k \notin \{\perp, \text{test}\}$ in the step 2 is $1 - \frac{2}{|G|}$. Under the condition $k \notin \{\perp, \text{test}\}$, $\mathbb{I}_{f,G}^k(k^*) = 1$ with probability 1, and for any $\tilde{k} \in \mathcal{K}_{\text{kem}} \setminus \{k^*\}$, $\mathbb{I}_{f,G}^k(\tilde{k}) = 1$ with probability $\frac{1}{|G|}$. Let $N := \mathbb{E}_{G \leftarrow \mathbb{S}\Omega_G}[|\{k \in \mathcal{K}_{\text{kem}} : \mathbb{I}_{f,G}^k(k) = 1\}|]$. Then, $N = \frac{1}{|G|}(|\mathcal{K}_{\text{kem}}| - 1) + 1$. Therefore, the probability of finding a $k \in \mathcal{K}_{\text{kem}}$ such that $\mathbb{I}_{f,G}^k(k) = 1$ is $p_0 = \Pr[\mathbb{I}_{f,G}^k(k) = 1 : k \in \mathcal{K}_{\text{kem}}] = \frac{N}{|\mathcal{K}_{\text{kem}}|} = \frac{1}{|G|} + \frac{1}{|\mathcal{K}_{\text{kem}}|} - \frac{1}{|G||\mathcal{K}_{\text{kem}}|} \leq \frac{1}{|G|} + \frac{1}{|\mathcal{K}_{\text{kem}}|}$. According to [64, Section 6.1.3], after applying q times Grover iterations (each requiring one query to the random oracle G), the probability of $\bar{k}$ output in the step 3 satisfying $\mathbb{I}_{f,G}^k(\bar{k}) = 1$ is $p_1 = \sin^2(\frac{2q+1}{2}\theta)$, where $\sin^2(\frac{\theta}{2}) = p_0$, and the probability of such $\bar{k} = k^*$ is $\frac{1}{N}$.

When $\sqrt{\frac{1}{|G|} + \frac{1}{|\mathcal{K}_{\text{kem}}|}} \leq \sin\left(\frac{\pi}{6q+3}\right)$, we have $(2q+1) \leq \frac{\pi}{3 \arcsin \sqrt{p_0}} = \frac{\pi}{\frac{3}{2}\theta}$, and hence $\frac{2q+1}{2}\theta \leq \frac{\pi}{3}$.

Define $g(x) = \sin(a+x) - \sin(a) - \frac{x}{2}$. Then $g(0) = 0$ and $g'(x) = \cos(a+x) - \frac{1}{2}$. When $0 \leq a+x \leq \frac{\pi}{3}$, $\cos(a+x) \geq \frac{1}{2}$, so $g'(x) \geq 0$. Therefore, for $0 \leq a+x \leq \frac{\pi}{3}$, we have $\sin(a+x) \geq \sin(a) + \frac{x}{2}$. Given that $0 \leq \frac{2q+1}{2}\theta \leq \frac{\pi}{3}$, it follows that

$$\begin{aligned} p_1 &= \sin^2\left(\frac{2q+1}{2}\theta\right) \geq \left(\sin\left(\frac{\theta}{2}\right) + \frac{2q\theta}{2}\right)^2 \stackrel{(c)}{\geq} \left(\sin\left(\frac{\theta}{2}\right) + q \sin\left(\frac{\theta}{2}\right)\right)^2 \\ &= (q+1)^2 \sin^2\left(\frac{\theta}{2}\right) = (q+1)^2 p_0, \end{aligned}$$

where (c) uses the inequality $x \geq \sin x$ for $x > 0$. Therefore, we have

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{B}) &\geq (1 - \frac{2}{|G|})p_1(\frac{1}{N}) \geq (1 - \frac{2}{|G|})(q+1)^2 p_0(\frac{1}{N}) \\ &= (1 - \frac{2}{|G|})(q+1)^2 \frac{N}{|\mathcal{K}_{\text{kem}}|}(\frac{1}{N}) = (1 - \frac{2}{|G|})(q+1)^2 \frac{1}{|\mathcal{K}_{\text{kem}}|} \\ &\stackrel{(d)}{\geq} (1 - \frac{2}{q+2})(q+1)^2 \frac{1}{|\mathcal{K}_{\text{kem}}|} = \left(\frac{q(q+1)}{q+2}\right) \frac{1}{|\mathcal{K}_{\text{kem}}|} \stackrel{(e)}{\geq} \frac{q^2}{|\mathcal{K}_{\text{kem}}|}, \end{aligned}$$

where (d) follows from $q \leq |G| - 2$, and (e) follows from the fact that $\frac{q(q+1)}{q+2} \geq q^2$ for $q \geq 0$. $\square$

Remark 5.1. For a λ -bit secure KEM with $|\mathcal{K}_{\text{kem}}| = 2^\lambda$, we require that the advantage of any PPT adversary against its security is at most $2^{-\lambda}$. By Theorem 5.2, the decapsulation oracle provided in the IND-1RCCA game enables an adversary to break the IND-1RCCA security of KEM_{tag} with advantage at least $\mathcal{O}(q \cdot 2^{-\lambda})$ in the ROM and $\mathcal{O}(q^2 \cdot 2^{-\lambda})$ in the QROM, implying that a $\mathcal{O}(q)$ loss factor in the ROM and a $\mathcal{O}(q^2)$ loss factor in the QROM are unavoidable for KEM_{tag} .

Remark 5.2. Strictly speaking, neither IND-1CCA-MAC* [81] nor IND-1CCA-1MAC defines the security of a KEM alone; rather, each characterizes the security of an integrated scheme comprising a KEM along with other components such as MAC. By combining Theorems 4.1, 4.2, and 5.1, we actually obtain a complete reduction from the IND-1CCA-MAC* security to the CPA security of the underlying KEM. Since Theorem 5.2 implies that the reduction in Theorem 5.1 is non-tight, the overall reduction must also be non-tight⁷. This clarifies why the security reductions in [52, 81] are inherently non-tight.

⁷We can also extend the attack strategy in the proof to directly use the decapsulation oracle in the IND-1CCA-MAC* game to attack the IND-1CCA-MAC* security and achieve a similar advantage. More details are available in Appendix F.

5.2 Reductions for KEM_{tag} in the ROM and the QROM

The attack in the proof of Theorem 5.2 that uses the decapsulation oracle motivates the introduction of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G, H']$, depicted in Fig. 7. The primary modification from KEM_{tag} is the expansion of the ciphertext: a $\text{tag} := H'(k)$ is appended to the ciphertext. The decapsulation algorithm now additionally requires valid verification of this tag . Consequently, for $\mathcal{B}$ to receive a meaningful response (i.e., other than $\perp$ or test) from the decapsulation oracle, as exploited in the proof of Theorem 5.2, it must be capable of computing a correct tag autonomously. This modification can restrict the use of the decapsulation oracle to $\mathcal{B}$. The security of KEM_{tag} in the ROM and the QROM is captured in Theorems 5.3 and 5.4, respectively.

Theorem 5.3 (Security of KEM_{tag} in the ROM). *Assume the key-derivation function G and the hash function H' in Fig. 7 are modeled as random oracles. For any PPT adversary $\mathcal{A}$ attacking the IND-1RCCA security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G, H']$, there exist a PPT adversary $\mathcal{B}$ attacking the OW-CPA security of KEM and a PPT adversary $\mathcal{B}'$ attacking the IND-CPA security of KEM s.t.*

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 3(q_G + q_{H'} + 1)\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + ((q_{H'} + 1)^2 + 1)/|H'|, \\ \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 6\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + \frac{(q_{H'} + 1)^2 + 1}{|H'|} + \frac{3(q_G + q_{H'} + 1)}{|\mathcal{K}_{\text{kem}}|}, \end{aligned}$$

where q_G (or $q_{H'}$) is the bounded number of the queries made by $\mathcal{A}$ to the random oracle G (or H'), $|H'|$ is the output space size of H' .

Proof Sketch. The key of the proof lies in simulating the decapsulation oracle without sk . We can employ a random guessing strategy: for query (c, tag) , the decapsulation oracle randomly guesses the outcome. Assume the guess is correct: if the result is test or $\perp$, the oracle returns it directly; otherwise, the adversary must have queried H' at $k := \text{Decaps}(\text{sk}, c)$ to obtain a valid tag , so the reduction can retrieve k from the query log of H' for $H'(k) = \text{tag}$, thereby enabling it to simulate the decapsulation oracle's response, with a loss factor of only $\mathcal{O}(1)$.

When reducing to the OW-CPA security of KEM, an additional loss factor of $\mathcal{O}(q)$ is incurred, as the reduction must search the random oracle query log for the OW-CPA solution. For the IND-CPA security, the reduction can maintain $\mathcal{O}(1)$ loss factor. The full proof is available in Appendix G. $\square$

Theorem 5.4 (Security of KEM_{tag} in the QROM). *Let the key-derivation function G and the hash function H' in Fig. 7 be modeled as quantum-accessible random oracles. For any adversary $\mathcal{A}$ attacking the IND-1RCCA security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G, H']$, there exist an adversary $\mathcal{B}$ attacking the OW-CPA security of KEM and an adversary $\mathcal{B}'$ attacking the IND-CPA security of KEM, s.t.*

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 2(q_G + q_{H'} + 2)\sqrt{3\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3}, \\ \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 2\sqrt{6\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3 + \frac{3(q_G + q_{H'} + 2)^2}{|\mathcal{K}_{\text{kem}}|}}, \end{aligned}$$

where $\epsilon_1 := \frac{40e^2(q_{H'} + 2)^3 + 2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$, $q_G, q_{H'}$ are the bounded numbers of the queries made by $\mathcal{A}$ to random oracles G and H' , respectively, and $|H'|$ is the output space size of H' .

Proof Sketch. The proof of Theorem 5.4 follows a strategy similar to that of Theorem 5.3, differing primarily in the use of the O2H lemma (Lemmas 3.1 and 3.2) in place of the difference lemma (Lemma 3.4), and in the application of Don et al.’s extractable RO-simulator technique [34] to efficiently extract the required k' from the adversary’s “quantum H' -queries” for decapsulation simulation. Complete details are provided in Appendix H. $\square$

Remark 5.3. The $\mathcal{O}(q)$ loss factor in the ROM for an OW-CPA secure KEM arises because the reduction must *randomly* search the random oracle query log for the OW-CPA solution, but for certain special KEMs, this loss can be further reduced. Note that a special KEM built from the rigid deterministic public-key encryption, called rigid DKEM or rigid D-OW-CPA KEM in [81], ensures a one-to-one correspondence between any key-ciphertext pair (k, c) from the encapsulation, which allows the reduction to *precisely* extract the OW-CPA solution from the query log, reducing the loss to $\mathcal{O}(1)$. Moreover, this correspondence enables simulation of the decapsulation oracle with only $\mathcal{O}(1)$ loss without using *tag*, so we can further eliminate the need for ciphertext expansion when building an IND-1RCCA secure *intermediate* KEM from a rigid DKEM. Proofs for this special case in both the ROM and the QROM are available in Appendix I.

6 Security of TLS 1.3 from CPA-Secure KEM

6.1 Reductions from IND-1CCA-1MAC Secure KEM to CPA-secure KEM

With the discussions in Sections 4 and 5, we can effectively analyze the reduction tightness of the IND-1CCA-1MAC secure KEM constructed from the CPA-secure KEM. Theorems 5.3 and 5.4 indicate that $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G, H']$ constructed from a OW-CPA/IND-CPA secure KEM is IND-1RCCA secure. Additionally, Theorems 4.1 and 4.2 indicate that an IND-1RCCA secure KEM achieves IND-1CCA-1MAC security when the other building blocks are secure. This leads us to the following theorem:

Theorem 6.1 (Security of KEM with Ciphertext Expansion). *Assume $\{H_i\}_{i \in \{1,2,3,4,D\}}$, G, H' are modeled as random oracles, and H_T is collision resistant, where $\{H_i\}_{i \in \{1,2,3,4,D,T\}}$ are the functions depicted in Fig. 5, and G, H' are the functions depicted in Fig. 7. For any PPT adversary $\mathcal{A}$ that attacks the IND-1CCA-1MAC security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G, H']$, there exists a PPT adversary $\mathcal{B}$ (or $\mathcal{B}'$) that attacks the OW-CPA (or IND-CPA) security of KEM, a PPT adversary $\mathcal{C}$ that breaks the collision resistance of H_T , and a PPT adversary $\mathcal{D}$ that breaks the EUF-0T security of MAC, such that*

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 6(q_G + q_{H'} + 1)\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) \\ &\quad + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) + \frac{2(q_1+q_2+q_3)+1}{|G|} + \frac{2(q_{H'}+1)^2+2}{|H'|} + \frac{q_4+q_D}{2^n}, \\ \text{(or } \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 12\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) \\ &\quad + \frac{2(q_1+q_2+q_3)+1}{|G|} + \frac{2(q_{H'}+1)^2+2}{|H'|} + \frac{q_4+q_D}{2^n} + \frac{6(q_G+q_{H'}+1)}{|\mathcal{K}_{\text{kem}}|}, \end{aligned}$$

where q_i ($i = 1, 2, 3, 4, D$), $q_G, q_{H'}$ are bounds on the number of H_i . G, H' random oracle queries made by $\mathcal{A}$, respectively, $|G|$ (or $|H'|$) is the output space size of G (or H'), and $\mathcal{K}_{\text{kem}}$ is the key space of the underlying KEM.

When the random oracles are quantum-accessible, we have

$$\begin{aligned}
& \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) \\
& \leq 4(q_G + q_{H'} + 2) \sqrt{3\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3 + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C})} \\
& \quad + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) + \frac{4(q_1+q_2+q_3)+1}{\sqrt{|G|}} + \frac{2(q_4+q_D)}{2^{n/2}}, \\
& \text{(or } \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) \leq 4\sqrt{6\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3 + \epsilon_4} \\
& \quad + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) + \frac{4(q_1+q_2+q_3)+1}{\sqrt{|G|}} + \frac{2(q_4+q_D)}{2^{n/2}}, \text{)}
\end{aligned}$$

where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$, $\epsilon_4 := \frac{3(q_G+q_{H'}+2)^2}{|\mathcal{K}_{\text{kem}}|}$.

Theorem 6.1. This is obtained directly by combining Theorems 4.1 and 5.3, and combining Theorems 4.2 and 5.4. $\square$

6.2 Security of TLS 1.3 from IND-1CCA-1MAC Secure KEM

This section will utilize the IND-1CCA-1MAC security of KEM to demonstrate the security of the full 1-RTT handshake of TLS 1.3 with KEM (as illustrated in Fig. 2) in the multi-stage model proposed by Dowling et al. [38]. The proof results are presented in Theorem 6.2.

Theorem 6.2 (Security of the Full 1-RTT Handshake of TLS 1.3 with KEM). *Assume that HKDF.Ext, HKDF.Exp_j ($j = 0, 4, 5, 6$), and HKDF.TK shown in Fig. 2 are modeled as random oracles. Let the hash function H used to compute the hashed transcripts T_i be a random oracle. For any PPT multi-stage adversary $\mathcal{A}$, there exist PPT adversaries $\{\mathcal{B}_i\}_{i \in \{0, \dots, 6\}}$, such that*

$$\text{Adv}_{\text{TLS1.3-1RTT}}^{\text{multi-stage}}(\mathcal{A}) \leq 6t_s \cdot \left(\text{Adv}_H^{\text{CR}}(\mathcal{B}_1) + t_u \cdot \text{Adv}_{\text{Sig}}^{\text{euf-cma}}(\mathcal{B}_2) + t_s \cdot \left(\text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{B}_3) + 2 \cdot \right. \right.$$

Adv_{HKDF.Ext}^{prf}($\mathcal{B}_4$) + Adv_{HKDF.Ext}^{prf}($\mathcal{B}_5$) + Adv_{HKDF.Exp}^{prf}($\mathcal{B}_6$)) $\left. \right)$, where t_s (or t_u) is the bounded number of sessions (or users), and Sig is the signature scheme used for the server authentication (not shown in Fig. 2).

Proof Sketch. The proof of Theorem 6.2 is the same as those of [52, Theorem 5] and [81, Theorem 6], except that the IND-1CCA-MAC(*) is replaced by IND-1CCA-1MAC. Note that both IND-1CCA-MAC(*) and IND-1CCA-1MAC are used to bound the advantage of distinguishing the secrets (CHTS, SHTS, dHS) from random secrets in the proof. The full proof is provided in Appendix K. $\square$

With the similar method as [52] and [81], we can also prove that IND-1CCA-1MAC secure KEM is sufficient to prove the security of the TLS 1.3 PSK-(EC)DHE 0-RTT handshake with KEM.

Theorem 6.3 (Security of the TLS 1.3 PSK-(EC)DHE 0-RTT Handshake with KEM). *The modified TLS 1.3 handshake in the pre-shared key (optional) 0-RTT mode with key exchange (i.e., TLS 1.3 PSK-(EC)-DHE 0-RTT) is secure in the multi-stage model if the KEM (as shown in the dashed box of Fig. 2) is IND-1CCA-1MAC (and signature, MAC, etc. are secure), in the sense of Dowling et al. [38].*

Theorem 6.3. The proof is identical to those of [52, Theorem 6] and [81, Theorem 7], except that IND-1CCA-MAC and IND-1CCA-MAC* are replaced with IND-1CCA-1MAC, as done in the proof of Theorem 6.2. $\square$

$\text{Gen}_{\parallel}(1^\lambda)$	$\text{Encaps}_{\parallel}(\text{pk}_1 \parallel \text{pk}_2)$	$\text{Decaps}_{\parallel}(\text{sk}_1 \parallel \text{sk}_2, c_1 \parallel c_2)$
1 : $(\text{pk}_1, \text{sk}_1) \leftarrow \text{Gen}_1(1^\lambda)$	1 : $(k'_1, c_1) \leftarrow \text{Encaps}_1(\text{pk}_1)$	1 : $k'_1 := \text{Decaps}_1(\text{sk}_1, c_1)$
$(\text{pk}_2, \text{sk}_2) \leftarrow \text{Gen}_2(1^\lambda)$	$(k'_2, c_2) \leftarrow \text{Encaps}_2(\text{pk}_2)$	$k'_2 := \text{Decaps}_2(\text{sk}_2, c_2)$
2 : $\text{pk}_{\parallel} := \text{pk}_1 \parallel \text{pk}_2$	2 : $k_{\parallel} := k'_1 \parallel k'_2$	2 : if $\perp \in \{k'_1, k'_2\}$ then
$\text{sk}_{\parallel} := \text{sk}_1 \parallel \text{sk}_2$	$c_{\parallel} := c_1 \parallel c_2$	return $\perp$
3 : return $(\text{pk}_{\parallel}, \text{sk}_{\parallel})$	3 : return $(k_{\parallel}, c_{\parallel})$	3 : return $k_{\parallel} := k'_1 \parallel k'_2$

Figure 10: $\text{KEM}_{\parallel} := \text{KEM}_1 \parallel \text{KEM}_2$, where $a \parallel b$ represents the concatenation of a and b .

Remark 6.1. Combining Theorem 6.1 and Theorems 6.2–6.3, we can obtain the overall security proof of TLS 1.3 from CPA-secure KEM in the ROM and the QROM, where the CPA-secure KEM is used to replace the (EC)DH key exchange. Hence, we finally obtain the post-quantum TLS 1.3 from CPA-secure KEM with tighter reduction than those proposed and proven by Huguenin-Dumittan and Vaudenay [52] and Zhou et al. [81].

7 Practical Considerations and Analysis

7.1 Security in the Hybrid Key Agreement

In previous sections, we show that ciphertext expansion can reduce the reduction loss for the TLS 1.3 handshake from a *single* KEM. We now explore its effect on *hybrid* key exchange. Note that the hybrid key exchange in the draft [75] combines traditional and post-quantum algorithms by a simple concatenation. Specifically, for $\text{KEM}_1 = (\text{Gen}_1, \text{Encaps}_1, \text{Decaps}_1)$ and $\text{KEM}_2 = (\text{Gen}_2, \text{Encaps}_2, \text{Decaps}_2)$, with algorithms like (EC)DH viewed as KEMs, the hybrid $\text{KEM}_{\parallel} := \text{KEM}_1 \parallel \text{KEM}_2$ is shown in Fig. 10, where $a \parallel b$ denotes the concatenation of a and b . We now analyze the IND-1CCA-1MAC security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\parallel}, G, H']$ under the assumption that exactly one of the underlying KEMs (w.l.o.g., KEM_1) is CPA-secure.

When KEM_1 is OW-CPA secure, the situation is relatively straightforward. We only need to show that $\text{KEM}_{\parallel}$ is OW-CPA secure.

Lemma 7.1 (OW-CPA Security of $\text{KEM}_{\parallel}$). *For any PPT adversary $\mathcal{A}$ against the OW-CPA security of $\text{KEM}_{\parallel} := \text{KEM}_1 \parallel \text{KEM}_2$, there exists a PPT adversary $\mathcal{B}$ against the OW-CPA security of KEM_1 , such that*

$$\text{Adv}_{\text{KEM}_{\parallel}}^{\text{OW-CPA}}(\mathcal{A}) \leq \text{Adv}_{\text{KEM}_1}^{\text{OW-CPA}}(\mathcal{B}).$$

Lemma 7.1. This proof is straightforward. Upon receiving (pk_1^*, c_1^*) from $\mathcal{B}$'s OW-CPA challenger, $\mathcal{B}$ computes $(\text{pk}_2^*, \text{sk}_2^*) \leftarrow \text{Gen}_2(1^\lambda)$ and $(k_2^*, c_2^*) \leftarrow \text{Encaps}_2(\text{pk}_2^*)$, and sends $(\text{pk}_1^* \parallel \text{pk}_2^*, c_1^* \parallel c_2^*)$ to $\mathcal{A}$. Once $\mathcal{A}$ outputs its OW-CPA solution $\hat{k}_1 \parallel \hat{k}_2$ for $(\text{pk}_1^* \parallel \text{pk}_2^*, c_1^* \parallel c_2^*)$, $\mathcal{B}$ directly outputs $\hat{k}_1$. We can see that, if $\hat{k}_1 \parallel \hat{k}_2 = \text{Decaps}_{\parallel}(\text{sk}_1^* \parallel \text{sk}_2^*, c_1^* \parallel c_2^*)$, we must have $\hat{k}_1 = \text{Decaps}_1(\text{sk}_1^*, c_1^*)$. Therefore, we have $\text{Adv}_{\text{KEM}_1}^{\text{OW-CPA}}(\mathcal{B}) \geq \text{Adv}_{\text{KEM}_{\parallel}}^{\text{OW-CPA}}(\mathcal{A})$. $\square$

When KEM_1 is IND-CPA secure, we can prove that $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\parallel}, G, H']$ is IND-1RCCA secure with loss factor of only $\mathcal{O}(1)$.

Lemma 7.2 (IND-1RCCA Security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\parallel}, G, H']$). *Assume G, H' in Fig. 7 are modeled as random oracles. For any PPT adversary $\mathcal{A}$ attacking the IND-1RCCA security*

of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_1 \parallel \text{KEM}_2, G, H']$, there exist a PPT adversary $\mathcal{B}'$ attacking the IND-CPA security of KEM_1 , such that

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 6\text{Adv}_{\text{KEM}_1}^{\text{IND-CPA}}(\mathcal{B}') + \frac{(q_{H'}+1)^2+1}{|H'|} + \frac{3(q_G+q_{H'}+1)}{|\mathcal{K}_{\text{KEM}_1}|},$$

where q_G (or $q_{H'}$) is the bounded number of $\mathcal{A}$'s queries to the random oracle G (or H'), $|H'|$ is the output space size of H' , and $\mathcal{K}_{\text{KEM}_1}$ is the key space of KEM_1 .

When the random oracles G and H' are quantum-accessible, we have

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 2\sqrt{6\text{Adv}_{\text{KEM}_1}^{\text{IND-CPA}}(\mathcal{B}') + \epsilon_1 + (q_{H'}+2)\epsilon_2 + 2\epsilon_3 + \frac{3(q_G+q_{H'}+2)^2}{|\mathcal{K}_{\text{KEM}_1}|}},$$

where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$.

Proof Sketch. The proof of Lemma 7.2 is exactly the same as the proofs of Theorem 5.3 and Theorem 5.4 for the case where KEM is IND-CPA secure with KEM replaced by $\text{KEM}_{\parallel}$, except that in the final game, the key k_1^* of KEM_1 , rather than the entire key $k_{\parallel}^* = k_1^* \parallel k_2^*$ of $\text{KEM}_{\parallel}$, is replaced by a random key. For details, see Appendices L.1 and L.2. $\square$

Now we can derive the IND-1CCA-1MAC security of the *intermediate* KEM in the hybrid key exchange setting.

Theorem 7.3 (IND-1CCA-1MAC Security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\parallel}, G, H']$). Assume that $\{H_i\}_{i \in \{1,2,3,4,D\}}$, G, H' are modeled as random oracles, and H_T is collision resistant, where $\{H_i\}_{i \in \{1,2,3,4,D,T\}}$, G, H' are the functions depicted in Fig. 5 and Fig. 7. For any PPT adversary $\mathcal{A}$ attacking the IND-1CCA-1MAC security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_1 \parallel \text{KEM}_2, G, H']$, there exists a PPT adversary $\mathcal{B}$ (or $\mathcal{B}'$) attacking the OW-CPA (or IND-CPA) security of KEM_1 , a PPT adversary $\mathcal{C}$ breaking the collision resistance of H_T , and a PPT adversary $\mathcal{D}$ breaking the EUF-0T security of MAC, s.t.

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 6(q_G + q_{H'} + 1)\text{Adv}_{\text{KEM}_1}^{\text{OW-CPA}}(\mathcal{B}) + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) \\ &\quad + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) + \frac{2(q_1+q_2+q_3)+1}{|G|} + \frac{2(q_{H'}+1)^2+2}{|H'|} + \frac{q_4+q_D}{2^n}, \end{aligned}$$

$$\begin{aligned} (\text{or } \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 12\text{Adv}_{\text{KEM}_1}^{\text{IND-CPA}}(\mathcal{B}') + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) \\ &\quad + \frac{2(q_1+q_2+q_3)+1}{|G|} + \frac{2(q_{H'}+1)^2+2}{|H'|} + \frac{q_4+q_D}{2^n} + \frac{6(q_G+q_{H'}+1)}{|\mathcal{K}_{\text{KEM}_1}|},) \end{aligned}$$

where q_i ($i = 1, 2, 3, 4, D$), $q_G, q_{H'}$ are bounds on the number of H_i, G, H' random oracle queries made by $\mathcal{A}$, respectively, $|G|$ (or $|H'|$) is the output space size of G (or H'), and $\mathcal{K}_{\text{KEM}_1}$ is the key space of KEM_1 .

When the random oracles are quantum-accessible, we have

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 4(q_G + q_{H'} + 2)\sqrt{3\text{Adv}_{\text{KEM}_1}^{\text{OW-CPA}}(\mathcal{B}) + \epsilon_1 + (q_{H'}+2)\epsilon_2 + 2\epsilon_3} + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) \\ &\quad + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) + \frac{4(q_1+q_2+q_3)+1}{\sqrt{|G|}} + \frac{1}{|G|} + \frac{2(q_4+q_D)}{2^{n/2}}, \end{aligned}$$

$$\begin{aligned} (\text{or } \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 4\sqrt{6\text{Adv}_{\text{KEM}_1}^{\text{IND-CPA}}(\mathcal{B}') + \epsilon_1 + (q_{H'}+2)\epsilon_2 + 2\epsilon_3} + \epsilon_4 \\ &\quad + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) + \frac{4(q_1+q_2+q_3)+1}{\sqrt{|G|}} + \frac{2(q_4+q_D)}{2^{n/2}},) \end{aligned}$$

where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$, $\epsilon_4 := \frac{3(q_G+q_{H'}+2)^2}{|\mathcal{K}_{\text{KEM}_1}|}$.

Theorem 7.3. This theorem follows from combining Lemma 7.1 with Theorem 6.1, and Lemma 7.2 with Theorems 4.1 and 4.2. $\square$

Remark 7.1. Theorem 7.3 shows that for hybrid key exchange with ciphertext expansion, as long as one of the component algorithms is CPA-secure, the *intermediate* KEM remains IND-1CCA-1MAC secure and maintains the same reduction bound as in Theorem 6.1 for the single-KEM case. Combining Theorem 7.3 with Theorems 6.2 and 6.3 yields a complete security proof for the TLS 1.3 handshake with hybrid key exchange and ciphertext expansion.

7.2 Practical Efficiency Implications

Tightness vs. Ciphertext Expansion. Our approach introduces $H'(k)$ as ciphertext expansion to achieve tighter reductions. This prompts the question: *is ciphertext expansion a viable trade-off for mitigating reduction losses?* While increasing security parameters could compensate for such losses, it introduces *broader* system costs in parameter selection, computation, and communication.

To quantify this, we follow [7] and benchmark four hybrid schemes (treated as KEMs), X25519MLKEM768, X448MLKEM1024, SecP256r1MLKEM768, and SecP384r1MLKEM1024, with the FO-like transformation [49] removed. The benchmarks were conducted under two representative settings: (i) OpenSSL 3.6.0 on an Ubuntu 24.04.3 LTS virtual machine with an Intel Core i7-1165G7 CPU and 16 GB RAM; and (ii) SIMD-optimized implementations (NEON on arm64 via liboqs 0.15.0 + oqsprovider 0.11.0 + OpenSSL 3.4.3⁸) on a system running macOS Sequoia 15.6 with an Apple M4 chip and 16 GB of memory. These modifications are for experimental comparison only and just serve to explore lighter alternatives, not implying that we intend to violate existing standards [1,62]. Experimental results are provided in Appendix M.

For X25519MLKEM768 or SecP256r1MLKEM768, to compensate for an $\mathcal{O}(q^2)$ reduction loss (e.g., $q = 2^{32}$), the security strength must be increased by 64 bits. This may prompt consideration of more secure alternatives such as X448MLKEM1024 or SecP384r1MLKEM1024, which increase key size by ≥ 400 bytes, ciphertext size by ≥ 500 bytes, and average encapsulation/decapsulation time by $2.4\times/3.7\times$ and $15.0\times/10.9\times$, respectively, under the optimized ARM implementation (similar multiplicative overheads can be observed in the Intel environment). In contrast, ciphertext expansion introduces only a slight computational overhead for $H'(k_{\parallel})$ ($< 1\mu s$) and a modest communication overhead (64 bytes for SHA3-512), which is practically acceptable.

It is also worth mentioning that, compared to Zhou et al.’s work [81], the MAC term in our QROM reduction is also tight (our $\epsilon_{R_{\text{mac}}} \approx \epsilon'_{\mathcal{A}}$ compared with their $\epsilon_{R_{\text{mac}}} \approx \mathcal{O}(1/q^4)\epsilon_{\mathcal{A}}^2$ when reduction to the underlying KEM’s OW-CPA security and $\epsilon_{R_{\text{mac}}} \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}^2$ when reduction to the underlying KEM’s IND-CPA security⁹), demonstrating broader applicability in quantum settings.

Further Discussion. Note that one important conclusion of this work is that the security of TLS 1.3 handshake can be guaranteed as long as the *intermediate* KEM (depicted in the dashed box in Fig. 2) is IND-1RCCA secure, and the ciphertext expansion proposed in this work serves merely as one way to construct an IND-1RCCA secure KEM with tighter reductions from a CPA-secure KEM. Thus, whether there exist alternative methods beyond ciphertext expansion

⁸OpenSSL 3.4.3 is used because with OpenSSL $\geq 3.5.0$, oqsprovider $\geq 0.9.0$ automatically disables certain liboqs algorithms. See <https://github.com/open-quantum-safe/oqs-provider/blob/main/README.md> for details.

⁹Here, $\epsilon_{R_{\text{mac}}}$ is the advantage of the reduction algorithm R_{mac} with respect to the EUF-OT security of the underlying MAC, $\epsilon_{\mathcal{A}}/\epsilon'_{\mathcal{A}}$ represents the advantage of the adversary $\mathcal{A}$ with respect to the IND-1CCA-MAC*/IND-1CCA-1MAC security of the obtained KEM, and q is the number of the random oracle queries made by $\mathcal{A}$.

for achieving tighter IND-1RCCA security from a CPA-secure KEM remains an interesting open problem.

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A Supporting Material: The (Quantum) Random Oracle Model

In cryptographic models, the random oracle model (ROM), introduced by Bellare and Rogaway [6], treats hash functions as idealized public random oracles. In this model, an adversary can only obtain the value of a hash function $H(x)$ by querying the random oracle on x , effectively treating H as a truly random function.

The quantum random oracle model (QROM) [13] extends the ROM by allowing adversaries to query the random oracle in a quantum superposition state. In the QROM, the random oracle H is represented as a unitary operator U_H , which transforms states of the form $|x, y\rangle$ to $|x, y \oplus H(x)\rangle$. Classical queries are also permissible in this model, which can be understood as querying the oracle with the input state $|x, 0\rangle$ and subsequently measuring the second register to obtain the classical output [34]. This model is essential for analyzing cryptographic protocols against quantum adversaries, as it accommodates the enhanced query capabilities inherent to quantum computing.

B Supporting Material: Cryptographic Primitives

Definition B.1 (Key Encapsulation Mechanism, KEM). A key encapsulation mechanism (KEM) is defined by three PPT algorithms with security parameter λ , a key space $\mathcal{K}$, and a ciphertext space $\mathcal{C}$: (1) The key generation algorithm Gen is a probabilistic algorithm that takes 1^λ as input and outputs a public/private key pair (pk, sk) . (2) The encapsulation algorithm Encaps is a probabilistic algorithm that takes pk as input and outputs a key-ciphertext pair $(k, c) \in \mathcal{K} \times \mathcal{C}$. (3) The decapsulation algorithm Decaps is deterministic, taking sk and $c \in \mathcal{C}$ as input, and returning $k \in \mathcal{K} \cup \perp$. We say that $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$ is defined over $(\mathcal{K}, \mathcal{C})$.

The correctness requirement of a KEM is that for all $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$ and all $(k, c) \leftarrow \text{Encaps}(\text{pk})$, it holds that $\text{Decaps}(\text{sk}, c) = k$. A KEM is said to be with explicit rejection if $\perp \notin \mathcal{K}$, or with implicit rejection if $\perp \in \mathcal{K}$ represents a random value.

Definition B.2 (OW-CPA Security of KEM). The OW-CPA security of a KEM $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$ defined over $(\mathcal{K}, \mathcal{C})$ is defined through an attack game between a challenger and an adversary $\mathcal{A}$, as shown in the left part of Fig. 11. The adversary's advantage is defined as $\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{A}) := \Pr[k^* = \hat{k}]$. If this advantage is negligible for all PPT adversaries, we say that KEM is OW-CPA secure. The key k^* and the ciphertext c^* generated by the challenger are referred to as the challenge key and the challenge ciphertext, respectively.

OW-CPA Game:	IND-CPA Game:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda), k_1 \leftarrow \mathcal{K}$
2 : $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$	2 : $(k_0, c^*) \leftarrow \text{Encaps}(\text{pk}), b \leftarrow \{0, 1\}$
3 : $\hat{k} \leftarrow \mathcal{A}(\text{pk}, c^*)$	3 : $\hat{b} \leftarrow \mathcal{A}(\text{pk}, c^*, k_b)$
4 : return $[k^* = \hat{k}]$	4 : return $[b = \hat{b}]$

Figure 11: Security games for OW-CPA and IND-CPA security of $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$.

Definition B.3 (IND-CPA Security of KEM). The IND-CPA security of a KEM $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$ defined over $(\mathcal{K}, \mathcal{C})$ is defined by an attack game between a challenger and an adversary $\mathcal{A}$, as shown in the right part of Fig. 11. The adversary's advantage is defined as $\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{A}) := |\Pr[b = \hat{b}] - 1/2|$. If this advantage is negligible for all PPT adversaries, we say that KEM is IND-CPA secure. The key k_b and the ciphertext c^* generated by the challenger are referred to as the challenge keys and the challenge ciphertext, respectively.

Malleability. In this paper, we say a $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$ defined over $(\mathcal{K}, \mathcal{C})$ is malleable, if for any $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$, there exists an algorithm B that takes (pk^*, c^*) as input and outputs (f, c') such that: (1) $f(k^*) = \text{Decaps}(\text{sk}^*, c') \notin \{k^*, \perp\}$, and (2) $f(\tilde{k}) \neq \text{Decaps}(\text{sk}^*, c')$ for all $\tilde{k} \in \mathcal{K} \setminus \{k^*\}$.

Definition B.4 (IND-1(R)CCA Security of KEM). The IND-1CCA (resp. IND-1RCCA) security of $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$ defined over $(\mathcal{K}, \mathcal{C})$ is defined via a game played between a challenger and an adversary $\mathcal{A}$ as shown in Fig. 12. The adversary's advantage is defined as $\text{Adv}_{\text{KEM}}^{\text{IND-1CCA}}(\mathcal{A}) := |\Pr[b = \hat{b}] - 1/2|$ (resp. $\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{A}) := |\Pr[b = \hat{b}] - 1/2|$). If this advantage is negligible for any PPT adversary, then KEM is said to be IND-1CCA (resp. IND-1RCCA) secure. The keys k_0, k_1 and ciphertext c generated by the challenger are referred to as the challenge keys and challenge ciphertext, respectively.

IND-1(R)CCA Game:	Decapsulation Oracle $O_{\text{Dec}}(c \neq c^*)$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	1 : if $\text{cnt} \neq 0$ then : return $\perp$
2 : $(k_0, c^*) \leftarrow \text{Encaps}(\text{pk})$	2 : $\text{cnt} := \text{cnt} + 1, k := \text{Decaps}(\text{sk}, c)$
3 : $k_1 \leftarrow \$\mathcal{K}, b \leftarrow \$\{0, 1\}$	3 : if $k = \perp$ then : return $\perp$
4 : $\text{cnt} := 0$	4 : if $k \in \{k_0, k_1\}$ then
5 : $\hat{b} \leftarrow \mathcal{A}^{O_{\text{Dec}}}(\text{pk}, c^*, k_b)$	return test // IND-1RCCA
6 : return $[b = \hat{b}]$	5 : return k

Figure 12: Security games for IND-1(R)CCA security of $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$.

Note that the IND-1RCCA security is strictly weaker than IND-1CCA security, as it permits anyone to generate new ciphertexts that decapsulate to the challenge keys. This mirrors the IND-RCCA security framework for the public-key encryption (PKE) schemes in [23, Fig. 3], adapted to the KEM setting.

Definition B.5 (Message Authentication Code, MAC). A message authentication code (MAC) system MAC is specified by two PPT algorithms with a key space $\mathcal{K}$, a message space $\mathcal{M}$, and a tag space $\mathcal{T}$: (1) The signing algorithm Sig is a probabilistic algorithm that takes a key $k \in \mathcal{K}$ and a message $m \in \mathcal{M}$ as input and outputs a tag $t \in \mathcal{T}$. (2) The verification algorithm Vrf is a deterministic algorithm that takes $(k, m, t) \in \mathcal{K} \times \mathcal{M} \times \mathcal{T}$ as input and outputs **accept** or **reject**. We say that $\text{MAC} = (\text{Sig}, \text{Vrf})$ is defined over $(\mathcal{K}, \mathcal{M}, \mathcal{T})$.

The correctness requirement of a MAC is that for all $k \in \mathcal{K}$, and all $m \in \mathcal{M}$, we have $\text{Vrf}(k, m, \text{Sig}(k, m)) = \text{accept}$.

Definition B.6 (EUF-0T Security of MAC [52, Definition 7]). We say a MAC system $\text{MAC} = (\text{Sig}, \text{Vrf})$, defined over $(\mathcal{K}, \mathcal{M}, \mathcal{T})$, is EUF-0T secure if for any PPT adversary $\mathcal{A}$, the advantage

$$\text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{A}) := \Pr[\text{Vrf}(k, m, t) = \text{accept} : (m, t) \leftarrow \mathcal{A}, k \leftarrow \$\mathcal{K}]$$

is negligible in the security parameter.

Definition B.7 (Collision Resistance of the Hash Function [14, Sec. 8.1]). We say a (keyless) hash function $H : \mathcal{M} \rightarrow \mathcal{T}$, with some (large) message space $\mathcal{M}$ and some (small) digest space $\mathcal{T}$, is collision resistant, if for any PPT adversary $\mathcal{A}$, the advantage

$$\text{Adv}_H^{\text{CR}}(\mathcal{A}) := \Pr[H(m_0) = H(m_1) : (m_0, m_1) \leftarrow \mathcal{A}]$$

is negligible in the security parameter.

C Supporting Material: Proof of Theorem 4.1

Theorem 4.1 (Security of the KEM in the ROM). *Let $\{H_i\}_{i \in \{1,2,3,4,D\}}$ be the key-derivation functions and H_T be the hash function as in Fig. 5. Assume $\{H_i\}_{i \in \{1,2,3,4,D\}}$ are modeled as random oracles, and H_T is collision resistant. For any PPT adversary $\mathcal{A}$ that breaks the IND-1CCA-1MAC security of KEM, there exists a PPT adversary $\mathcal{B}$ that breaks the IND-1RCCA security of KEM, a PPT adversary $\mathcal{C}$ that breaks the collision resistance of H_T , and a PPT adversary $\mathcal{D}$ that breaks the EUF-0T security of MAC, such that*

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 2\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{B}) + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) \\ &\quad + (2(q_1 + q_2 + q_3) + 1)/|\mathcal{K}_{\text{kem}}| + (q_4 + q_D)/2^n, \end{aligned}$$

Games G_0 to G_6 :

```

1 :  $(pk, sk) \leftarrow \text{Gen}(1^\lambda), (k^*, c^*) \leftarrow \text{Encaps}(pk), HS^* \leftarrow \mathcal{K}_{\text{kem}}, n^* \leftarrow \{0, 1\}^n$ 
2 :  $HS^* := k^* \quad // \quad G_0 - G_1$ 
3 :  $(CHTS_0, SHTS_0, dHS_0) \leftarrow \{0, 1\}^{3n}, (CHTS_1, SHTS_1, dHS_1) \leftarrow \{0, 1\}^{3n}$ 
4 :  $CHTS_0 := H_1(HS^*, H_T(c^*, n^*)), SHTS_0 := H_2(HS^*, H_T(c^*, n^*)),$ 
    $dHS_0 := H_3(HS^*) \quad // \quad G_0 - G_4$ 
5 :  $b \leftarrow \{0, 1\}, cnt_0, cnt_1 := 0$ 
6 :  $\hat{b} \leftarrow \mathcal{A}^{\{H_i\}_{i \in \{1, 2, 3, 4, D\}}, O_{\text{Dec}}, O_{\text{MAC}}^{\text{Dec}}}(pk, c^*, n^*, (CHTS_b, SHTS_b, dHS_b))$ 
7 : return  $[b = \hat{b}]$ 

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Figure 13: Games G_0 to G_6 for the proof of Theorem 4.1.

where q_i ($i = 1, 2, 3, 4, D$) is a bound on the number of H_i random oracle queries made by $\mathcal{A}$, and $\mathcal{K}_{\text{kem}}$ is the key space of KEM.

Theorem 4.1. We define the challenger for the adversary $\mathcal{A}$ in game G_j , where $j = 0, \dots, 6$, as shown in Fig. 13 and Fig. 14. In each game, b is the random bit sampled by the challenger, while $\hat{b}$ is the bit output by $\mathcal{A}$ at the end of the game. We define W_j be the event that $b = \hat{b}$ in game G_j .

Game G_0 . This game is identical to the IND-1CCA-1MAC game of KEM, except that the values HS^* , $(CHTS_0, SHTS_0, dHS_0)$ are sampled randomly at the beginning of the game, but they are reset to the right values immediately. Note that since IND-1CCA-1MAC security game restricts $\mathcal{A}$ to make the MAC oracle query only after making the decapsulation oracle query, and the input components (c, n) must be the same, HS' and SHTS obtained in the MAC oracle are consistent with that in the decapsulation oracle. Thus, the MAC oracle can directly use HS' and SHTS obtained in the decapsulation oracle to answer $\mathcal{A}$'s MAC oracle query. These modifications do not affect the challenger's behavior from $\mathcal{A}$'s view. Therefore, we have

$$|\Pr[W_0] - 1/2| = \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}).$$

Game G_1 . This game is the same as game G_0 , except that in the decapsulation oracle, when the input component c satisfying $\text{Decaps}(sk, c) = k^*$, it resets $HS' := HS^*$. When $\text{Decaps}(sk, c) = k^*$, this is equivalent to setting $HS' := HS^* = k^* = \text{Decaps}(sk, c)$ as in game G_0 . Thus,

$$\Pr[W_1] = \Pr[W_0].$$

Game G_2 . In this game, we delete the step $HS^* := k^*$, i.e., line 2 in Fig. 13, in the initialization phase of the challenger in game G_1 . At this point, HS^* is a random value. We claim that if $\mathcal{A}$ can detect this modification, we can obtain a new adversary $\mathcal{B}$ to break the IND-1RCCA security of the underlying KEM. In particular, upon receiving $(pk^*, c^*, k_{b'})$ from $\mathcal{B}$'s IND-1RCCA game challenger, where b' denotes the random bit sampled by $\mathcal{B}$'s challenger, $\mathcal{B}$ first sets $HS^* := k_{b'}$, then obtains $(n^*, b, CHTS_b, SHTS_b, dHS_b)$ in the same way as in game G_1 , and finally sends $(pk^*, c^*, n^*, (CHTS_b, SHTS_b, dHS_b))$ to $\mathcal{A}$. To simulate the decapsulation oracle and the MAC oracle, we make the following modifications:

- Decapsulation oracle: It forward the queried c to the decapsulation oracle of $\mathcal{B}$'s IND-1RCCA game. If it returns **test**, implying that $\text{Decaps}(sk^*, c) \in (k_0, k_1)$, it sets $HS' := HS^*$; otherwise, it obtains $k' := \text{Decaps}(sk^*, c)$ and sets $HS' := k'$. Other values can be obtained in the same way as in game G_1 .

Decapsulation Oracle $O_{\text{Dec}}((c, n) \neq (c^*, n^*))$:
1 : if $\text{cnt}_0 \neq 0$ then : return $\perp$ 2 : $\text{cnt}_0 := \text{cnt}_0 + 1$, $\text{HS}' := \text{Decaps}(\text{sk}, c)$ 3 : if $\text{Decaps}(\text{sk}, c) = k^*$ then : $\text{HS}' := \text{HS}^*$ $\parallel G_1 - G_6$ 4 : if $\text{HS}' = \perp$ then : return $\perp$ 5 : if $H_T(c, n) = H_T(c^*, n^*)$ then : return $\perp$ $\parallel G_3 - G_6$ 6 : $\text{CHTS} := H_1(\text{HS}', H_T(c, n))$, $\text{SHTS} := H_2(\text{HS}', H_T(c, n))$ 7 : if $\text{Decaps}(\text{sk}, c) = k^*$ then : $(\text{CHTS}, \text{SHTS}) \leftarrow \{0, 1\}^{2n}$ $\parallel G_5 - G_6$ 8 : $tk_C := H_D(\text{CHTS})$, $tk_S := H_D(\text{SHTS})$ 9 : if $\text{Decaps}(\text{sk}, c) = k^*$ then : $tk_S \leftarrow \{0, 1\}^n$ $\parallel G_6$ 10 : return (tk_C, tk_S)
MAC Oracle $O_{\text{MAC}}^{\text{Dec}}((c, n) \neq (c^*, n^*), \text{tag}, \text{txt})$:
11 : if $\text{cnt}_1 \neq 0$ then : return $\perp$ 12 : if $O_{\text{MAC}}^{\text{Dec}}(c, n, \text{tag}, \text{txt})$ is not invoked after $O_{\text{Dec}}(c, n)$ or (c, n) in the two oracles are distinct then : return $\perp$ 13 : $\text{cnt}_1 := \text{cnt}_1 + 1$, $fk_S := H_4(\text{SHTS})$ 14 : if $\text{Decaps}(\text{sk}, c) = k^*$ then : $fk_S \leftarrow \mathcal{K}_{\text{mac}}$ $\parallel G_6$ 15 : if $\text{Decaps}(\text{sk}, c) = k^*$ then : return $\perp$ $\parallel G_4 - G_6$ 16 : if $\text{HS}' = \perp$ then : return $\perp$ 17 : if $\text{Vrf}(fk_S, \text{txt}, \text{tag}) = \text{accept}$ then : return HS' 18 : return $\perp$

Figure 14: The decapsulation and the MAC oracles used in the proof of Theorem 4.1.

- MAC oracle: There is no modification since the simulation in game G_1 does not rely on sk .

After $\mathcal{A}$ outputs $\hat{b}$, $\mathcal{B}$ outputs $[b = \hat{b}]$.

When $b' = 1$, i.e., $k_{b'}$ is a random key, then $\text{HS}^* := k_1$ is a random value. In the modified decapsulation oracle, if $\text{Decaps}(\text{sk}^*, c) = k_0$, it directly sets $\text{HS}' := \text{HS}^*$; if $\text{Decaps}(\text{sk}^*, c) = k_1$, it is equivalent to setting $\text{HS}' := \text{HS}^* = k_1 = \text{Decaps}(\text{sk}^*, c)$. In this case, we can see that $\mathcal{B}$'s behavior is consistent with the challenger's in game G_2 . Thus, we have $\Pr[[b = \hat{b}] = 1 | b' = 1] = \Pr[W_2]$.

But when $b' = 0$, i.e., $k_{b'}$ is the real key from $(k_0, c^*) \leftarrow \text{Encaps}(\text{pk}^*)$, the situation becomes a little different. In the modified decapsulation oracle, if $\text{Decaps}(\text{sk}^*, c) = k_1$, it sets $\text{HS}' := \text{HS}^* = k_0$, which is contrary to the decapsulation oracle in game G_1 , where it sets $\text{HS}' := \text{Decaps}(\text{sk}^*, c) = k_1$. However, since k_1 is not used to public computations, it remains uniformly random and independent of $\mathcal{A}$'s view. Let E be the event $\text{Decaps}(\text{sk}^*, c) = k_1$. We have $\Pr[E] \leq 1/|\mathcal{K}_{\text{kem}}|$. Conditioned on $\bar{E}$, $\mathcal{B}$'s behavior is consistent with the challenger's in game G_1 . Thus, by Lemma 3.4, we have $|\Pr[[b = \hat{b}] = 1 | b' = 0] - \Pr[W_1]| \leq \Pr[E] \leq 1/|\mathcal{K}_{\text{kem}}|$.

Define W as the event $[b = \hat{b}] = b'$. Then we have

$$\begin{aligned}
\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{B}) &\geq |\Pr[W] - 1/2| \\
&= |\Pr[W|b' = 0] \Pr[b' = 0] + \Pr[W|b' = 1] \Pr[b' = 1] - 1/2| \\
&= \left| \frac{\Pr[[b=\hat{b}]=0|b'=0]}{2} + \frac{\Pr[[b=\hat{b}]=1|b'=1]}{2} - 1/2 \right| \\
&= \left| \frac{(1 - \Pr[[b=\hat{b}]=1|b'=0]) + \Pr[[b=\hat{b}]=1|b'=1]}{2} - 1/2 \right| \\
&= \left| -\Pr[[b = \hat{b}] = 1|b' = 0] + \Pr[[b = \hat{b}] = 1|b' = 1] \right|/2 \\
&= |\Pr[W_2] - \Pr[[b = \hat{b}] = 1|b' = 0]|/2 .
\end{aligned}$$

By the triangle inequality, we have

$$\begin{aligned}
|\Pr[W_2] - \Pr[W_1]| &\leq |\Pr[W_2] - \Pr[[b = \hat{b}] = 1|b' = 0]| \\
&\quad + |\Pr[[b = \hat{b}] = 1|b' = 0] - \Pr[W_1]| \\
&\leq 2\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{B}) + 1/|K_{\text{kem}}| .
\end{aligned}$$

Game G_3 . In this game, we assume that in the decapsulation oracle, the event $H_T(c, n) = H_T(c^*, n^*)$ would not occur. If this event occurs, we can construct an adversary $\mathcal{C}$ to break the collision resistance of H_T . In particular, $\mathcal{C}$ can simulate the challenger for $\mathcal{A}$ in the same way as described in game G_2 . When $\mathcal{A}$ makes the decapsulation oracle query $(c, n) \neq (c^*, n^*)$, $\mathcal{C}$ can output the pair $((c, n), (c^*, n^*))$ as a collision for H_T . Thus,

$$|\Pr[W_3] - \Pr[W_2]| \leq \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) .$$

Game G_4 . In this game, we make a modification to the MAC oracle by introducing a new step, where if the query component c satisfies $\text{Decaps}(\text{sk}, c) = k^*$, this oracle simply returns $\perp$ (see line 15 in Fig. 14). Denote V_j as the event that the MAC oracle query components (tag, txt) satisfy $\text{Vrf}(fk_S, \text{txt}, \text{tag}) = \text{accept}$ in the case $\text{Decaps}(\text{sk}, c) = k^*$ in game G_j . We can see that unless V_4 occurs, games G_3 and G_4 proceed identically. By Lemma 3.4,

$$|\Pr[W_4] - \Pr[W_3]| \leq \Pr[V_4] .$$

Game G_5 . In this game, we remove the step resetting $(\text{CHTS}_0, \text{SHTS}_0, \text{dHS}_0)$ to the right values in the initialization phase, i.e., line 4 in Fig. 13, and in the decapsulation oracle, we introduce a new step where if $\text{Decaps}(\text{sk}, c) = k^*$ we reset $(\text{CHTS}, \text{SHTS}) \leftarrow \$\{0, 1\}^{2n}$, i.e., line 7 in Fig. 14.

Let Z be the event that $\mathcal{A}$ has made one of the random oracle queries $H_1(\text{HS}^*, H_T(c^*, n^*))$, $H_2(\text{HS}^*, H_T(c^*, n^*))$, $H_3(\text{HS}^*)$, $H_1(\text{HS}^*, H_T(c, n))$, $H_2(\text{HS}^*, H_T(c, n))$, where (c, n) is $\mathcal{A}$'s decapsulation oracle query. Now we suppose that we run games G_4 and G_5 but Z does not occur. Since $\mathcal{A}$ has not made the above random oracle queries, and we have assumed that $H_T(c, n) \neq H_T(c^*, n^*)$, then $(\text{CHTS}_0, \text{SHTS}_0, \text{dHS}_0)$ and $(\text{CHTS}, \text{SHTS})$, if $\mathcal{A}$ makes the decapsulation query on (c, n) where $\text{Decaps}(\text{sk}, c) = k^*$, are mutually random and independent from $\mathcal{A}$'s view. Thus, games G_3 and G_4 proceed identically unless Z occurs. In particular, we have $\Pr[W_5 \wedge \bar{Z}] = \Pr[W_4 \wedge \bar{Z}]$, and $\Pr[V_5 \wedge \bar{Z}] = \Pr[V_4 \wedge \bar{Z}]$. Applying Lemma 3.4, we have

$$|\Pr[W_5] - \Pr[W_4]| \leq \Pr[Z] \text{ and } |\Pr[V_5] - \Pr[V_4]| \leq \Pr[Z] .$$

Define Z_j be the event that $\mathcal{A}$ makes a H_1, H_2 , or H_3 random oracle query about HS^* in game G_j , i.e., HS^* is in the query. We have

$$\Pr[Z] \leq \Pr[Z_5] .$$

Note that the random HS^* is not used to obtain any public values in game G_5 . Therefore, HS^* is random and independent of $\mathcal{A}$'s view. Assume q_1, q_2 and q_3 are the bound on the number of H_1, H_2 and H_3 random oracle queries made by $\mathcal{A}$, respectively. We have

$$\Pr[Z_5] \leq (q_1 + q_2 + q_3)/|\mathcal{K}_{\text{kem}}|.$$

Since in game G_5 , $(\text{CHTS}_0, \text{SHTS}_0, \text{dHS}_0)$ and $(\text{CHTS}_1, \text{SHTS}_1, \text{dHS}_1)$ are all randomly sampled from $\{0, 1\}^{3n}$, and they are independent of all the oracles and all other public values. Thus, b is random and independent of $\mathcal{A}$'s view. Therefore,

$$\Pr[W_5] = 1/2.$$

Game G_6 . This game is the same as game G_5 , except that we make the following modifications to the decapsulation oracle and the MAC oracle:

- In the decapsulation oracle: We introduce a new step, where if $\text{Decaps}(\text{sk}, c) = k^*$, we reset $tk_S \leftarrow \$ \{0, 1\}^n$ (see line 9 in Fig. 14).
- In the MAC oracle: We introduce a new step, where if $\text{Decaps}(\text{sk}, c) = k^*$, we reset $fk_S \leftarrow \$ \mathcal{K}_{\text{mac}}$ (see line 14 in Fig. 14). $\mathcal{K}_{\text{mac}}$ is the key space of MAC.

Let Z' be the event that the decapsulation oracle query component c satisfies $\text{Decaps}(\text{sk}, c) = k^*$ and $\mathcal{A}$ makes the random oracle query on SHTS obtained in the decapsulation oracle to H_4 or H_D . If Z' does not occur, tk_S and fk_S in game G_5 are random from $\mathcal{A}$'s point of view, like in game G_6 . Thus, $\Pr[V_6 \wedge \bar{Z}'] = \Pr[V_5 \wedge \bar{Z}']$. Therefore, by Lemma 3.4,

$$|\Pr[V_6] - \Pr[V_5]| \leq \Pr[Z'].$$

Note that in game G_6 , if the decapsulation oracle query component c satisfies $\text{Decaps}(\text{sk}, c) = k^*$, SHTS is random and independent of all the public values. Since $\text{SHTS} \in \{0, 1\}^n$, we have

$$\Pr[Z'] \leq (q_4 + q_D)/2^n,$$

where q_4 and q_D are the bound on the number of H_4 random oracle queries and H_D random oracle queries made by $\mathcal{A}$, respectively.

Now, we want to claim that if the event V_6 occurs, we can construct a new adversary $\mathcal{D}$ to break the EUF-OT security of MAC. In particular, $\mathcal{D}$ can use the key pair $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$ to obtain the input of $\mathcal{A}$ and simulate all the oracles with sk in the same way as in game G_6 . Upon receiving $\mathcal{A}$'s MAC oracle query on $(c, n, \text{tag}, \text{txt})$, $\mathcal{D}$ outputs (txt, tag) as a forgery. Therefore, we have

$$\Pr[V_6] \leq \text{Adv}_{\text{MAC}}^{\text{EUF-OT}}(\mathcal{D}).$$

To summarize, we obtain

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 2\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{B}) + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) + \text{Adv}_{\text{MAC}}^{\text{EUF-OT}}(\mathcal{D}) \\ &\quad + (2(q_1 + q_2 + q_3) + 1)/|\mathcal{K}_{\text{kem}}| + (q_4 + q_D)/2^n. \end{aligned}$$

□

Games G_0 to G_8 :
1 : $(pk, sk) \leftarrow \text{Gen}(1^\lambda), (k^*, c^*) \leftarrow \text{Encaps}(pk), HS^* \leftarrow \mathcal{K}_{\text{kem}}, n^* \leftarrow \{0, 1\}^n$
2 : $HS^* := k^* \quad // \quad G_0 - G_1$
3 : $(CHTS_0, SHTS_0, dHS_0) \leftarrow \{0, 1\}^{3n}, (CHTS_1, SHTS_1, dHS_1) \leftarrow \{0, 1\}^{3n}$
4 : $CHTS_0 := H_1(HS^*, H_T(c^*, n^*)), SHTS_0 := H_2(HS^*, H_T(c^*, n^*)),$ $dHS_0 := H_3(HS^*) \quad // \quad G_0 - G_5$
5 : $b \leftarrow \{0, 1\}, cnt_0, cnt_1 := 0$
6 : use $H_{123} : \mathcal{K} \times \{\text{Domain}(H_T) \cup \perp\} \rightarrow \text{Range}(H_1) \times \text{Range}(H_2) \times \text{Range}(H_3)$ to simulate the random oracles $H_1, H_2, H_3 \quad // \quad G_5 - G_8$
7 : use $H_{4D} : \{0, 1\}^n \rightarrow \text{Range}(H_4) \times \text{Range}(H_D)$ to simulate the random oracles $H_4, H_D \quad // \quad G_7 - G_8$
8 : $\hat{b} \leftarrow \mathcal{A}^{\{H_i\}_{i \in \{1, 2, 3, 4, D\}}, O_{\text{Dec}}, O_{\text{MAC}}^{\text{Dec}}}(pk, c^*, n^*, (CHTS_b, SHTS_b, dHS_b))$
9 : return $[b = \hat{b}]$

Figure 15: Games G_0 to G_8 for the proof of Theorem 4.2.

D Supporting Material: Proof of Theorem 4.2

Theorem 4.2 (Security of the KEM in the QROM). *Let $\{H_i\}_{i \in \{1, 2, 3, 4, D\}}$ be the key-derivation functions and H_T be the hash function as in Fig. 5. Assume $\{H_i\}_{i \in \{1, 2, 3, 4, D\}}$ are modeled as quantum-accessible random oracles, and H_T is collision resistant. For any adversary $\mathcal{A}$ that attacks the IND-1CCA-1MAC security of KEM with quantum access to the random oracles $\{H_i\}_{i \in \{1, 2, 3, 4, D\}}$, there exists an adversary $\mathcal{B}$ that breaks the IND-1RCCA security of KEM, an adversary $\mathcal{C}$ that breaks the collision resistance of H_T , and an adversary $\mathcal{D}$ that breaks the EUF-0T security of MAC, such that*

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 2\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{B}) + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) \\ &\quad + (4(q_1 + q_2 + q_3) + 1)/\sqrt{|\mathcal{K}_{\text{kem}}|} + 2(q_4 + q_D)/2^{n/2}, \end{aligned}$$

where q_i is the bounded number of the queries made by $\mathcal{A}$ to the H_i random oracles for $i \in \{1, 2, 3, 4, D\}$, and $\mathcal{K}_{\text{kem}}$ is the key space of KEM.

Theorem 4.2. For $j = 0, \dots, 8$, the simulation of the challenger for the adversary $\mathcal{A}$ in game G_j is defined in Fig. 15 and Fig. 16. In each game, we define b to be the random bit sampled by the challenger, while $\hat{b}$ to be the bit output by $\mathcal{A}$. We also define W_j to be the event that $b = \hat{b}$ in game G_j .

Game G_0 . This game is the same as the IND-1CCA-1MAC game of KEM, except that we additionally samples HS^* and $(CHTS_0, SHTS_0, dHS_0)$ from their probability spaces, respectively, but they are reset to the corresponding values as defined in the IND-1CCA-1MAC game immediately. Since $\mathcal{A}$ is restricted to make the MAC oracle query only after having made the decapsulation oracle query and the components (c, n) in these queries are the same, the MAC oracle can directly use the HS' and the SHTS obtained in the decapsulation oracle to answer $\mathcal{A}$'s query. Thus, we can obtain

$$|\Pr[W_0] - 1/2| = \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}).$$

Game G_1 . In the decapsulation oracle, if the query component c satisfying $\text{Decaps}(sk, c) = k^*$, we let $HS' := HS^*$ instead of $HS' := \text{Decaps}(sk, c)$. Since $HS^* = k^*$, we have

$$\Pr[W_1] = \Pr[W_0].$$

Decapsulation Oracle $O_{\text{Dec}}((c, n) \neq (c^*, n^*))$:
1 : if $\text{cnt}_0 \neq 0$ then : return $\perp$
2 : $\text{cnt}_0 := \text{cnt}_0 + 1$, $\text{HS}' := \text{Decaps}(\text{sk}, c)$
3 : if $\text{Decaps}(\text{sk}, c) = k^*$ then : $\text{HS}' := \text{HS}^*$ $\parallel G_1 - G_8$
4 : if $\text{HS}' = \perp$ then : return $\perp$
5 : if $H_T(c, n) = H_T(c^*, n^*)$ then : return $\perp$ $\parallel G_3 - G_8$
6 : $\text{CHTS} := H_1(\text{HS}', H_T(c, n))$, $\text{SHTS} := H_2(\text{HS}', H_T(c, n))$
7 : if $\text{Decaps}(\text{sk}, c) = k^*$ then : $(\text{CHTS}, \text{SHTS}) \leftarrow \{0, 1\}^{2n}$ $\parallel G_6 - G_8$
8 : $tk_C := H_D(\text{CHTS})$, $tk_S := H_D(\text{SHTS})$
9 : if $\text{Decaps}(\text{sk}, c) = k^*$ then : $tk_S \leftarrow \{0, 1\}^n$ $\parallel G_8$
10 : return (tk_C, tk_S)
MAC Oracle $O_{\text{MAC}}^{\text{Dec}}((c, n) \neq (c^*, n^*), \text{tag}, \text{txt})$:
11 : if $\text{cnt}_1 \neq 0$ then : return $\perp$
12 : if $O_{\text{MAC}}^{\text{Dec}}(c, n, \text{tag}, \text{txt})$ is not invoked after $O_{\text{Dec}}(c, n)$ or (c, n) in the two oracles are distinct then : return $\perp$
13 : $\text{cnt}_1 := \text{cnt}_1 + 1$, $fk_S := H_4(\text{SHTS})$
14 : if $\text{Decaps}(\text{sk}, c) = k^*$ then : $fk_S \leftarrow \mathcal{K}_{\text{mac}}$ $\parallel G_8$
15 : if $\text{Decaps}(\text{sk}, c) = k^*$ then : return $\perp$ $\parallel G_4 - G_8$
16 : if $\text{HS}' = \perp$ then : return $\perp$
17 : if $\text{Vrf}(fk_S, \text{txt}, \text{tag}) = \text{accept}$ then : return HS'
18 : return $\perp$

Figure 16: The decapsulation and the MAC oracles used in the proof of Theorem 4.2.

Game G_2 . This game is the same as game G_1 , except that we remove line 2 in Fig. 15, i.e., $\text{HS}^* := k^*$. We can claim that $|\Pr[W_2] - \Pr[W_1]|$ is bounded by $2\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{B}) + 1/|\mathcal{K}_{\text{kem}}|$ for a new adversary $\mathcal{B}$.

In particular, when $\mathcal{B}$ receives (pk^*, c^*, k^*) from his IND-1RCCA game challenger for KEM, he can set $\text{HS}^* := k^*$, compute n^* and $(\text{CHTS}_b, \text{SHTS}_b, \text{dHS}_b)$ in the same way as in game G_1 , and send $(\text{pk}^*, c^*, n^*, (\text{CHTS}_b, \text{SHTS}_b, \text{dHS}_b))$ to $\mathcal{A}$. For the decapsulation oracle query $(c, n) \neq (c^*, n^*)$ and the MAC oracle query $((c, n) \neq (c^*, n^*), \text{tag}, \text{txt})$, $\mathcal{B}$ queries c to his IND-1RCCA game's decapsulation oracle: if it returns **test**, $\mathcal{B}$ directly sets $\text{HS}' := \text{HS}^*$ and compute the responses by himself; otherwise, he can use the decapsulated value to obtain the right HS' and compute the responses correctly. After $\mathcal{A}$ outputs $\hat{b}$, $\mathcal{B}$ outputs $[b = \hat{b}]$. Then, following the same analysis to the game G_2 in the proof of Theorem 4.1, we can obtain that

$$|\Pr[W_2] - \Pr[W_1]| \leq 2\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{B}) + 1/|\mathcal{K}_{\text{kem}}|.$$

Game G_3 . This game is the same as game G_2 , except that we assume the decapsulation query (c, n) made by $\mathcal{A}$ satisfies $H_T(c, n) \neq H_T(c^*, n^*)$. Otherwise, we can construct a new adversary $\mathcal{C}$ to break the collision resistance of H_T . In particular, $\mathcal{C}$ simulates the challenger as described in game G_2 and interacts with $\mathcal{A}$. Upon receiving $\mathcal{A}$'s decapsulation query $(c, n) \neq (c^*, n^*)$, $\mathcal{C}$ directly outputs $((c, n), (c^*, n^*))$ as a collision of H_T . Thus,

$$|\Pr[W_3] - \Pr[W_2]| \leq \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}).$$

Game G_4 . This game is the same as game G_3 , except that in the MAC oracle, if $\text{Decaps}(\text{sk}, c) = k^*$, it returns $\perp$. Let V_j denote the event that the MAC oracle query components (tag, txt)

satisfy $\text{Vrf}(fk_S, \text{txt}, \text{tag}) = \text{accept}$ in the case $\text{Decaps}(\text{sk}, c) = k^*$ in game G_j . We can observe that unless Z_4 occurs, games G_4 and G_3 proceed identically. Thus,

$$|\Pr[W_4] - \Pr[W_3]| \leq \Pr[V_4] .$$

Game G_5 . The domains of H_1 , H_2 , and H_3 can be seen as the same, i.e., $\mathcal{K} \times \{\text{Domain}(H_T) \cup \perp\}$. In this game, we can use a new random function $H_{123} : \mathcal{K} \times \{\text{Domain}(H_T) \cup \perp\} \rightarrow \text{Range}(H_1) \times \text{Range}(H_2) \times \text{Range}(H_3)$ to simulate H_1 , H_2 , and H_3 random oracles.

Using the query from $\mathcal{A}$ to the random oracle H_1 as an example, the challenger can prepare two additional output registers, initialized to a uniform superposition of all states, for the random oracles H_2 and H_3 . Then, the challenger applies $U_{H_{123}}$ to $\mathcal{A}$'s input and output registers with the two additional output registers. It can be verified that the states of the two additional registers remain unchanged during this operation, so they can be discarded. From $\mathcal{A}$'s point of view, this is equivalent to making the H_1 random oracle query as in game G_4 . For $\mathcal{A}$'s H_2 or H_3 random oracle queries, the challenger can deal with it in a similar way. This trick is widely used to ignore part of the output of an oracle [15, 55, 76, 79]. The total number of the invocation of H_{123} is bounded by $q_1 + q_2 + q_3$, and we have

$$\Pr[W_5] = \Pr[W_4] \quad \text{and} \quad \Pr[V_5] = \Pr[V_4] .$$

Game G_6 . This game is the same as game G_5 , except that we remove line 4 in Fig. 15, i.e., resetting $(\text{CHTS}_0, \text{SHTS}_0, \text{dHS}_0)$, and we introduce line 7 in Fig. 16, i.e., resetting $(\text{CHTS}, \text{SHTS})$ to random values when the query component c satisfying $\text{Decaps}(\text{sk}, c) = k^*$. It is equivalent to reprogramming H_{123} at the points $(\text{HS}^*, H_T(c^*, n^*))$, $(\text{HS}^*, \perp)$, and $(\text{HS}^*, H_T(c, n))$, where (c, n) is $\mathcal{A}$'s decapsulation query satisfying $\text{Decaps}(\text{sk}, c) = k^*$, to random values. By applying the regular O2H lemma (Lemma 3.1), we can obtain $|\Pr[W_6] - \Pr[W_5]| \leq 2(q_1 + q_2 + q_3)\sqrt{\Pr[Z]}$, where Z denotes the event that in game G_6 , the challenger randomly picks $i \leftarrow \{1, \dots, q_1 + q_2 + q_3\}$, runs $\mathcal{A}$ until (just before) the i -th H_{123} oracle query, measures the input register in the computational basis, and outputs the measurement outcome that has at least one of the above points. Note that, in game G_6 , the randomly sampled HS^* is irrelevant to all the oracles and all the public values. Therefore, HS^* is random and independent of $\mathcal{A}$'s view, which means the probability of the measurement outcome containing HS^* is bounded by $1/|\mathcal{K}_{\text{kem}}|$, where $\mathcal{K}_{\text{kem}}$ is the key space of KEM. Thus, we have $\Pr[Z] \leq 1/|\mathcal{K}_{\text{kem}}|$. Therefore,

$$|\Pr[W_6] - \Pr[W_5]| \leq 2(q_1 + q_2 + q_3)\sqrt{1/|\mathcal{K}_{\text{kem}}|} .$$

Similarly, we can obtain

$$|\Pr[V_6] - \Pr[V_5]| \leq 2(q_1 + q_2 + q_3)\sqrt{1/|\mathcal{K}_{\text{kem}}|} .$$

Since both $(\text{CHTS}_0, \text{SHTS}_0, \text{dHS}_0)$ and $(\text{CHTS}_1, \text{SHTS}_1, \text{dHS}_1)$ are randomly sampled, and independent of all the oracles and all other public values, b is random and independent of $\mathcal{A}$'s view. Thus,

$$\Pr[W_6] = 1/2 .$$

Game G_7 . This game is the same as game G_6 , except that we use a new random function $H_{4D} : \{0, 1\}^n \rightarrow \text{Range}(H_4) \times \text{Range}(H_D)$ to simulate H_4 and H_D random oracles. Like the analysis to game G_5 where we use the random function H_{123} to simulate H_1 , H_2 , and H_3 random oracles, we have

$$\Pr[V_7] = \Pr[V_6] .$$

Game G_8 . In this game, we introduce two operations in decapsulation oracle and MAC oracle, respectively, i.e., if the query component c satisfying $\text{Decaps}(\text{sk}, c) = k^*$, we reset $tk_S \leftarrow \{0, 1\}^n$

and $fk_S \leftarrow \$ \mathcal{K}_{\text{mac}}$, where $\mathcal{K}_{\text{mac}}$ is the key space of MAC (see lines 9 and 14 in Fig. 16). It is equivalent to reprogramming H_{4D} at the point SHTS to a random values when $\text{Decaps}(\text{sk}, c) = k^*$. By applying the regular O2H lemma (Lemma 3.1), we can obtain $|\Pr[V_8] - \Pr[V_7]| \leq 2(q_4 + q_D)\sqrt{\Pr[Z']}$, where Z' denotes the event that in game G_8 with $\text{Decaps}(\text{sk}, c) = k^*$, the challenger samples $i \leftarrow \$ \{1, \dots, q_4 + q_D\}$, runs $\mathcal{A}$ until (just before) the i -th H_{4D} oracle queries, measures the input query, and the measure outcome $\hat{\text{SHTS}} = \text{SHTS}$. Since when $\text{Decaps}(\text{sk}, c) = k^*$, SHTS is random from $\mathcal{A}$'s view, $\Pr[Z']$ is bounded by $1/2^n$. Thus, we have

$$|\Pr[V_8] - \Pr[V_7]| \leq 2(q_4 + q_D)/2^{n/2}.$$

Since fk_S in game G_8 is random and independent of $\mathcal{A}$'s view, if V_8 occurs, we can construct a new adversary $\mathcal{D}$ to break the EUF-0T security of MAC. In particular, $\mathcal{D}$ can interact with $\mathcal{A}$ in the same way as the challenger in game G_8 until $\mathcal{A}$ makes the MAC oracle query on $(c, n, \text{tag}, \text{txt})$. Then $\mathcal{D}$ can output (txt, tag) as a forger to the EUF-0T game of MAC. Thus, we have

$$\Pr[V_8] \leq \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}).$$

To summarize, we have

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{A}) &\leq 2\text{Adv}_{\text{KEM}}^{\text{IND-1RCCA}}(\mathcal{B}) + \text{Adv}_{H_T}^{\text{CR}}(\mathcal{C}) + \text{Adv}_{\text{MAC}}^{\text{EUF-0T}}(\mathcal{D}) \\ &\quad + \frac{4(q_1 + q_2 + q_3) + 1}{\sqrt{|\mathcal{K}_{\text{kem}}|}} + \frac{2(q_4 + q_D)}{2^{n/2}}. \end{aligned}$$

□

E Supporting Material: Proof of Theorem 5.1

Theorem 5.1 (IND-1RCCA Security of KEM_{tag}). *Assume the key-derivation function G depicted in Fig. 6 is modeled as a random oracle. For any PPT adversary $\mathcal{A}$ attacking the IND-1RCCA security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G]$, there exist a PPT adversary $\mathcal{B}$ attacking the OW-CPA security of KEM and a PPT adversary $\mathcal{B}'$ attacking the IND-CPA security of KEM s.t.*

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq q_G(q_G + 3)\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}), \\ \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 2(q_G + 3)\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + q_G(q_G + 3)/|\mathcal{K}_{\text{kem}}|, \end{aligned}$$

where q_G is the bounded number of the random oracle G queries made by $\mathcal{A}$, and $\mathcal{K}_{\text{kem}}$ is the key space of the underlying KEM.

When the random oracle G is quantum-accessible, we have

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 10(q_G + 1)^2 \sqrt{\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + 1/|G|}, \\ \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 10(q_G + 1) \sqrt{2\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + (q_G + 1)^2/|\mathcal{K}_{\text{kem}}| + 1/|G|}, \end{aligned}$$

where $|G|$ is the output space size of G .

E.1 Proof of the ROM Case in Theorem 5.1

ROM Case in Theorem 5.1. We define games G_j (for $j = 0, \dots, 4$) played between the challenger and the adversary $\mathcal{A}$, as shown in Fig. 17. In each game, b is the random bit sampled by the challenger, and $\hat{b}$ is the bit output by $\mathcal{A}$. Let W_j be the event that $b = \hat{b}$ in game G_j .

Games G_0 to G_4 :	Decapsulation Oracle $O_{\text{Dec}}(c \neq c^*)$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda), (k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$	13 : if $\text{cnt} \neq 0$ then : return $\perp$
2 : $k^* \leftarrow \$ \mathcal{K}_{\text{kem}} \quad // \quad G_4$	14 : $\text{cnt} := \text{cnt} + 1$
3 : initialize an empty associative array $\text{Map} : \mathcal{K}_{\text{kem}} \rightarrow \text{Range}(G)$	15 : $k' := \text{Decaps}(\text{sk}, c) \quad // \quad G_0 - G_2$
4 : $k_0, k_1 \leftarrow \$ \text{Range}(G), \bar{k} \leftarrow \$ \text{Range}(G)$	16 : if $k' = k^*$ then $// \quad G_1 - G_2$
5 : $\text{Map}[k^*] := k_0 \quad // \quad G_0 - G_1$	17 : if $i^* = q_G + 1$ then $// \quad G_3 - G_4$
6 : $b \leftarrow \$ \{0, 1\}, \text{cnt} := 0$	18 : return test $// \quad G_1 - G_4$
7 : $i^* \leftarrow \$ \{0, \dots, q_G + 2\} \quad // \quad G_3 - G_4$	19 : if $k' = \perp$ then $// \quad G_0 - G_2$
8 : $\hat{b} \leftarrow \mathcal{A}^{G, O_{\text{Dec}}}(\text{pk}, c^*, k_b)$	20 : if $i^* = q_G + 2$ then $// \quad G_3 - G_4$
9 : return $[b = \hat{b}]$	21 : return $\perp$
Random Oracle $G(k)$:	22 : $k := G(k') \quad // \quad G_0 - G_2$
10 : if k is the i^* -th new query then $\text{Map}[k] := \bar{k} \quad // \quad G_3 - G_4$	23 : $k := \bar{k} \quad // \quad G_3 - G_4$
11 : if $k \notin \text{Domain}(\text{Map})$ then $\text{Map}[k] \leftarrow \$ \text{Range}(G)$	24 : if $k \in \{k_0, k_1\}$ then return test
12 : return $\text{Map}[k]$	25 : return k

Figure 17: Games G_0 to G_4 for the proof of the ROM case in Theorem 5.1.

Game G_0 . This game is defined in Fig. 17. In the initialization phase, the challenger explicitly initialize an empty associative array $\text{Map} : \mathcal{K}_{\text{kem}} \rightarrow \text{Range}(G)$ to implement the random oracle G , where $\mathcal{K}_{\text{kem}}$ is the key space of KEM, and $\text{Range}(G)$ is the output space of G . Note that the challenger sampled $k_0 \leftarrow \$ \text{Range}(G)$ at the beginning, but it immediately sets $\text{Map}[k^*] := k_0$, which is equivalent to setting the random oracle $G(k^*) := k_0$. The challenger also samples $\bar{k} \leftarrow \$ \text{Range}(G)$, though it is not used in this game. From $\mathcal{A}$'s view, the behavior of the challenger is the same as that in the IND-1RCCA game of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G]$. Thus, we have

$$|\Pr[W_0] - 1/2| = \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) .$$

Game G_1 . This game is identical to G_0 , except that the decapsulation oracle now explicitly handles the case where $\text{Decaps}(\text{sk}, c) = k^*$: in such case, it directly returns **test**. Note that this modification does not change the output behavior of O_{Dec} compared to G_0 . Hence,

$$\Pr[W_1] = \Pr[W_0] .$$

Game G_2 . This game removes the step $\text{Map}[k^*] := k_0$, i.e., line 5 in Fig. 17. Let Z_j be the event that in game G_j , adversary $\mathcal{A}$ queries k^* to the random oracle G . Since $G(k^*)$ remains uniformly random from the perspective of $\mathcal{A}$ unless a query for k^* has been made, the modification remains undetectable by $\mathcal{A}$ unless Z_2 occurs. By the difference lemma (Lemma 3.4), we have

$$|\Pr[W_2] - \Pr[W_1]| \leq \Pr[Z_2] .$$

At this point, both k_0 and k_1 are uniformly random and independent of all other values. Therefore, the bit b is random and independent of the adversary's view, implying that

$$\Pr[W_2] = 1/2 .$$

Game G_3 . Note that in the decapsulation oracle, for $k' := \text{Decaps}(\text{sk}, c)$, if $k' \notin \{k^*, \perp\}$, the oracle first compute $k := G(k')$ (i.e., line 22 in Fig. 17), and then use k to determine the final response. Since if $\mathcal{A}$ has not queried k' to the random oracle G , k is uniformly random from $\mathcal{A}$'s view. Thus, in the decapsulation oracle, we can directly use a pre-sampled random value $\bar{k}$ as the value of k when $k' \notin \{k^*, \perp\}$, while letting the random oracle return $\bar{k}$ when queried on k' to ensure consistency in the oracle simulations.

However, the G cannot know the value of k' in advance to decide whether to respond with $\bar{k}$, since $\mathcal{A}$ might query k' to G before invoking the decapsulation oracle. Since $\mathcal{A}$ makes at most q_G queries to G , we can assert that if $\mathcal{A}$ ever queries k' , it must be one of these q_G queries. Thus, the challenger can randomly choose one of these q_G queries to respond with $\bar{k}$.

In G_3 , the challenger adopts the following strategy to simulate both the decapsulation oracle and the random oracle G : it first samples $i^* \leftarrow \$ \{0, \dots, q_G + 2\}$ and $\bar{k} \leftarrow \$ \text{Range}(G)$. If $i^* \in \{1, \dots, q_G\}$, the challenger guesses that $\mathcal{A}$ will query k' on the i^* -th new query to G , and that $k' \notin \{k^*, \perp\}$. In this case, the decapsulation oracle directly uses $k = \bar{k}$ to compute the response, and for the i^* -th new query to G , it sets $\text{Map}[k] := \bar{k}$ and returns $\bar{k}$. If $i^* = 0$, the challenger still assumes $k' \notin \{k^*, \perp\}$, but that $\mathcal{A}$ does not query k' to G ; hence, the decapsulation oracle simply uses $k := \bar{k}$ without modifying G . If $i^* = q_G + 1$ or $q_G + 2$, the challenger guesses that $k' = k^*$ or $k' = \perp$, respectively, and the decapsulation oracle returns **test** or $\perp$ accordingly. Note that this simulation strategy does not require the challenger to compute the precise value of k' . Thus the step $k' := \text{Decaps}(\text{sk}, c)$ (i.e., line 15 in Fig. 17) can also be removed. The resulting game G_3 is illustrated in Fig. 17.

From the above analysis, when the challenger correctly guesses $\mathcal{A}$'s behavior, the oracle responses in G_3 are consistent with those in G_2 . Therefore,

$$\Pr[Z_3 \mid \text{correct guess}] = \Pr[Z_2] .$$

Since the probability of a correct guess is $1/(q_G + 3)$, it follows that

$$\Pr[Z_3] \geq \Pr[Z_3 \wedge \text{correct guess}] = \frac{\Pr[Z_3 \mid \text{correct guess}]}{q_G + 3} .$$

Since the challenger in G_3 does not require k^* and sk to interact with $\mathcal{A}$, if Z_3 occurs, we can construct an adversary $\mathcal{B}$ to attack the OW-CPA security of the underlying KEM. Specifically, given (pk^*, c^*) from the OW-CPA game challenger, $\mathcal{B}$ randomly selects $k, \bar{k} \leftarrow \$ \text{Range}(G)$, sends (pk^*, c^*, k) to $\mathcal{A}$, and then simulates the random oracle G and the decapsulation oracle to interact with $\mathcal{A}$ following the same strategy of G_3 . At the end of the game, if Z_3 occurs, $\mathcal{B}$ randomly selects a value k^* from $\text{Domain}(\text{Map})$ and outputs it. Since there are at most q_G queries to G , we have

$$\frac{\Pr[Z_3]}{q_G} \leq \text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) .$$

In summary, we have

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq q_G(q_G + 3)\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) .$$

Game G_4 . In this game, the challenger resamples $k^* \leftarrow \$ \mathcal{K}_{\text{kem}}$ at the beginning (i.e., line 2 in Fig. 17). Since k^* is independent of all other values in G_4 , we have

$$\Pr[Z_4] \leq \frac{q_G}{|\mathcal{K}_{\text{kem}}|} .$$

We now construct an adversary $\mathcal{B}'$ against the IND-CPA security of KEM. Specifically, upon receiving $(\text{pk}^*, c^*, k_{b'})$ from the IND-CPA challenger, where b' is the random bit sampled by the

Games G_0 to G_3 :	Decapsulation Oracle $O_{\text{Dec}}(c \neq c^*)$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$	9 : if $\text{cnt} \neq 0$ then : return $\perp$
2 : $G \leftarrow \Omega_G$, $k_0, k_1 \leftarrow \text{Range}(G)$	10 : $\text{cnt} := \text{cnt} + 1$
3 : $k_0 := G(k^*)$ // $G_0 - G_1$	11 : $k' := \text{Decaps}(\text{sk}, c)$
4 : $b \leftarrow \{0, 1\}$, $\text{cnt} := 0$	12 : if $k' = k^*$ then
5 : $\hat{b} \leftarrow \mathcal{A}^{(G), O_{\text{Dec}}}(\text{pk}, c^*, k_b)$	return test // $G_1 - G_3$
6 : return $[b = \hat{b}]$	13 : if $k' = \perp$ then : return $\perp$
Random Oracle $G(k)$:	14 : $k := G(k')$
7 : if $k = k^*$ then : return k_0 // G_3	15 : if $k \in \{k_0, k_1\}$ then
8 : return $G(k)$	return test
	16 : return k

Figure 18: Games G_0 to G_3 for the proof of the QROM case in Theorem 5.1.

challenger, $\mathcal{B}'$ sets $k^* := k_{b'}$, samples $k, \bar{k} \leftarrow \text{Range}(G)$, sends (pk^*, c^*, k) to $\mathcal{A}$, and simulates the random oracle G and the decapsulation oracle identically to G_3 to interactive with $\mathcal{A}$. At the end of the game, $\mathcal{B}'$ checks whether $k_{b'} \in \text{Domain}(\text{Map})$: if so, it outputs $\hat{b}' = 1$; otherwise, it outputs $\hat{b}' = 0$.

Note that when $b' = 0$, $k^* := k_{b'}$ is the real key, so $\mathcal{B}'$ behaves exactly as the challenger in G_3 ; when $b' = 1$, $k^* := k_{b'}$ is a random key, so $\mathcal{B}'$ behaves as in G_4 . Therefore, $\Pr[\hat{b}' = 1 | b' = 0] = \Pr[Z_3]$, and $\Pr[\hat{b}' = 1 | b' = 1] = \Pr[Z_4]$. Hence,

$$\begin{aligned}
\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') &\geq \left| \Pr[b' = \hat{b}'] - \frac{1}{2} \right| \\
&= \left| \Pr[b' = \hat{b}' \wedge b' = 0] + \Pr[b' = \hat{b}' \wedge b' = 1] - \frac{1}{2} \right| \\
&= \frac{1}{2} \left| \Pr[\hat{b}' = 0 | b' = 0] + \Pr[\hat{b}' = 1 | b' = 1] - 1 \right| \\
&= \frac{1}{2} \left| \Pr[\hat{b}' = 1 | b' = 1] - \Pr[\hat{b}' = 1 | b' = 0] \right| \\
&= \frac{1}{2} |\Pr[Z_4] - \Pr[Z_3]|.
\end{aligned}$$

In summary, we can obtain

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 2(q_G + 3)\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + q_G(q_G + 3)/|\mathcal{K}_{\text{kem}}|.$$

□

E.2 Proof of the QROM Case in Theorem 5.1

QROM Case in Theorem 5.1. We define a sequence of games G_j (where $j = 0, \dots, 3$) played between a challenger and an adversary $\mathcal{A}$, as illustrated in Fig. 18. In each game G_j , b is a randomly chosen bit by the challenger, and $\hat{b}$ is the bit output by $\mathcal{A}$. Let W_j denote the event that $b = \hat{b}$ in game G_j .

Game G_0 . This game is defined as shown in Fig. 18. Note that, during the initialization phase, the challenger randomly selects a function $G \in \Omega_G$ to simulate the random oracle G , and randomly samples a $k_0 \leftarrow \mathcal{K}_{\text{kem}}$, which is immediately reset to the correct value. It can be

observed that the behavior of the challenger in G_0 is consistent with that in the IND-1RCCA game of KEM_{tag} . Therefore, we have

$$|\Pr[W_0] - 1/2| = \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) .$$

Game G_1 . This game is identical to G_0 , except that the case $\text{Decaps}(\text{sk}, c) = k^*$ is handled explicitly in the decapsulation oracle (i.e., line 12 in Fig. 18). Note that this modification does not change the responses of the random oracle. Thus,

$$\Pr[W_1] = \Pr[W_0] .$$

Game G_2 . This game deletes the reset operation on k_0 during the initialization phase (i.e., line 3 in Fig. 18). Since both k_0 and k_1 are chosen uniformly at random, the bit b is independent of $\mathcal{A}$'s view. Hence, we have

$$\Pr[W_2] = 1/2 .$$

Game G_3 . This game reprograms the random oracle G at the point k^* to output k_0 . Specifically, whenever the adversary $\mathcal{A}$ queries the random oracle G on a value k such that $k = k^*$, the oracle responds with k_0 . This is equivalent to simulating the random oracle G using a new function G' , where $G'(k) = G(k)$ for all $k \neq k^*$, and $G'(k^*) = k_0$. It can be observed that the behavior of the challenger in this game is consistent with that in G_1 . Therefore, we have

$$\Pr[W_3] = \Pr[W_1] .$$

Combining the results from games G_0 to G_3 , we obtain

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) = |\Pr[W_3] - \Pr[W_2]| .$$

Based on the analysis in Appendix E.3, we can further obtain

$$|\Pr[W_3] - \Pr[W_2]| \leq 10(q_G + 1)^2 \sqrt{\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + 1/|G|} ,$$

and

$$|\Pr[W_3] - \Pr[W_2]| \leq 10(q_G + 1) \sqrt{2\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + (q_G + 1)^2/|\mathcal{K}_{\text{kem}}| + 1/|G|} ,$$

where q_G is the bounded numbers of G -queries made by $\mathcal{A}$, and $|G|$ is the output space size of G . Thus, we have

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 10(q_G + 1)^2 \sqrt{\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + 1/|G|} ,$$

and

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 10(q_G + 1) \sqrt{2\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + (q_G + 1)^2/|\mathcal{K}_{\text{kem}}| + 1/|G|} .$$

□

E.3 Bound on $|\Pr[W_3] - \Pr[W_2]|$ in Appendix E.2

We now use Lemmas E.2 and E.3 to derive the upper bounds on $|\Pr[W_3] - \Pr[W_2]|$ for the case when KEM is OW-CPA secure and IND-CPA secure, respectively. The security proofs of Lemma E.2 and Lemma E.3 rely on the use of Lemma E.1.

Lemma E.1 (Single-Classical-Query Measure-and-Reprogram [54, Lemma 3.1], adapted). *Let $A^{[H]}$ be an oracle quantum algorithm that makes at most q (quantum) queries to the random function $H : \mathcal{X} \rightarrow \mathcal{Y}$, and outputs some classical $x \in \mathcal{X}$ and a (possibly quantum) output z . Particularly, the i^* -th query input state of $A^{[H]}$ is $|x\rangle$, where $i^* \in \{1, \dots, q\}$, and $|x\rangle$ is a classical state and identical with the x output by $A^{[H]}$.*

Let $S^A(\Theta)$ be an oracle quantum algorithm that samples a random pair $(i, b_0) \in (\{1, \dots, q\} \setminus \{i^\} \times \{0, 1\}) \cup \{(q+1, 0)\}$, runs $A^{[H_i^{i^*}]}$ to output z , where $H_i^{i^*}$ is an oracle that returns Θ for A 's i^* -th H -query, measures A 's i -th H -query input to obtain x , returns A 's l -th H -query (where $l < (i + b_0)$ and $l \neq i^*$) using H , and returns A 's l -th H -query (where $l \geq (i + b_0)$ and $l \neq i^*$) using $H_{x\Theta}$ (where $H_{x\Theta}(x) = \Theta$ and $H_{x\Theta}(x') = H(x')$ for all $x' \neq x$).*

Let $S_1^A(\Theta)$ be an oracle quantum algorithm that samples a random pair $(j, b_1) \in (\{i^+1, \dots, q\} \times \{0, 1\}) \cup \{(q+1, 0)\}$, runs $A^{[H_j]}$ to output z , where H_j is an oracle that measures A 's j -th H -query input to obtain x , returns A 's l -th H -query (where $l < (j + b_1)$) using H , and returns A 's l -th H -query (where $l \geq (j + b_1)$) using $H_{x\Theta}$.*

Then, for any $x_0 \in \mathcal{X}$, $i^ \in \{1, \dots, q\}$, and any predicate V , we have*

$$\begin{aligned} \Pr_H[x = x_0 \wedge V(x, H(x), z) = 1 : (x, z) \leftarrow A^{[H]}] \\ \leq 2(2q-1)^2 \Pr_{H, \Theta}[x = x_0 \wedge V(x, \Theta, z) = 1 : (x, z) \leftarrow S^A] \\ + 8q^2 \Pr_{H, \Theta}[x = x_0 \wedge V(x, \Theta, z) = 1 : (x, z) \leftarrow S_1^A], \end{aligned}$$

where the subscript $\{H, \Theta\}$ in $\Pr_H$ and $\Pr_{H, \Theta}$ indicates that the probability is taken over the random choices of H and Θ . Moreover, if $V = V_1 \wedge V_2$, where $V_1(x, y, z) = 1$ if and only if y is returned in response to the i^* -th H -query of A , then

$$\sum_{x_0} \Pr_{H, \Theta}[x = x_0 \wedge V(x, \Theta, z) = 1 : (x, z) \leftarrow S_1^A] \leq \frac{1}{|\mathcal{Y}|}.$$

Lemma E.2. *For the events W_3 and W_2 and the other notation defined in Appendix E.2, there exists an adversary $\mathcal{B}$ against the OW-CPA security of KEM such that*

$$|\Pr[W_3] - \Pr[W_2]| \leq 10(q_G + 1)^2 \sqrt{\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + 1/|G|},$$

where q_G is the bounded number of G -queries made by $\mathcal{A}$, and $|G|$ is the output space size of G .

Lemma E.2. We begin by constructing two oracle algorithms, $A^{[G], O_{\text{Dec}}}(z)$ and $A^{[G'], O_{\text{Dec}}}(z)$, which simulate games G_2 and G_3 , respectively. Here, $z := (\text{pk}, c^*, k_0, k_1, b)$, where $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$, $k_0, k_1 \leftarrow \text{Range}(G)$, and $b \leftarrow \{0, 1\}$. Let $G \leftarrow \Omega_G$, and define $G' \in \Omega_G$ such that $G'(k) = G(k)$ for all $k \neq k^*$, and $G'(k^*) = k_0$.

The algorithm $A^{[G], O_{\text{Dec}}}(z)$ (or $A^{[G'], O_{\text{Dec}}}(z)$) simulates the random oracle G as described in G_2 (or G_3), runs $\hat{b} \leftarrow A^{[G], O_{\text{Dec}}}(\text{pk}, c^*, k_b)$, and finally outputs $[b = \hat{b}]$. Clearly, we have

$$\Pr[1 \leftarrow A^{[G], O_{\text{Dec}}}(z)] = \Pr[W_2] \quad \text{and} \quad \Pr[1 \leftarrow A^{[G'], O_{\text{Dec}}}(z)] = \Pr[W_3].$$

Now define an oracle algorithm $B^{[G], O_{\text{Dec}}}(z)$ that takes z as input, randomly samples $i^* \leftarrow \{1, \dots, q_G + 1\}$ ¹⁰, executes $A^{[G], O_{\text{Dec}}}(z)$ until (just before) the i^* -th query to G , measures its input register in the computational basis, and outputs the measurement outcome.

¹⁰ $q_G + 1$ is the total number of G random oracle queries made by $A^{[G], O_{\text{Dec}}}(z)$: q_G queries from $\mathcal{A}$ and one additional query from the simulation of O_{Dec} .

Games G_4^A to G_5^A :	Decapsulation Oracle $O_{\text{Dec}}(c \neq c^*)$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda), (k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$	9 : if $\text{cnt} \neq 0$ then : return $\perp$
2 : $k_0, k_1 \leftarrow \$ \text{Range}(G), b \leftarrow \$ \{0, 1\}$	10 : $\text{cnt} := \text{cnt} + 1$
3 : $G \leftarrow \$ \Omega_G, \text{cnt} := 0$	11 : $k' := \text{Decaps}(\text{sk}, c) \quad // G_4^A$
4 : $l := 0, \text{flag} \leftarrow \$ \{0, 1, 2\}, \bar{k} \leftarrow \$ \text{Range}(G)$ $(i, b) \leftarrow \$ (\{1, \dots, q_G\} \times \{0, 1\}) \cup \{(q_G + 1, 0)\}$	12 : if $k' = k^*$ then $// G_4^A$
5 : $\hat{k} \leftarrow B^{ \mathcal{G}\rangle, O_{\text{Dec}}}(\text{pk}, c^*, k_0, k_1, b)$	13 : if $\text{flag} = 0$ then $// G_5^A$
6 : return $[k^* = \hat{k}]$	14 : return test
Random Oracle $G(k)$:	15 : if $k' = \perp$ then $// G_4^A$
7 : if $\text{flag} = 2$ then if $l \geq (i + b)$ and $k = k_i$ then $// k_i$ is the measurement outcome on $// B$'s i -th G -query input register return $\bar{k}$ $l = l + 1 \quad // G_5^A$	16 : if $\text{flag} = 1$ then $// G_5^A$
8 : return $G(k)$	17 : return $\perp$
	18 : $k := G(k') \quad // G_4^A$
	19 : $k := \bar{k} \quad // G_5^A$
	20 : if $k \in \{k_0, k_1\}$ then return test
	21 : return k

Figure 19: Games G_4^A to G_5^A for the proof of Lemma E.2.

By the regular O2H lemma¹¹ (Lemma 3.1), we obtain:

$$\begin{aligned} & \left| \Pr[1 \leftarrow A^{|\mathcal{G}\rangle, O_{\text{Dec}}}(z)] - \Pr[1 \leftarrow A^{|\mathcal{G}'\rangle, O_{\text{Dec}}}(z)] \right| \\ & \leq 2(q_G + 1) \sqrt{\Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{|\mathcal{G}\rangle, O_{\text{Dec}}}(z)]} . \end{aligned}$$

Next, we define games G_j^A for $j = 4, 5$, played between a challenger and the oracle algorithm B , as illustrated in Fig. 19. In each game, k^* is the key generated by the challenger, and $\hat{k}$ is the key output by B . Let Z_j denote the event that $k^* = \hat{k}$ in game G_j^A .

Game G_4^A . This game is defined as shown in Fig. 19. Although the challenger generates several values in line 4 of Fig. 19, these values are not used anywhere. Note that the random oracle G accessed by $B^{|\mathcal{G}\rangle, O_{\text{Dec}}}(z)$ in G_4^A is simulated by G . Hence, we have

$$\Pr[Z_4] = \Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{|\mathcal{G}\rangle, O_{\text{Dec}}}(z)] = \Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{|\mathcal{G}\rangle, O_{\text{Dec}}}(z)] .$$

Game G_5^A . In this game, the decapsulation oracle is simulated using a random guessing strategy. The challenger randomly selects $\text{flag} \leftarrow \$ \{0, 1, 2\}$. For a decapsulation query $c \neq c^*$, let $k' := \text{Decaps}(\text{sk}, c)$: if $\text{flag} = 0$, the challenger guesses that $k' = k^*$, and O_{Dec} returns **test**; if $\text{flag} = 1$, the challenger guesses that $k' = \perp$, and O_{Dec} returns $\perp$; if $\text{flag} = 2$, the challenger guesses that $k' \notin \{k^*, \perp\}$ and then, in O_{Dec} , it assigns a predetermined random value $\bar{k}$ to k (line 19 in Fig. 19). If $k \in \{k_0, k_1\}$, the oracle returns **test**; otherwise, it returns k .

In the case where $\text{flag} = 2$ (i.e., the challenger guesses $k' \notin \{k^*, \perp\}$), the challenger further modifies the simulation of the random oracle G : it first reprograms G based on a pre-sampled index $(i, b) \leftarrow \$ (\{1, \dots, q_G\} \times \{0, 1\}) \cup \{(q_G + 1, 0)\}$, where q_G is the bounded number of G -query made by B . When B makes its i -th query to G , the challenger measures the input register to

¹¹When applying the regular O2H lemma, we treat O_{Dec} as an internal oracle of A .

obtain k_i and reprograms all responses of G at point k_i starting from the $(i + b)$ -th G -query to $\bar{k}$ (i.e., line 7 in Fig. 19).

Note that when the challenger guesses correctly:

1. If $k' = k^*$, the oracle O_{Dec} in both G_4^A and G_5^A returns **test**, so $\Pr[Z_4 | k' = k^*] = \Pr[Z_5 | k' = k^* \wedge \text{flag} = 0]$.
2. If $k' = \perp$, O_{Dec} in both games returns $\perp$, hence $\Pr[Z_4 | k' = \perp] = \Pr[Z_5 | k' = \perp \wedge \text{flag} = 1]$.
3. If $k' \notin \{k^*, \perp\}$, we will use Lemma E.1 to relate $\Pr[Z_4 | k' \notin \{k^*, \perp\}]$ and $\Pr[Z_5 | k' \notin \{k^*, \perp\} \wedge \text{flag} = 2]$ as follows.

Conditioned on $k' \notin \{k^*, \perp\}$, we first construct an oracle algorithm $C^{(G)}$ that generates $(\text{pk}, \text{sk}, k^*, c^*, k_0, k_1, b)$ and then runs $B^{(G), O_{\text{Dec}}}(\text{pk}, c^*, k_0, k_1, b)$ as the challenger does in G_4^A . Let $\bar{c}$ be the decapsulation query made by B , and let $k' = \text{Decaps}(\text{sk}, \bar{c})$. Define $x := k'$, $y := G(x)$, and $z := (z_1, z_2, z_3) = (k, k^*, \hat{k})$, where k is used in line 20 to determine the final response in O_{Dec} . Define predicates $V_1(x, y, z) := [y = z_1]$ and $V_2(x, y, z) := [z_2 = z_3]$. Let the predicate V in Lemma E.1 be $V := V_1 \wedge V_2$. Since V_1 always holds in G_4^A when $k' \notin \{k^*, \perp\}$, we have

$$\begin{aligned} \Pr[Z_4 \mid k' \notin \{k^*, \perp\}] &= \Pr_G[V(x, y, z) = 1 : (x, z) \leftarrow C^{(G)}] \\ &= \sum_{x_0 \in \mathcal{K}_{\text{kem}}} \Pr_G[x = x_0 \wedge V(x, y, z) = 1 : (x, z) \leftarrow C^{(G)}] . \end{aligned}$$

Note that C implicitly calls $G(k')$ once when simulating O_{Dec} , so C makes a total of $q_G + 1$ queries to G . Unless specified, all references to G -queries below exclude this implicit query. Let $S^C(\Theta)$ be an oracle algorithm that always returns Θ for C 's implicit query $G(k')$. Algorithm S^C randomly selects $(i, b) \leftarrow \$ (\{1, \dots, q_G\} \times \{0, 1\}) \cup \{(q_G + 1, 0)\}$, runs $C^{(G)}$ until (just before) the i -th G -query (excluding the implicit query), measures the input register to obtain x , continues running $C^{(G)}$ until the $(i + b)$ -th G -query, reprograms G to $G_{x\Theta}$ where $G_{x\Theta}(x) = \Theta$ and $G_{x\Theta}(x') = G(x')$ for all $x' \neq x$, and finally continues running $C^{(G_{x\Theta})}$ to output z . Let $y := \Theta$ and $z := (z_1, z_2, z_3) = (k, k^*, \hat{k})$, where k is used in line 20 to determine the final response in O_{Dec} . We can see that $V_1(x, y, z) = [\Theta = z_1] = 1$ still holds. When $\Theta \leftarrow \$ \text{Range}(G)$, $S^C(\Theta)$ perfectly simulates G_5^A under the condition $k' \notin \{k^*, \perp\}$ and $\text{flag} = 2$. Therefore,

$$\begin{aligned} \Pr[Z_5 \mid k' \notin \{k^*, \perp\} \wedge \text{flag} = 2] &= \Pr_{G, \Theta}[V(x, \Theta, z) = 1 : (x, z) \leftarrow S^C(\Theta)] \\ &= \sum_{x_0 \in \mathcal{K}_{\text{kem}}} \Pr_{G, \Theta}[x = x_0 \wedge V(x, \Theta, z) = 1 : (x, z) \leftarrow S^C(\Theta)] . \end{aligned}$$

By Lemma E.1, we have

$$\begin{aligned} &\sum_{x_0} \Pr_G[x = x_0 \wedge V(x, H(x), z) = 1 : (x, z) \leftarrow C^{(G)}] \\ &\leq 2(2q_G + 1)^2 \sum_{x_0} \Pr_{G, \Theta}[x = x_0 \wedge V(x, \Theta, z) = 1 : (x, z) \leftarrow S^C] + \frac{8(q_G + 1)^2}{|G|} , \end{aligned}$$

where $|G|$ denotes the size of the output space of G . Hence,

$$\Pr[Z_4 \mid k' \notin \{k^*, \perp\}] \leq 8(q_G + 1)^2 (\Pr[Z_5 \mid k' \notin \{k^*, \perp\} \wedge \text{flag} = 2] + 1/|G|) .$$

Therefore, we can conclude

$$\begin{aligned}
\Pr[Z_5] &\geq \Pr[Z_5 \wedge k' = k^* \wedge \text{flag} = 0] + \Pr[Z_5 \wedge k' = \perp \wedge \text{flag} = 1] \\
&\quad + \Pr[Z_5 \wedge k' \notin \{k^*, \perp\} \wedge \text{flag} = 2] \\
&= \frac{\Pr[Z_5 | k' = k^* \wedge \text{flag} = 0] \Pr[k' = k^*] + \Pr[Z_5 | k' = \perp \wedge \text{flag} = 1] \Pr[k' = \perp]}{3} \\
&\quad + \frac{\Pr[Z_5 | k' \notin \{k^*, \perp\} \wedge \text{flag} = 2] \Pr[k' \notin \{k^*, \perp\}]}{3} \\
&\geq \frac{\Pr[Z_4 | k' = k^*] \Pr[k' = k^*] + \Pr[Z_4 | k' = \perp] \Pr[k' = \perp]}{3} \\
&\quad + \frac{\Pr[Z_4 | k' \notin \{k^*, \perp\}] \Pr[k' \notin \{k^*, \perp\}]}{24(q_G + 1)^2} - \frac{1}{|G|} \\
&\geq \Pr[Z_4]/24(q_G + 1)^2 - 1/|G|.
\end{aligned}$$

At this point, we can see that an OW-CPA adversary $\mathcal{B}$ for KEM can perfectly simulate game G_5^A , since all public values and oracle simulations are independent of k^* and sk . Therefore, when event Z_5 occurs, $\mathcal{B}$ can use $\hat{k}$ output by B to win its OW-CPA game. Hence, we have $\Pr[Z_5] \leq \text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B})$.

In summary, we obtain

$$|\Pr[W_3] - \Pr[W_2]| \leq 10(q_G + 1)^2 \sqrt{\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + 1/|G|}.$$

□

Lemma E.3. *For the events W_3 and W_2 and the other notation defined in Appendix E.2, there exists an adversary $\mathcal{B}'$ against the IND-CPA security of KEM such that*

$$|\Pr[W_3] - \Pr[W_2]| \leq 10(q_G + 1) \sqrt{2\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + (q_G + 1)^2/|\mathcal{K}_{\text{kem}}| + 1/|G|},$$

where q_G is the bounded number of G -queries made by $\mathcal{A}$, and $|G|$ is the output space size of G .

Lemma E.3. We begin by constructing two oracle algorithms, $A^{|\mathcal{G}\rangle, O_{\text{Dec}}}(z)$ and $A^{|\mathcal{G}'\rangle, O_{\text{Dec}}}(z)$, identical to those used in the proof of Lemma E.2, such that

$$\Pr[1 \leftarrow A^{|\mathcal{G}\rangle, O_{\text{Dec}}}(z)] = \Pr[W_2] \quad \text{and} \quad \Pr[1 \leftarrow A^{|\mathcal{G}'\rangle, O_{\text{Dec}}}(z)] = \Pr[W_3],$$

where $z := (\text{pk}, c^*, k_0, k_1, b)$ (with $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$, $k_0, k_1 \leftarrow \$ \text{Range}(G)$, $b \leftarrow \$ \{0, 1\}$), $\mathcal{G} \leftarrow \$ \Omega_G$, and $\mathcal{G}' \in \Omega_G$ satisfies $\mathcal{G}'(k) = \mathcal{G}(k)$ for all $k \neq k^*$, and $\mathcal{G}'(k^*) = k_0$.

Since $\mathcal{G}$ and $\mathcal{G}'$ differ only at the point k^* , by the double-sided O2H lemma (Lemma 3.2), there exists an oracle algorithm $B^{|\mathcal{G}\rangle, |\mathcal{G}'\rangle, O_{\text{Dec}}}(z)$ such that

$$\begin{aligned}
&\left| \Pr[1 \leftarrow A^{|\mathcal{G}\rangle, O_{\text{Dec}}}(z)] - \Pr[1 \leftarrow A^{|\mathcal{G}'\rangle, O_{\text{Dec}}}(z)] \right| \\
&\leq 2 \sqrt{\Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{|\mathcal{G}\rangle, |\mathcal{G}'\rangle, O_{\text{Dec}}}(z)]}.
\end{aligned}$$

Next, we define games G_j^B for $j = 4, 5, 6$, played between a challenger and the oracle algorithm B , as illustrated in Fig. 20. In each game, k^* is the key generated by the challenger, and $\hat{k}$ is the key output by B . Let Z_j denote the event that $k^* = \hat{k}$ in game G_j^B .

Game G_4^B . This game is defined as shown in Fig. 20. The challenger selects a random function $\mathcal{G} \leftarrow \$ \Omega_G$ to simulate the random oracles G and G' : a query k to G returns $\mathcal{G}(k)$; a query k to

Games G_4^B to G_6^B :	Decapsulation Oracle $O_{\text{Dec}}(c \neq c^*)$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda), (k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$	10 : if $\text{cnt} \neq 0$ then : return $\perp$
2 : $k^* \leftarrow \$ \mathcal{K}_{\text{kem}} \quad // G_6^B$	11 : $\text{cnt} := \text{cnt} + 1$
3 : $k_0, k_1 \leftarrow \$ \text{Range}(G), b \leftarrow \$ \{0, 1\}$	12 : $k' := \text{Decaps}(\text{sk}, c) \quad // G_4^B$
4 : $G \leftarrow \$ \Omega_G, \text{cnt} := 0$	13 : if $k' = k^*$ then $// G_4^B$
5 : $l := 0, \text{flag} \leftarrow \$ \{0, 1, 2\}, \bar{k} \leftarrow \$ \text{Range}(G)$	14 : if $\text{flag} = 0$ then $// G_5^B - G_6^B$
$(i, b) \leftarrow \$ (\{1, \dots, q_G\} \times \{0, 1\}) \cup \{(q_G + 1, 0)\}$	15 : return test
6 : $\hat{k} \leftarrow B^{ \mathcal{G} , \mathcal{G}' , O_{\text{Dec}}}(\text{pk}, c^*, k_0, k_1, b)$	16 : if $k' = \perp$ then $// G_4^B$
7 : return $[k^* = \hat{k}]$	17 : if $\text{flag} = 1$ then $// G_5^B - G_6^B$
	18 : return $\perp$
Random Oracle $G(k)$:	19 : $k := G(k') \quad // G_4^B$
8 : if $\text{flag} = 2$ then	20 : $k := \bar{k} \quad // G_5^B - G_6^B$
if $l \geq (i + b)$ and $k = k_i$ then	21 : if $k \in \{k_0, k_1\}$ then
$// k_i$ is the measurement outcome on	return test
$// B$'s i -th G -query input register	22 : return k
return $\bar{k}$	Random Oracle $G'(k)$:
$l = l + 1 \quad // G_5^B - G_6^B$	23 : if $k = k^*$ then : return k_0
9 : return $G(k)$	24 : return $G(k)$

Figure 20: Games G_4^B to G_6^B for the proof of Lemma E.3.

$G'(k)$ returns k_0 if $k = k^*$, and $G(k)$, i.e., $G(k)$, otherwise. This setup is equivalent to having direct access to G and G' . Therefore, we have

$$\Pr[Z_4] = \Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{|\mathcal{G}|, |\mathcal{G}'|, O_{\text{Dec}}}(z)] = \Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{|\mathcal{G}|, |\mathcal{G}'|, O_{\text{Dec}}}(z)] .$$

Game G_5^B . In this game, the decapsulation oracle is simulated using a guessing strategy. The challenger randomly selects $\text{flag} \leftarrow \$ \{0, 1, 2\}$ to guess the value of $k' := \text{Decaps}(\text{sk}, c)$ during the simulation of $O_{\text{Dec}}(c \neq c^*)$: if $\text{flag} = 0$, the challenger guesses that $k' = k^*$, and O_{Dec} returns **test**; if $\text{flag} = 1$, the challenger guesses that $k' = \perp$, and O_{Dec} returns $\perp$; if $\text{flag} = 2$, the challenger guesses that $k' \notin \{k^*, \perp\}$. In the last case, O_{Dec} assigns a predetermined random value $\bar{k}$ to k (line 20 in Fig. 20), and returns **test** if $k \in \{k_0, k_1\}$, and returns k otherwise.

When $\text{flag} = 2$ (i.e., the challenger guesses $k' \notin \{k^*, \perp\}$), the challenger further reprograms the random oracle G conditioned on (i, b) , where $(i, b) \leftarrow \$ (\{1, \dots, q_G\} \times \{0, 1\}) \cup \{(q_G + 1, 0)\}$ and q_G denotes the upper bound on the number of G -queries made by B .

Using Lemma E.1 in the same way as in game G_5^A in the proof of Lemma E.2, we can derive

$$\Pr[Z_5] \geq \Pr[Z_4]/24(q_G + 1)^2 - 1/|G| .$$

Game G_6^B . In this game, k^* is resampled from $\mathcal{K}_{\text{kem}}$ (i.e., line 2 in Fig. 20). Since k^* is now independent of the oracles G and O_{Dec} , as well as the input to the oracle algorithm B , by Lemma 3.3, we have

$$\Pr[Z_6] \leq (q_G + 1)^2/|\mathcal{K}_{\text{kem}}| .$$

Note that the simulation of the decapsulation oracle O_{Dec} in both G_5^B and G_6^B no longer requires sk . We now construct an adversary B' against the IND-CPA security of KEM. Specifically, upon receiving $(\text{pk}^*, c^*, k_{b'})$ from the IND-CPA challenger (where b' is a random bit chosen

by the challenger), $\mathcal{B}'$ sets $k^* := k_{b'}$, defines $z := (\mathbf{pk}^*, c^*, k_0, k_1, b)$ with $k_0, k_1 \leftarrow \$ \text{Range}(G)$ and $b \leftarrow \$ \{0, 1\}$, and then runs $B^{(G), (G'), O_{\text{Dec}}}(z)$. Here, $\mathcal{B}'$ simulates all the oracles exactly as in G_5^B without using $\mathbf{sk}^*$. Once $B^{(G), (G'), O_{\text{Dec}}}(z)$ outputs $\hat{k}$, $\mathcal{B}'$ returns $\hat{b}' := [k^* = \hat{k}]$.

Observe that if $b' = 0$, then $k^* = k_{b'}$ is the real key, and $\mathcal{B}'$ perfectly simulates the challenger in G_5^B ; if $b' = 1$, then k^* is random, and $\mathcal{B}'$ behaves as the challenger in G_6^B . Therefore, we have

$$\Pr[\hat{b}' = 1 \mid b' = 0] = \Pr[Z_5] \quad \text{and} \quad \Pr[\hat{b}' = 1 \mid b' = 1] = \Pr[Z_6] .$$

It follows that

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') &\geq \left| \Pr[b' = \hat{b}'] - \frac{1}{2} \right| \\ &= \left| \Pr[b' = \hat{b}' \wedge b' = 0] + \Pr[b' = \hat{b}' \wedge b' = 1] - \frac{1}{2} \right| \\ &= \frac{1}{2} \left| \Pr[\hat{b}' = 0 \mid b' = 0] + \Pr[\hat{b}' = 1 \mid b' = 1] - 1 \right| \\ &= \frac{1}{2} \left| \Pr[\hat{b}' = 1 \mid b' = 1] - \Pr[\hat{b}' = 1 \mid b' = 0] \right| \\ &= \frac{1}{2} |\Pr[Z_6] - \Pr[Z_5]| . \end{aligned}$$

To summarize, we have

$$|\Pr[W_3] - \Pr[W_2]| \leq 10(q_G + 1) \sqrt{2\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + (q_G + 1)^2/|\mathcal{K}_{\text{kem}}| + 1/|G|} .$$

□

F Supporting Material: Tightness of the Reduction for IND-1CCA-MAC*

By Theorem F.1 and an analysis similar to Remark 5.1, we can conclude that any security reduction from IND-1CCA-MAC* security to CPA security must incur a loss factor of at least $\mathcal{O}(q)$ in the ROM and $\mathcal{O}(q^2)$ in the QROM in general.

Theorem F.1. *Let $\text{KEM} = (\text{Gen}, \text{Encaps}, \text{Decaps})$ be a KEM with the malleability property. There exists an adversary $\mathcal{B}$ that can win the IND-1CCA-MAC* game (Fig. 3) of KEM in the ROM (resp. QROM) with advantage*

$$\text{Adv}_{\text{KEM}}^{\text{IND-1CCA-MAC}^*}(\mathcal{B}) \geq \frac{q}{20.1|\mathcal{K}_{\text{kem}}|} \quad \left(\text{resp.} \quad \geq \frac{q^2}{|\mathcal{K}_{\text{kem}}|} \right) ,$$

where q is the number of G -queries made by $\mathcal{B}$ with $\sqrt{\frac{1}{|\mathcal{K}_{\text{kem}}|} + \frac{3}{|F|}} \leq \sin\left(\frac{\pi}{6q+3}\right)$ and $q \leq |F| - 2$, $|F| = \min\{|G|, |H_1|, |H_D|\} > 3$, and $|G|, |H_1|, |H_D|$ denote the output space size of the random oracles G, H_1, H_D , respectively.

Theorem F.1. Let $(\mathbf{pk}^*, c^*, n^*, (\text{CHTS}_b, \text{SHTS}_b, \text{dHS}_b))$ be the IND-1CCA-MAC* challenge for KEM, where $(\mathbf{pk}^*, \mathbf{sk}^*) \leftarrow \text{Gen}(1^\lambda)$, $(k^*, c^*) \leftarrow \text{Encaps}(\mathbf{pk}^*)$, $\text{HS}^* := G(k^*)$, $n^* \leftarrow \$ \{0, 1\}^n$, $(\text{CHTS}_0, \text{SHTS}_0, \text{dHS}_0) := (H_1(\text{HS}^*, H_T(c^*, n^*)), H_2(\text{HS}^*, H_T(c^*, n^*)), H_3(\text{HS}^*))$, $(\text{CHTS}_1, \text{SHTS}_1, \text{dHS}_1) \leftarrow \$ \{0, 1\}^{3n}$, and $b \leftarrow \$ \{0, 1\}$. WLOG, we assume $\text{dHS}_0 \neq \text{dHS}_1$. Since KEM is malleable, there exists an algorithm $\bar{B}$ that takes $(\mathbf{pk}^*, c^*)$ as input and outputs (f, c') such that: (1) $f(k^*) = \text{Decaps}(\mathbf{sk}^*, c') \notin \{k^*, \perp\}$, and (2) $f(\tilde{k}) \neq \text{Decaps}(\mathbf{sk}^*, c')$ for all $\tilde{k} \in \mathcal{K}_{\text{kem}} \setminus \{k^*\}$.

Define an indicator function $\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')} : \mathcal{K}_{\text{kem}} \rightarrow \{0, 1\}$ as

$$\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(k) = \begin{cases} 1 & \text{if } H_D(H_1(G(f(k)), H_T(c', n'))) = \text{tk}_C, \\ 0 & \text{otherwise.} \end{cases}$$

We first consider the case in the ROM. Let $\mathcal{B}^{G,H_1,H_D,O_{\text{Dec}}}(\text{pk}^*, c^*, k_b)$ be a ROM adversary that proceeds as follows:

1. Run $\bar{B}(\text{pk}^*, c^*)$ to obtain (f, c') .
2. Sample $n' \leftarrow \$ \{0, 1\}^n$ and query O_{Dec} with (c', n') to obtain $(\text{tk}_C, \text{tk}_S)$. If $(\text{tk}_C, \text{tk}_S) = \perp$, output $\perp$.
3. Randomly sample q different $\bar{k}_1, \dots, \bar{k}_q \leftarrow \$ \mathcal{K}_{\text{kem}}$, and for each $i \in \{1, \dots, q\}$, compute $\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(\bar{k}_i)$ by querying G, H_1, H_D .
4. If there exists a $\bar{k}_i$ such that $\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(\bar{k}_i) = 1$, output $1 - [H_3(G(\bar{k}_i)) = \text{dHS}_b]$. Otherwise, output $\perp$.

Since $f(k^*) \neq \perp$, $(\text{tk}_C, \text{tk}_S) = O_{\text{Dec}}(c', n') \neq \perp$ with probability 1. Under the condition $(\text{tk}_C, \text{tk}_S) \neq \perp$, we have $\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(k^*) = 1$ with probability 1, and for any $\tilde{k} \in \mathcal{K}_{\text{kem}} \setminus \{k^*\}$, $\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(\tilde{k}) = 1$ with probability at most $\frac{1}{|G|} + \frac{1}{|H_1|} + \frac{1}{|H_D|} \leq \frac{3}{|F|}$. Note also that $\Pr[k^* \in \{\bar{k}_1, \dots, \bar{k}_q\}] = \frac{q}{|\mathcal{K}_{\text{kem}}|}$. Therefore, we can obtain

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-MAC}^*}(\mathcal{B}) &\geq 1 - \left(\frac{q}{|\mathcal{K}_{\text{kem}}|}\right) \left(1 - \frac{3}{|F|}\right)^{q-1} \stackrel{(a)}{\geq} \left(1 - \frac{3}{|F|}\right)^{|F|-3} \left(\frac{q}{|\mathcal{K}_{\text{kem}}|}\right) \\ &= \left(\left(1 - \frac{3}{|F|}\right)^{\frac{|F|}{3}-1}\right)^3 \left(\frac{q}{|\mathcal{K}_{\text{kem}}|}\right) \stackrel{(b)}{\geq} \left(\frac{1}{e}\right)^3 \left(\frac{q}{|\mathcal{K}_{\text{kem}}|}\right) \geq \frac{q}{20.1|\mathcal{K}_{\text{kem}}|}, \end{aligned}$$

where (a) is derived from $q \leq |F| - 2$ and (b) uses the inequality $(1 - 1/x)^{x-1} \geq 1/e$ which holds for all $x > 1$.

Next, we consider the case in the QROM. Let $\mathcal{B}^{[G], [H_1], [H_D], O_{\text{Dec}}}$ be a QROM adversary that proceeds as follows:

1. Run $\bar{B}(\text{pk}^*, c^*)$ to obtain (f, c') .
2. Sample $n' \leftarrow \$ \{0, 1\}^n$ and query O_{Dec} with (c', n') to obtain $(\text{tk}_C, \text{tk}_S)$. If $(\text{tk}_C, \text{tk}_S) = \perp$, output $\perp$.
3. Use Grover's algorithm [41] to search k^* . It applies q times Grover iterations, and each Grover iteration uses the oracle $O : |k\rangle \mapsto (-1)^{\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(k)} |k\rangle$.
4. After Grover's algorithm outputs $\bar{k}$, if $\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(\bar{k}) = 1$, then output $1 - [H_3(G(\bar{k})) = \text{dHS}_b]$, and output $\perp$ otherwise.

As in the ROM case, the probability that $(\text{tk}_C, \text{tk}_S) \neq \perp$ in the step 2 is 1. Under the condition $(\text{tk}_C, \text{tk}_S) \neq \perp$, $\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(k^*) = 1$ with probability 1, and for any $\tilde{k} \in \mathcal{K}_{\text{kem}} \setminus \{k^*\}$, $\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(\tilde{k}) = 1$ with probability at most $\frac{1}{|G|} + \frac{1}{|H_1|} + \frac{1}{|H_D|} \leq \frac{3}{|F|}$. Let $N := \mathbb{E}|\{k \in \mathcal{K}_{\text{kem}} : \mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(k) = 1\}|$. Thus, the probability of finding a $k \in \mathcal{K}_{\text{kem}}$ such that $\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(k) = 1$ is $p_0 = \Pr[\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C,H_T(c',n')}(k) = 1 : k \in \mathcal{K}_{\text{kem}}] = \frac{N}{|\mathcal{K}_{\text{kem}}|} \leq \frac{1}{|\mathcal{K}_{\text{kem}}|} + \frac{3}{|F|}$. According to [64, Section

6.1.3], after applying q times Grover iterations (each requiring one query to the random oracles G, H_1, H_D , respectively), the probability of $\bar{k}$ output in the step 3 satisfying $\mathbb{I}_{f,G,H_1,H_D}^{\text{tk}_C, H_T(c', n')}(\bar{k}) = 1$ is $p_1 = \sin^2(\frac{2q+1}{2}\theta)$, where $\sin^2(\frac{\theta}{2}) = p_0$, and the probability of such $\bar{k} = k^*$ is $\frac{1}{N}$.

When $\sqrt{\frac{1}{|\mathcal{K}_{\text{kem}}|} + \frac{3}{|F|}} \leq \sin\left(\frac{\pi}{6q+3}\right)$, we have $(2q+1) \leq \frac{\pi}{3 \arcsin \sqrt{p_0}} = \frac{\pi}{\frac{3}{2}\theta}$, and hence $\frac{2q+1}{2}\theta \leq \frac{\pi}{3}$.

Define $g(x) = \sin(a+x) - \sin(a) - \frac{x}{2}$. Then $g(0) = 0$ and $g'(x) = \cos(a+x) - \frac{1}{2}$. When $0 \leq a+x \leq \frac{\pi}{3}$, $\cos(a+x) \geq \frac{1}{2}$, so $g'(x) \geq 0$. Therefore, for $0 \leq a+x \leq \frac{\pi}{3}$, we have $\sin(a+x) \geq \sin(a) + \frac{x}{2}$. Given that $0 \leq \frac{2q+1}{2}\theta \leq \frac{\pi}{3}$, it follows that

$$\begin{aligned} p_1 &= \sin^2\left(\frac{2q+1}{2}\theta\right) \geq \left(\sin\left(\frac{\theta}{2}\right) + \frac{\frac{2q}{2}\theta}{2}\right)^2 \stackrel{(c)}{\geq} \left(\sin\left(\frac{\theta}{2}\right) + q \sin\left(\frac{\theta}{2}\right)\right)^2 \\ &= (q+1)^2 \sin^2\left(\frac{\theta}{2}\right) = (q+1)^2 p_0, \end{aligned}$$

where (c) uses the inequality $x \geq \sin x$ for $x > 0$. Therefore, we have

$$\text{Adv}_{\text{KEM}}^{\text{IND-1CCA-MAC}^*}(\mathcal{B}) \geq p_1\left(\frac{1}{N}\right) \geq (q+1)^2 p_0\left(\frac{1}{N}\right) \geq q^2\left(\frac{N}{|\mathcal{K}_{\text{kem}}|}\right)\left(\frac{1}{N}\right) \geq \frac{q^2}{|\mathcal{K}_{\text{kem}}|}.$$

□

G Supporting Material: Proof for Theorem 5.3

Theorem 5.3 (Security of KEM_{tag} in the ROM). *Assume the key-derivation function G and the hash function H' in Fig. 7 are modeled as random oracles. For any PPT adversary $\mathcal{A}$ attacking the IND-1RCCA security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G, H']$, there exist a PPT adversary $\mathcal{B}$ attacking the OW-CPA security of KEM and a PPT adversary $\mathcal{B}'$ attacking the IND-CPA security of KEM s.t.*

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 3(q_G + q_{H'} + 1)\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + ((q_{H'} + 1)^2 + 1)/|H'|, \\ \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 6\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + \frac{(q_{H'} + 1)^2 + 1}{|H'|} + \frac{3(q_G + q_{H'} + 1)}{|\mathcal{K}_{\text{kem}}|}, \end{aligned}$$

where q_G (or $q_{H'}$) is the bounded number of the queries made by $\mathcal{A}$ to the random oracle G (or H'), $|H'|$ is the output space size of H' .

Theorem 5.3. Fig. 21 shows the challenger for the adversary $\mathcal{A}$ in game G_j for $j = 0, \dots, 6$. In each game, b is the random bit sampled by the challenger, while $\hat{b}$ is the bit output by $\mathcal{A}$ at the end of the game. The event that $b = \hat{b}$ in game G_j is defined by W_j .

Game G_0 . In this game, the challenger explicitly initializes two empty associative arrays $\text{Map} : \mathcal{K}_{\text{kem}} \rightarrow \text{Range}(G)$ and $\text{Map}' : \mathcal{K}_{\text{kem}} \rightarrow \text{Range}(H')$ to implement the random oracles G and H' , respectively, where $\mathcal{K}_{\text{kem}}$ is the key space of KEM, and $\text{Range}(G)$ (or $\text{Range}(H')$) is the output space of G (or H'). Note that $k_0 \leftarrow \$ \text{Range}(G)$ and $\text{tag}^* \leftarrow \$ \text{Range}(H')$ are sampled at the beginning, but it immediately sets $\text{Map}[k^*] := k_0$ and $\text{Map}'[k^*] := \text{tag}^*$, which is equivalent to setting the random oracles $G(k^*) := k_0$ and $H'(k^*) := \text{tag}^*$. From the view of $\mathcal{A}$, the behavior of challenger is consistent with that in the IND-1RCCA game of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G, H']$. Thus, we have

$$|\Pr[W_0] - 1/2| = \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}).$$

Game G_1 . In this game, we explicitly handle the case where $\text{Decaps}(\text{sk}, c) = k^*$ in the decapsulation oracle: if $\text{Decaps}(\text{sk}, c) = k^*$, the oracle returns $\perp$ when $\text{tag} \neq \text{tag}^*$, and returns **test**

Games G_0 to G_6 :	Decapsulation Oracle $O_{\text{Dec}}(c, \text{tag})$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	15 : if $\text{cnt} \neq 0$ then : return $\perp$
2 : $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$	16 : if $(c, \text{tag}) = (c^*, \text{tag}^*)$ then
3 : $k^* \leftarrow \$ \mathcal{K}_{\text{kem}} \quad // G_6$	return $\perp$
4 : initialize two empty associative arrays	17 : $\text{cnt} := \text{cnt} + 1$
$\text{Map} : \mathcal{K}_{\text{kem}} \rightarrow \text{Range}(G)$	18 : if $\text{Decaps}(\text{sk}, c) = k^*$ then $// G_1 - G_4$
$\text{Map}' : \mathcal{K}_{\text{kem}} \rightarrow \text{Range}(H')$	19 : if $\text{flag} = 0$ then $// G_5 - G_6$
5 : $k_0, k_1 \leftarrow \$ \text{Range}(G)$	20 : if $\text{tag} \neq \text{tag}^*$ then : return $\perp$
$\text{tag}^* \leftarrow \$ \text{Range}(H')$	else : return test $// G_1 - G_6$
6 : $\text{Map}[k^*] := k_0, \text{Map}'[k^*] := \text{tag}^*$	21 : if COLL then : return $\perp // G_3 - G_6$
$// G_0 - G_1$	22 : if $\nexists k' \in \text{Domain}(\text{Map}')$ such that
7 : $b \leftarrow \$ \{0, 1\}, \text{cnt} := 0$	$\text{Map}'[k'] = \text{tag}$ then
8 : $\text{flag} \leftarrow \$ \{0, 1, 2\} \quad // G_5 - G_6$	return $\perp \quad // G_4 - G_6$
9 : $\hat{b} \leftarrow \mathcal{A}^{G, H', O_{\text{Dec}}}(\text{pk}, (c^*, \text{tag}^*), k_b)$	23 : $k' := \text{Decaps}(\text{sk}, c) \quad // G_0 - G_4$
10 : return $[b = \hat{b}]$	24 : if $k' = \perp$ or $H'(k') \neq \text{tag}$ then
Random Oracle $G(k)$:	$// G_0 - G_4$
11 : if $k \notin \text{Domain}(\text{Map})$ then	25 : if $\text{flag} = 1$ then $// G_5 - G_6$
$\text{Map}[k] \leftarrow \$ \text{Range}(G)$	26 : return $\perp$
12 : return $\text{Map}[k]$	27 : Find $k' \in \text{Domain}(\text{Map}')$ such that
Random Oracle $H'(k)$:	$\text{Map}'[k'] = \text{tag} \quad // G_5 - G_6$
13 : if $k \notin \text{Domain}(\text{Map}')$ then	28 : if $G(k') \in \{k_0, k_1\}$
$\text{Map}'[k] \leftarrow \$ \text{Range}(H')$	return test
14 : return $\text{Map}'[k]$	29 : return $k := G(k')$

Figure 21: Games G_0 to G_6 for the proof of Theorem 5.3.

when $\text{tag} = \text{tag}^*$, i.e., lines 18 and 20 in Fig. 21. We can see that this modification does not change the simulation of the decapsulation oracle. Thus,

$$\Pr[W_1] = \Pr[W_0] .$$

Game G_2 . In this game, we remove the step setting $\text{Map}[k^*] := k_0$ and $\text{Map}'[k^*] := \text{tag}^*$, i.e., line 6 in the challenger's initialization phase. Let Z_j be the event that $\mathcal{A}$ queries k^* to the random oracles G or H' in game G_j . We can see that the behaviors of the challengers in games G_2 and G_1 are the same from the view of $\mathcal{A}$ unless Z_2 occurs. Thus, by Lemma 3.4, we have

$$|\Pr[W_2] - \Pr[W_1]| \leq \Pr[Z_2] .$$

Since now k_0 and k_1 are random and independent of all other values, the bit b is random and independent of the adversary's view. Thus,

$$\Pr[W_2] = 1/2 .$$

Game G_3 . In this game, we assume the event COLL does not occur, where COLL denotes that there exist two distinct k and k' in $\text{Domain}(\text{Map}')$ s.t. $\text{Map}'[k] = \text{Map}'[k']$. Since $\text{Map}'[k]$ is randomly sampled for any k , COLL occurs with probability of at most $(q_{H'} + 1)^2 / |H'|$, where

$q_{H'}$ is the bounded number of H' random oracle queries made by $\mathcal{A}$, and $|H'|$ is the output space size of H' . Thus,

$$|\Pr[Z_3] - \Pr[Z_2]| \leq (q_{H'} + 1)^2 / |H'|.$$

Game G_4 . Upon receiving the decapsulation oracle query on (c, tag) where $\text{Decaps}(\text{sk}, c) \neq k^*$, we let the challenger first check whether there exists $k' \in \text{Domain}(\text{Map}')$ such that $\text{Map}'[k'] = \text{tag}$. If not, it directly returns $\perp$. Games G_4 and G_3 are the same unless $H'(\text{Decaps}(\text{sk}, c)) = \text{tag}$. From the simulation of random oracle H' , we can see that in this case, $H'(\text{Decaps}(\text{sk}, c)) = \text{tag}$ with the probability of at most $1/|H'|$. Thus,

$$|\Pr[Z_4] - \Pr[Z_3]| \leq 1/|H'|.$$

Game G_5 . In game G_5 , the challenger simulates the decapsulation oracle by random guessing. For a decapsulation oracle query (c, tag) , let E_0 be the event that $\text{Decaps}(\text{sk}, c) = k^*$. Given $\bar{E}_0$, let F be the event that there does not exist $k' \in \text{Domain}(\text{Map}')$ such that $\text{Map}'[k'] = \text{tag}$. Further conditioning on $\bar{E}_0$ and the existence of $k' \in \text{Domain}(\text{Map}')$ satisfying $\text{Map}'[k'] = \text{tag}$, let E_1 and E_2 be the events that $k' \neq \text{Decaps}(\text{sk}, c)$ and $k' = \text{Decaps}(\text{sk}, c)$, respectively. Clearly, the events $\{E_0, E_1, E_2, F\}$ form a partition of the probability space, i.e., they are mutually exclusive and satisfy $\sum_{i=0}^2 \Pr[E_i] + \Pr[F] = 1$.

In game G_5 , upon receiving a decapsulation oracle query on $(c, \text{tag}) \neq (c^*, \text{tag}^*)$, the challenger guesses whether event E_0 occurs by randomly sampling $\text{flag} \leftarrow \{0, 1, 2\}$, where $\text{flag} = 0$ indicates that E_0 is guessed to occur. If the challenger guesses $\bar{E}_0$ and there exists $k' \in \text{Domain}(\text{Map}')$ such that $\text{Map}'[k'] = \text{tag}$, the value of flag (1 or 2) is further used to guess which of the events E_1 and E_2 occurs. If $\text{flag} = 1$, it returns $\perp$; otherwise, it returns test or $G(k')$ depending on whether $G(k') \in \{k_0, k_1\}$.

Since we have assumed COLL does not occur, there exists at most one $k' \in \text{Domain}(\text{Map}')$ such that $\text{Map}'[k'] = \text{tag}$. Therefore, we can see that, if the challenger correctly identifies the occurring event, its behavior in G_5 matches that in G_4 . Specifically, we have $\Pr[Z_4 \wedge E_i] = \Pr[Z_5 \wedge E_i | \text{flag} = i]$ for $i = 0, 1, 2$, and $\Pr[Z_4 \wedge F] = \Pr[Z_5 \wedge F | \text{flag} \neq 0]$. Thus, we can obtain

$$\begin{aligned} \Pr[Z_5] &\geq \sum_{i=0}^2 \Pr[Z_5 \wedge E_i \wedge \text{flag} = i] + \Pr[Z_5 \wedge F \wedge \text{flag} \neq 0] \\ &= \sum_{i=0}^2 \Pr[Z_5 \wedge E_i | \text{flag} = i]/3 + \Pr[Z_5 \wedge F | \text{flag} \neq 0](2/3) \\ &\geq (\sum_{i=0}^2 \Pr[Z_5 \wedge E_i | \text{flag} = i] + \Pr[Z_5 \wedge F | \text{flag} \neq 0])/3 \\ &= (\sum_{i=0}^2 \Pr[Z_4 \wedge E_i] + \Pr[Z_4 \wedge F])/3 \\ &= \Pr[Z_4]/3. \end{aligned}$$

We can see that k^* and sk are not used in the simulation of the challenger in game G_5 . Thus, if Z_5 occurs, we can construct an adversary $\mathcal{B}$ to attack the OW-CPA security of the underlying KEM. In particular, upon receiving (pk^*, c^*) from $\mathcal{B}$'s challenger in the OW-CPA game of KEM, he can sample $k \leftarrow \text{Range}(G)$ and $\text{tag}^* \leftarrow \text{Range}(H')$, and send $(\text{pk}^*, (c^*, \text{tag}^*), k)$ to $\mathcal{A}$. $\mathcal{B}$ can simulate the random oracles and decapsulation oracle as described in game G_5 to interact with $\mathcal{A}$. At the end of the game, when Z_5 occurs, $\mathcal{B}$ can randomly sample a key k^* from $\text{Domain}(\text{Map}) \cup \text{Domain}(\text{Map}')$ and output it. Since there are at most $q_G + 1$ queries to G and

$q_{H'}$ queries to H' , we have

$$\frac{1}{q_G + q_{H'} + 1} \Pr[Z_5] \leq \text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) .$$

To summarize, we have

$$\text{Adv}_{\text{KEM}_{tag}}^{\text{IND-IRCCA}}(\mathcal{A}) \leq 3(q_G + q_{H'} + 1)\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + \frac{(q_{H'} + 1)^2 + 1}{|H'|} .$$

Game G_6 . In the initialization phase, the challenger resamples $k^* \leftarrow \$ \mathcal{K}_{\text{kem}}$ (i.e., line 3 in Fig. 21). We argue that if $\mathcal{A}$ detects this change, we can construct an adversary $\mathcal{B}'$ that breaks the IND-CPA security of KEM. Specifically, upon receiving $(\text{pk}^*, c^*, k_{b'})$ from the IND-CPA challenger, where b' is the random bit chosen by the challenger, $\mathcal{B}'$ sets $k^* := k_{b'}$, samples $k \leftarrow \$ \text{Range}(G)$ and $tag^* \leftarrow \$ \text{Range}(H')$, and sends $(\text{pk}^*, (c^*, tag^*), k)$ to $\mathcal{A}$. The oracles are simulated by $\mathcal{B}'$ exactly as in G_5 . At the end, $\mathcal{B}'$ checks whether $k_{b'} \in \text{Domain}(Map) \cup \text{Domain}(Map')$: if so, it outputs $\hat{b}' := 1$; otherwise, $\hat{b}' := 0$.

Note that if $b' = 0$, then $k_{b'}$ is the real key and $\mathcal{B}'$ behaves as in G_5 ; if $b' = 1$, then $k_{b'}$ is random and $\mathcal{B}'$ behaves as in G_6 . Therefore, $\Pr[\hat{b}' = 1 | b' = 0] = \Pr[Z_5]$ and $\Pr[\hat{b}' = 1 | b' = 1] = \Pr[Z_6]$. Hence,

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') &\geq \left| \Pr[b' = \hat{b}'] - \frac{1}{2} \right| \\ &= \left| \Pr[b' = \hat{b}' \wedge b' = 0] + \Pr[b' = \hat{b}' \wedge b' = 1] - \frac{1}{2} \right| \\ &= \frac{1}{2} \left| \Pr[\hat{b}' = 0 | b' = 0] + \Pr[\hat{b}' = 1 | b' = 1] - 1 \right| \\ &= \frac{1}{2} \left| \Pr[\hat{b}' = 1 | b' = 1] - \Pr[\hat{b}' = 1 | b' = 0] \right| \\ &= \frac{1}{2} |\Pr[Z_6] - \Pr[Z_5]| . \end{aligned}$$

Since k^* is independent of all oracles and public values in G_6 , we have

$$\Pr[Z_6] \leq \frac{(q_G + q_{H'} + 1)}{|\mathcal{K}_{\text{kem}}|} .$$

In summary, we have

$$\text{Adv}_{\text{KEM}_{tag}}^{\text{IND-CPA}} \leq 6\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + \frac{(q_{H'} + 1)^2 + 1}{|H'|} + \frac{3(q_G + q_{H'} + 1)}{|\mathcal{K}_{\text{kem}}|} .$$

□

H Supporting Material: Proof for Theorem 5.4

This section first outlines some key concepts necessary for the subsequent proofs, and then presents a security analysis of $\text{KEM}_{tag} := T_{tag}[\text{KEM}, G, H']$.

COLL ($\mathcal{A}$)
1 : $(x_1, t_1), \dots, (x_m, t_m) \leftarrow \mathcal{A}^{\text{S.R.O}}$ 2 : for $i \in \{1, \dots, m\}$ then : $t'_i := \text{S.R.O}(x_i)$ 3 : for $i \in \{1, \dots, m\}$ then : $\hat{x}_i := \text{S.Ext}_f(t_i)$ 4 : if $\exists i : \hat{x}_i \neq x_i$ and $t_i = t'_i$ then : return 1 5 : else : return 0

Figure 22: Collision game for Definition H.1 [51].

H.1 The Extractable RO-Simulator

The extractable RO-simulator is a power proof technique proposed by Don et al. [34], based on the compressed oracle technique introduced by Zhandry [80]. Roughly speaking, the extractable RO-simulator S simulates the random oracle H using the compressed standard oracle S.R.O defined in [80] and provides an extraction-interface S.Ext_f that can output an x satisfying $f(x, H(x)) = t$ for a given input t and a function f .

This subsection briefly reviews the definition of the extractable RO-simulator. We adopt Definition 8 from [51], in which the simulator is treated as a black-box tool, abstracting away details irrelevant to the subsequent proofs, and some descriptions have been appropriately modified to align with the proof requirements. Furthermore, in the definition, we say two queries are *independent* if the input of one query does not rely on the output of the other. It should note that the function f is defined through the assignment $f(x, H(x)) := H(x)$ in this definition, and the output space of H is $\{0, 1\}^n$. For further details on the extractable RO-simulator and the compressed oracle technique, please refer to the original definitions in [34] and [80].

Definition H.1 (Extractable RO-Simulator [51, Definition 8]). The extractable RO-simulator is defined as a tuple $\text{S} = (\text{S.R.O}, \text{S.Ext}_f)$, where S.R.O represents the compressed standard oracle, and S.Ext_f denotes the extraction-interface. The simulator S satisfies the following properties.

1. If S.Ext_f is never used, S.R.O is perfectly indistinguishable from the (*standard*) random oracle.
2. Any two subsequent independent queries to S.R.O commute.
3. Any two subsequent independent queries to S.Ext_f commute.
4. Any two subsequent independent queries to S.R.O and S.Ext_f $8\sqrt{2/2^n}$ -almost commute.
5. Making multiple *identical* classical queries to S.R.O (or S.Ext_f) has the same effect on the state of S as making only *one* of these queries.
6. Let $\hat{x} := \text{S.Ext}_f(t)$ and $\hat{t} := \text{S.R.O}(\hat{x})$ be two subsequent classical queries, then

$$\Pr[\hat{t} \neq t \wedge \hat{x} \neq \perp] \leq 2/2^n .$$

7. Let $t := \text{S.R.O}(x)$ and $\hat{x} := \text{S.Ext}_f(t)$ be two subsequent classical queries, then

$$\Pr[\hat{x} = \perp] \leq 2/2^n .$$

8. We consider the collision game outlined in Fig. 22. For any adversary $\mathcal{A}$ making at most q queries to S.R.O and outputting m tuples, we have

$$\Pr[1 \leftarrow \text{COLL}(\mathcal{A})] \leq \frac{40e^2(q + m + 1)^3 + 2}{2^n} .$$

Games G_0 to G_4 :	Decapsulation Oracle $O_{\text{Dec}}(c, \text{tag})$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	13 : if $\text{cnt} \neq 0$ then : return $\perp$
2 : $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$	14 : if $(c, \text{tag}) = (c^*, \text{tag}^*)$ then
3 : $G \leftarrow \$ \Omega_G, H' \leftarrow \$ \Omega_{H'}$	return $\perp$
4 : $H_{12} \leftarrow \$ \Omega_{H_{12}} \quad // \quad G_2 - G_4$	15 : $\text{cnt} := \text{cnt} + 1, k' := \text{Decaps}(\text{sk}, c)$
5 : $k_0, k_1 \leftarrow \$ \text{Range}(G)$	16 : if $k' = k^*$ then
$\text{tag}^* \leftarrow \$ \text{Range}(H')$	if $\text{tag} \neq \text{tag}^*$ then : return $\perp$
6 : $k_0 := G(k^*), \text{tag}^* := H'(k^*)$	else : return test $// G_1 - G_4$
$// G_0 - G_2$	17 : if $k' = \perp$ or $H'(k') \neq \text{tag}$ then
7 : $b \leftarrow \$ \{0, 1\}, \text{cnt} := 0$	return $\perp$
8 : $\hat{b} \leftarrow \mathcal{A}^{ \mathcal{G} , H' , O_{\text{Dec}}}(\text{pk}, (c^*, \text{tag}^*), k_b)$	18 : if $G(k') \in \{k_0, k_1\}$ then
9 : return $[b = \hat{b}]$	return test
Random Oracle $G(k)$:	19 : return $k := G(k')$
10 : if $k = k^*$ then	Random Oracle $H'(k)$:
return $k_0 \quad // G_4$	20 : if $k = k^*$ then
11 : $(G(k), \cdot) := H_{12}(k) \quad // G_2 - G_4$	return $\text{tag}^* \quad // G_4$
12 : return $G(k)$	21 : $(\cdot, H'(k)) := H_{12}(k) \quad // G_2 - G_4$
	22 : return $H'(k)$

Figure 23: Games G_0 to G_4 for the proof of Theorem 5.4.

H.2 Omitted Proof for Theorem 5.4

Theorem 5.4 (Security of KEM_{tag} in the QROM). *Let the key-derivation function G and the hash function H' in Fig. 7 be modeled as quantum-accessible random oracles. For any adversary $\mathcal{A}$ attacking the IND-1RCCA security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}, G, H']$, there exist an adversary $\mathcal{B}$ attacking the OW-CPA security of KEM and an adversary $\mathcal{B}'$ attacking the IND-CPA security of KEM, s.t.*

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 2(q_G + q_{H'} + 2) \sqrt{3 \text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3},$$

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) \\ \leq 2 \sqrt{6 \text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}) + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3 + \frac{3(q_G + q_{H'} + 2)^2}{|\mathcal{K}_{\text{kem}}|}}, \end{aligned}$$

where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$, $q_G, q_{H'}$ are the bounded numbers of the queries made by $\mathcal{A}$ to random oracles G and H' , respectively, and $|H'|$ is the output space size of H' .

Theorem 5.4. We define a series of games played between the challenger and the adversary $\mathcal{A}$, denoted as games G_j for $j = 0, \dots, 4$, as shown in Fig. 23. In each game G_j , b is a random bit sampled by the challenger, and $\hat{b}$ is the bit output by $\mathcal{A}$. We define W_j be the event that $b = \hat{b}$ in game G_j .

Game G_0 . At the beginning, the challenger randomly samples two functions $G \leftarrow \$ \Omega_G$ and $H' \leftarrow \$ \Omega_{H'}$ to model the random oracles G and H' , respectively, and randomly samples k_0 and tag^* , but k_0 and tag^* are reset to the right values immediately. We can see that the challenger's

behavior in G_0 is identical to that in the original IND-1RCCA game. Thus,

$$|\Pr[W_0] - 1/2| = \text{Adv}_{\text{KEM}_{tag}}^{\text{IND-1RCCA}}(\mathcal{A}) .$$

Game G_1 . This game is identical to G_0 , except that the decapsulation oracle explicitly handles the case where $k' = k^*$ (line 16 in Fig. 23): if $k' = k^*$, it returns $\perp$ when $tag \neq tag^*$, and returns **test** otherwise. Note that this modification does not change the responses of the decapsulation oracle. Hence,

$$\Pr[W_1] = \Pr[W_0] .$$

Game G_2 . This game uses a random function $H_{12}(k) := (G(k), H'(k))$ to simulate the random oracles G and H' , following the same approach used in the proof of Theorem 4.2 where the function H_{123} is used to simulate the random oracles H_1 , H_2 , and H_3 . Since this modification does not change the challenger's behavior, we have

$$\Pr[W_2] = \Pr[W_1] .$$

Game G_3 . This game deletes the step $k_0 := G(k^*)$, $tag^* := H'(k^*)$, i.e., line 6 in Fig. 23. Since the keys k_0, k_1 are random from $\mathcal{A}$'s view, we have

$$\Pr[W_3] = 1/2 .$$

Game G_4 . This game reprograms $H_{12}(k)$ at the point k^* to (k_0, tag^*) . In particular, when the query k made by $\mathcal{A}$ to the random oracle G (or H') satisfies $k = k^*$, the random oracle responds with k_0 (or tag^*) instead of invoking H_{12} . It is equivalent to using a new function G_{12} to simulate the random oracles G and H' , where $G_{12}(k) = H_{12}(k)$ for all $k \neq k^*$, and $G_{12}(k^*) = (k_0, tag^*)$. Note that the behavior of the challenger in this game is the same as that in game G_2 . Thus, we have

$$\Pr[W_4] = \Pr[W_2] .$$

Combining the results from games G_0 to G_4 , we obtain

$$\text{Adv}_{\text{KEM}_{tag}}^{\text{IND-1RCCA}}(\mathcal{A}) = |\Pr[W_4] - \Pr[W_3]| .$$

Based on the analysis in Appendix H.3, we further have

$$|\Pr[W_4] - \Pr[W_3]| \leq 2(q_G + q_{H'} + 2)\sqrt{3\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3} ,$$

and

$$\begin{aligned} |\Pr[W_4] - \Pr[W_3]| &\leq 2\sqrt{6\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3 + \frac{3(q_G + q_{H'} + 2)^2}{|\mathcal{K}_{\text{kem}}|}} , \end{aligned}$$

where $\epsilon_1 := \frac{40e^2(q_{H'} + 2)^3 + 2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$, $q_G, q_{H'}$ are the bounded numbers of the queries made by $\mathcal{A}$ to random oracles G and H' , respectively, and $|H'|$ is the output space size of H' . Thus, we have

$$\text{Adv}_{\text{KEM}_{tag}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 2(q_G + q_{H'} + 2)\sqrt{3\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3} ,$$

and

$$\begin{aligned} \text{Adv}_{\text{KEM}_{tag}}^{\text{IND-1RCCA}}(\mathcal{A}) &\leq 2\sqrt{6\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3 + \frac{3(q_G + q_{H'} + 2)^2}{|\mathcal{K}_{\text{kem}}|}} . \end{aligned}$$

□

H.3 Bound on $|\Pr[W_4] - \Pr[W_3]|$ in Appendix H.2

We now use Lemmas H.1 and H.2 to derive upper bounds on $|\Pr[W_4] - \Pr[W_3]|$ for the cases where KEM is OW-CPA secure and IND-CPA secure, respectively.

Lemma H.1. *For the events W_4 and W_3 , and the other notation defined in Appendix H.2, there exists an adversary $\mathcal{B}$ against the OW-CPA security of KEM, such that*

$$|\Pr[W_4] - \Pr[W_3]| \leq 2(q_G + q_{H'} + 2)\sqrt{3\text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3},$$

where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$, q_G (or $q_{H'}$) is the bounded number of the queries made by $\mathcal{A}$ to random oracles G (or H'), and $|H'|$ is the output space size of H' .

Lemma H.1. We can construct two oracle algorithms $A^{|H_{12}\rangle, O_{\text{Dec}}}(z)$ and $A^{|G_{12}\rangle, O_{\text{Dec}}}(z)$ to simulate games G_3 and G_4 , respectively, and then use the regular O2H lemma (Lemma 3.1) to bound $|\Pr[W_4] - \Pr[W_3]|$. Let $z := (\text{pk}, (c^*, \text{tag}^*), k_0, k_1, b)$, where $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$, $k_0, k_1 \leftarrow \text{Range}(G)$, $\text{tag}^* \leftarrow \text{Range}(H')$, and $b \leftarrow \{0, 1\}$. $A^{|H_{12}\rangle, O_{\text{Dec}}}(z)$ (or $A^{|G_{12}\rangle, O_{\text{Dec}}}(z)$) simulates the oracles in the same way as in game G_3 (or G_4), runs $\hat{b} \leftarrow \mathcal{A}^{|G|, |H'|, O_{\text{Dec}}}(\text{pk}, (c^*, \text{tag}^*), k_b)$, and outputs $[b = \hat{b}]$. It is easy to see that

$$\Pr[1 \leftarrow A^{|H_{12}\rangle, O_{\text{Dec}}}(z)] = \Pr[W_3] \quad \text{and} \quad \Pr[1 \leftarrow A^{|G_{12}\rangle, O_{\text{Dec}}}(z)] = \Pr[W_4].$$

Let $B^{|H_{12}\rangle, O_{\text{Dec}}}(z)$ be an oracle algorithm that takes z as input, randomly picks $i \leftarrow \{1, \dots, q_G + q_{H'} + 2\}$ ¹², runs $A^{|H_{12}\rangle, O_{\text{Dec}}}(z)$ until (just before) the i -th query, measures the input register in the computational basis, and outputs the measurement outcomes. By applying the regular O2H lemma (Lemma 3.1), we have¹³

$$\begin{aligned} & \left| \Pr[1 \leftarrow A^{|H_{12}\rangle, O_{\text{Dec}}}(z)] - \Pr[1 \leftarrow A^{|G_{12}\rangle, O_{\text{Dec}}}(z)] \right| \\ & \leq 2(q_G + q_{H'} + 2)\sqrt{\Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{|H_{12}\rangle, O_{\text{Dec}}}(z)]}. \end{aligned}$$

We next define a series of games G_j^A ($j = 5, \dots, 12$) played between the challenger and the oracle algorithm B , as shown in Fig. 24. In each game, k^* denotes the key generated by the challenger, and $\hat{k}$ represents the key output by B . We define Z_j as the event that $k^* = \hat{k}$ in game G_j^A .

Note that the transitions between games G_j^A (for $j = 5, \dots, 12$) are analogous to those between the games Υ_j ($j = 0, \dots, 7$) in the proof of Theorem 2 in [51], with the exception that O'_{Dec} uses a probabilistic simulation strategy. Consequently, the probability analysis of event Z_j follows a similar line of reasoning as in [51].

Game G_5^A . This game is defined in Fig. 24. The random oracles G, H' are simulated by H_{12} as in G_3 in Fig. 23. With the above analysis, it is easy to see that

$$\Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{|H_{12}\rangle, O_{\text{Dec}}}(z)] = \Pr[Z_5].$$

From now on, let $(\bar{c}, \overline{\text{tag}})$ denote the decapsulation oracle query made by B , and let $k' := \text{Decaps}(\text{sk}, \bar{c})$.

¹²The total number of queries to H_{12} in $A^{|H_{12}\rangle, O_{\text{Dec}}}(z)$ comprises the q_G queries to the random oracle G and the $q_{H'}$ queries to the random oracle H' made by $\mathcal{A}$, with two additional queries required by the decapsulation oracle.

¹³Since O_{Dec} in game G_3 is identical to that in game G_4 , it can be treated as an internal oracle of the oracle algorithm A when applying the regular O2H lemma.

Games G_5^A to G_{12}^A :	Random Oracles $G(k), H'(k)$:
1 : $(pk, sk) \leftarrow \text{Gen}(1^\lambda), (k^*, c^*) \leftarrow \text{Encaps}(pk)$	17 : use H_{12} to simulate G
2 : $k_0, k_1 \leftarrow \$ \text{Range}(G), tag^* \leftarrow \$ \text{Range}(H')$	18 : use H_{12} to simulate H' $\parallel G_5^A$
3 : $b \leftarrow \$ \{0, 1\}, cnt := 0, H_{12} \leftarrow \$ \Omega_{H_{12}}$	19 : use S.RO to simulate H'
4 : $\hat{k} \leftarrow B^{(G), H' , O_{\text{Dec}}}(pk, (c^*, tag^*), k_0, k_1, b)$ $\parallel G_5^A - G_{10}^A$	$\parallel G_6^A - G_{12}^A$
5 : $\hat{k} \leftarrow B^{(G), H' , O'_{\text{Dec}}}(pk, (c^*, tag^*), k_0, k_1, b)$ $\parallel G_{11}^A - G_{12}^A$	Decapsulation Oracle $O_{\text{Dec}}(\bar{c}, \overline{tag})$:
6 : return $[k^* = \hat{k}]$	20 : if $cnt \neq 0$ or $(\bar{c}, \overline{tag}) = (c^*, tag^*)$ then : return $\perp$
7 : $\hat{k} := \text{S.Ext}_f(\overline{tag}) \parallel G_6^A - G_7^A$	21 : $cnt := cnt + 1, k' := \text{Decaps}(sk, \bar{c})$
Decapsulation Oracle $O'_{\text{Dec}}(\bar{c}, \overline{tag})$:	22 : if $k' = k^*$ then if $\overline{tag} \neq tag^*$ then : return $\perp$ else : return test
8 : if $cnt \neq 0$ or $(\bar{c}, \overline{tag}) = (c^*, tag^*)$ then : return $\perp$	23 : $\hat{k} := \text{S.Ext}_f(\overline{tag}) \parallel G_{10}^A$
9 : $cnt := cnt + 1, flag \leftarrow \$ \{0, 1, 2\}$	24 : if $k' = \perp$ or $H'(k') \neq \overline{tag}$ then return $\perp$
10 : $k' := \text{Decaps}(sk, \bar{c}) \parallel G_{11}^A$	25 : $\hat{k} := \text{S.Ext}_f(\overline{tag}) \parallel G_8^A - G_9^A$
11 : if $flag = 0$ then if $\overline{tag} \neq tag^*$ then : return $\perp$ else : return test	26 : if $\hat{k} = \perp$ then $bad' := \text{true}; \perp \parallel G_9^A - G_{10}^A$
12 : $\hat{k} := \text{S.Ext}_f(\overline{tag})$	27 : if $k' \neq \hat{k} \neq \perp$ then $bad := \text{true}; \perp \parallel G_7^A - G_{10}^A$
13 : $tag' := H'(k') \parallel G_{11}^A$	28 : if $G(k') \in \{k_0, k_1\}$ then return test
14 : if $\hat{k} = \perp$ or $flag = 1$ then : return $\perp$	29 : return $k := G(k')$
15 : if $G(\hat{k}) \in \{k_0, k_1\}$ then : return test	
16 : return $k := G(\hat{k})$	

Figure 24: Games G_5^A to G_{12}^A for the proof of Lemma H.1.

Game G_6^A . In this game, the random oracle H' is substituted with the compressed oracle S.RO from the extractable RO-simulator S. At the end of the game, S.Ext_f is invoked on $\overline{tag}$ to extract $\hat{k}$, where $(\bar{c}, \overline{tag})$ is the decapsulation query made by B . By Property 1 in Definition H.1, and since the final invocation of S.Ext_f does not affect the probability of B , we have

$$\Pr[Z_6] = \Pr[Z_5] .$$

Game G_7^A . In this game, we assume the event **bad** does not occur, where **bad** denotes the event where, for the query $(\bar{c}, \overline{tag})$, the decapsulation oracle does not output $\perp$ (i.e., $H'(k') = \overline{tag}$), but at the end, $\hat{k} \neq \perp$ and $\hat{k} \neq k'$. Based on Property 8 of Definition H.1, we have $\Pr[\text{bad}] \leq \epsilon_1$, where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$. Therefore,

$$|\Pr[Z_7] - \Pr[Z_6]| \leq \epsilon_1 .$$

Note that in Fig. 24, we check whether **bad** occurs in the decapsulation oracle (i.e., line 27) for simplicity. This is technically inaccurate, as the values $\hat{k}$ are not yet defined at this point. This issue will be addressed in the next game.

Game G_8^A . In this game, we move the invocation of S.Ext_f into the decapsulation oracle, just after the verification of $H'(k')$. This move requires swapping S.Ext_f and S.RO at most $(q_{H'} + 1)$

times. Therefore, based on Property 4 of Definition H.1, we have

$$|\Pr[Z_8] - \Pr[Z_7]| \leq (q_{H'} + 1)\epsilon_2 ,$$

where $\epsilon_2 := 8\sqrt{2/|H'|}$.

Game G_9^A . Games G_9^A and G_8^A are the same, except that in G_9^A , we assume the event **bad'** does not occur, where **bad'** denotes the event that, for the query $(\bar{c}, \overline{tag})$, the decapsulation oracle does not output $\perp$ (i.e., $H'(k') = \overline{tag}$), but $\hat{k} = \perp$. Based on Property 7 of Definition H.1, we have $\Pr[\mathbf{bad}'] \leq \epsilon_3$, where $\epsilon_3 := 2/|H'|$. Thus, we have

$$|\Pr[Z_9] - \Pr[Z_8]| \leq \epsilon_3 .$$

Game G_{10}^A . In this game, we moves the call to $\mathbf{S.Ext}_f$ before the classical query to H' used for $\overline{tag}$ verification in the decapsulation oracle. Based on Property 4 of Definition H.1, we have

$$|\Pr[Z_{10}] - \Pr[Z_9]| \leq \epsilon_2 .$$

Game G_{11}^A . In this game, the decapsulation oracle employs a probabilistic guessing strategy: it randomly samples $\mathbf{flag} \leftarrow_{\$} \{0, 1, 2\}$ to determine whether $\mathbf{Decaps}(\mathbf{sk}, \bar{c}) = k^*$ and, provided that $\hat{k} \neq \perp$, whether $\hat{k} \neq \mathbf{Decaps}(\mathbf{sk}, \bar{c})$.

If $\mathbf{flag} = 0$, the oracle assumes $\mathbf{Decaps}(\mathbf{sk}, \bar{c}) = k^*$ and returns **test** if $\overline{tag} = \mathbf{tag}^*$, and $\perp$ otherwise. If $\mathbf{flag} \neq 0$ (it assumes $\mathbf{Decaps}(\mathbf{sk}, \bar{c}) \neq k^*$) and $\hat{k} = \perp$, it returns $\perp$. When $\hat{k} \neq \perp$ and $\mathbf{flag} = 1$, it assumes $\hat{k} \neq \mathbf{Decaps}(\mathbf{sk}, \bar{c}) \neq k^*$, and also returns $\perp$. When $\hat{k} \neq \perp$ and $\mathbf{flag} = 2$, it assumes $\hat{k} = \mathbf{Decaps}(\mathbf{sk}, \bar{c}) \neq k^*$, and returns **test** if $G(\hat{k}) \in \{k_0, k_1\}$, or $G(\hat{k})$ otherwise.

Note that conditioned on the correct guess:

1. For the case where $O_{\text{Dec}}(\bar{c}, \overline{tag}) \neq \perp$ in game G_{10}^A , since **bad** and **bad'** are assumed not to occur, which implies $\hat{k} = k'$, $O'_{\text{Dec}}(\bar{c}, \overline{tag})$ in game G_{11}^A will return the same value as in G_{10}^A .
2. Now consider the case where $O_{\text{Dec}}(\bar{c}, \overline{tag}) = \perp$ in game G_{10}^A . This implies either $k' = \perp$ or $H'(k') \neq \overline{tag}$.
 - If $k' = \perp$: if $\hat{k} = \perp$, then O'_{Dec} in G_{11}^A will return $\perp$; if $\hat{k} \neq \perp$, since $\hat{k} \neq k'$, it also returns $\perp$.
 - If $k' \neq \perp$ but $H'(k') \neq \overline{tag}$, suppose for contradiction that the output of $O'_{\text{Dec}}(\bar{c}, \overline{tag})$ in G_{11}^A is not $\perp$, which means $\hat{k} = \mathbf{Decaps}(\mathbf{sk}, \bar{c})$, implying $\hat{k} = k'$. By Property 6 in Definition H.1, we have $H'(k') = H'(\hat{k}) = \overline{tag}$ except with probability at most ϵ_3 , contradicting the assumption that $H'(k') \neq \overline{tag}$.

Thus, we have $|\Pr[Z_{11} \mid \text{correct guess}] - \Pr[Z_{10}]| \leq \epsilon_3$. Furthermore, we have

$$\begin{aligned}
\Pr[Z_{11}] &\geq \Pr[Z_{11} \mid \text{correct guess}] \Pr[\text{correct guess}] \\
&= \Pr[Z_{11} \mid \text{correct guess}] \left(\begin{aligned} &\Pr[\text{flag} = 0 \wedge k' = k^*] \\ &+ \Pr[\text{flag} \neq 0 \wedge k' \neq k^* \wedge \hat{k} = \perp] \\ &+ \Pr[\text{flag} = 1 \wedge k' \neq k^* \wedge \hat{k} \neq \perp \wedge \hat{k} \neq k'] \\ &+ \Pr[\text{flag} = 2 \wedge k' \neq k^* \wedge \hat{k} \neq \perp \wedge \hat{k} = k'] \end{aligned} \right) \\
&= \Pr[Z_{11} \mid \text{correct guess}] \left(\frac{1}{3} \Pr[k' = k^*] + \frac{2}{3} \Pr[k' \neq k^* \wedge \hat{k} = \perp] \right. \\
&\quad \left. + \frac{1}{3} \Pr[k' \neq k^* \wedge \hat{k} \neq \perp \wedge \hat{k} \neq k'] + \frac{1}{3} \Pr[k' \neq k^* \wedge \hat{k} \neq \perp \wedge \hat{k} = k'] \right) \\
&\geq \Pr[Z_{11} \mid \text{correct guess}] \left(\Pr[k' = k^*] + \Pr[k' \neq k^* \wedge \hat{k} = \perp] \right. \\
&\quad \left. + \Pr[k' \neq k^* \wedge \hat{k} \neq \perp \wedge \hat{k} \neq k'] + \Pr[k' \neq k^* \wedge \hat{k} \neq \perp \wedge \hat{k} = k'] \right) / 3 \\
&= \Pr[Z_{11} \mid \text{correct guess}] / 3 .
\end{aligned}$$

At this point, it is easy to see that the original tag verification and logic can be removed, as these steps do not affect $\Pr[Z_{11}]$.

Game G_{12}^A . Based on the previous analysis, the computations used to obtain k' and tag' in the decapsulation oracle can be removed without affecting $\Pr[Z_{11}]$. Additionally, this game moves the oracle query to H' used for computing tag' in the decapsulation oracle to the end of the game. Note that this move does not result in any swaps between $S.RO$ and $S.Ext_f$. Furthermore, since this query to H' (now moved to the end) is classical and independent of k^* , it can be removed without affecting $\Pr[Z_{11}]$. Thus,

$$\Pr[Z_{12}] = \Pr[Z_{11}] .$$

Now, it is clear that an OW-CPA adversary $\mathcal{B}$ for KEM can perfectly simulate game G_{12} , as both the challenge key k_b and the tag tag^* are random, and all the oracles can be simulated without k^* and sk . Then if Z_{12} occurs, $\mathcal{B}$ can use the $\hat{k}$ output by the oracle algorithm B to win its OW-CPA game. Therefore, we have

$$\Pr[Z_{12}] \leq \text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) .$$

In summary, we can obtain

$$|\Pr[W_4] - \Pr[W_3]| \leq 2(q_G + q_{H'} + 2) \sqrt{3 \text{Adv}_{\text{KEM}}^{\text{OW-CPA}}(\mathcal{B}) + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3} ,$$

where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$, q_G (or $q_{H'}$) is the bounded number of the queries made by $\mathcal{A}$ to the G (or H') random oracle, and $|H'|$ is the output space size of H' . $\square$

When the underlying KEM is IND-CPA secure, a tighter bound can be obtained via Lemma H.2.

Lemma H.2. *For the events W_4 and W_3 , and the other notation defined in Appendix H.2, there exists an adversary $\mathcal{B}'$ against the IND-CPA security of KEM, such that*

$$\begin{aligned}
|\Pr[W_4] - \Pr[W_3]| &\leq 2 \sqrt{6 \text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3 + \frac{3(q_G + q_{H'} + 2)^2}{|\mathcal{K}_{\text{kem}}|}} ,
\end{aligned}$$

where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$, q_G (or $q_{H'}$) is the bounded number of the queries made by $\mathcal{A}$ to random oracles G (or H'), and $|H'|$ is the output space size of H' .

Lemma H.2. First, we can construct two oracle algorithms, $A^{|H_{12}\rangle, O_{\text{Dec}}}(z)$ and $A^{|G_{12}\rangle, O_{\text{Dec}}}(z)$, as in the proof of Lemma H.1, to simulate games G_3 and G_4 , respectively, such that

$$\Pr[1 \leftarrow A^{|H_{12}\rangle, O_{\text{Dec}}}(z)] = \Pr[W_3] \quad \text{and} \quad \Pr[1 \leftarrow A^{|G_{12}\rangle, O_{\text{Dec}}}(z)] = \Pr[W_4] ,$$

where $z = (\text{pk}, (c^*, \text{tag}^*), k_0, k_1, b)$, $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$, $k_0, k_1 \leftarrow \$ \text{Range}(G)$, $\text{tag}^* \leftarrow \$ \text{Range}(H')$, and $b \leftarrow \$ \{0, 1\}$.

Let $G_{\neq k^*}$ and $H'_{\neq k^*}$ be the portions of the random oracles G and H' corresponding to queries $k \neq k^*$, respectively. Since $H_{12}(k) = G_{12}(k)$ for all $k \neq k^*$, the behaviors of the simulated $G_{\neq k^*}$ and $H'_{\neq k^*}$ in both $A^{|H_{12}\rangle, O_{\text{Dec}}}(z)$ and $A^{|G_{12}\rangle, O_{\text{Dec}}}(z)$ are identical. Thus, we can use the same random function (e.g., H_{12}) to simulate $G_{\neq k^*}$ and $H'_{\neq k^*}$. Note that the simulation of O_{Dec} does not involve any random oracle query to k^* , so its behavior remains the same in both $A^{|H_{12}\rangle, O_{\text{Dec}}}(z)$ and $A^{|G_{12}\rangle, O_{\text{Dec}}}(z)$. By the double-sided O2H lemma (Lemma 3.2), we can derive an oracle algorithm $B^{|H_{12}\rangle, |G_{12}\rangle, O_{\text{Dec}}}(z)$ satisfying¹⁴

$$\begin{aligned} & |\Pr[1 \leftarrow A^{|H_{12}\rangle, O_{\text{Dec}}}(z)] - \Pr[1 \leftarrow A^{|G_{12}\rangle, O_{\text{Dec}}}(z)]| \\ & \leq 2\sqrt{\Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{|H_{12}\rangle, |G_{12}\rangle, O_{\text{Dec}}}(z)]} . \end{aligned}$$

Next, we define a sequence of games G_j^B (where $j = 5, \dots, 13$), played between the challenger and the oracle algorithm B , as illustrated in Fig. 25. In each game, k^* denotes the key generated by the challenger, and $\hat{k}$ denotes the key output by B . We define the event Z_j as $k^* = \hat{k}$ in game G_j^B .

It is worth noting that the transitions from G_5^B to G_{12}^B mirror exactly those from G_5^A to G_{12}^A in Fig. 24, focusing on modifications to the decapsulation oracle O_{Dec} aimed at making its simulation independent of sk . The difference between G_{12}^B and G_{13}^B lies in whether k^* is generated via $\text{Encaps}(\text{pk})$ or sampled randomly from $\mathcal{K}_{\text{kem}}$. We reduce B 's advantage in distinguishing between G_{12}^B and G_{13}^B to the IND-CPA security of KEM. Thus, in the following analysis, we briefly summarize the changes from G_5^B to G_{12}^B (directly applying the conclusions from G_5^A to G_{12}^A), and then focus on the transition from G_{12}^B to G_{13}^B .

Game G_5^B . This game is defined as in Fig. 25. Note that the simulation of O_{Dec} does not involve any random oracle query to k^* , and the portions of the random oracles G and H' for queries $k \neq k^*$ (denoted as $G_{\neq k^*}$ and $H'_{\neq k^*}$) are uniformly simulated using H_{12} . By the above analysis, we have

$$\Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{|H_{12}\rangle, |G_{12}\rangle, O_{\text{Dec}}}(z)] = \Pr[Z_5] .$$

In the following games, we use $(\bar{c}, \overline{\text{tag}})$ to denote the decapsulation oracle query made by B , and k' to denote $\text{Decaps}(\text{sk}, \bar{c})$.

Game G_6^B . In this game, we use the compressed oracle S.RO from the extractable RO-simulator S to implement the $H'_{\neq k^*}$, and introduce a new step $\hat{k} := S.\text{Ext}_f(\overline{\text{tag}})$ at the end of the game. Based on Property 1 of Definition H.1, and since calling $S.\text{Ext}_f$ at the end of the game does not affect the probability of event Z_5 , we have

$$\Pr[Z_6] = \Pr[Z_5] .$$

¹⁴Since the simulation of O_{Dec} remains identical in both games, when applying the double-sided O2H lemma, O_{Dec} can be treated as an internal oracle of A .

Games G_5^B to G_{13}^B :	Random Oracles $G(k), H'(k)$:
1 : $(pk, sk) \leftarrow \text{Gen}(1^\lambda), (k^*, c^*) \leftarrow \text{Encaps}(pk)$	18 : use H_{12} to simulate $G_{\neq k^*}$
2 : $k^* \leftarrow \$ \mathcal{K}_{\text{kem}} \quad // \quad G_{13}^B$	19 : use H_{12} to simulate $H'_{\neq k^*} \quad // \quad G_5^B$
3 : $k_0, k_1 \leftarrow \$ \text{Range}(G), tag^* \leftarrow \$ \text{Range}(H')$	20 : use S.RO to simulate $H'_{\neq k^*} \quad // \quad G_6^B - G_{13}^B$
4 : $b \leftarrow \$ \{0, 1\}, \text{cnt} := 0$	
5 : $\hat{k} \leftarrow B^{ H_{12} , G_{12} , O_{\text{Dec}}}(pk, (c^*, tag^*), k_0, k_1, b) \quad // \quad G_5^B - G_{10}^B$	
6 : $\hat{k} \leftarrow B^{ H_{12} , G_{12} , O'_{\text{Dec}}}(pk, (c^*, tag^*), k_0, k_1, b) \quad // \quad G_{11}^B - G_{13}^A$	
7 : return $[k^* = \hat{k}]$	
8 : $\hat{k} := \text{S.Ext}_f(\overline{tag}) \quad // \quad G_6^B - G_7^B$	
Decapsulation Oracle $O'_{\text{Dec}}(\bar{c}, \overline{tag})$:	Decapsulation Oracle $O_{\text{Dec}}(\bar{c}, \overline{tag})$:
9 : if $\text{cnt} \neq 0$ or $(\bar{c}, \overline{tag}) = (c^*, tag^*)$ then : return $\perp$	21 : if $\text{cnt} \neq 0$ or $(\bar{c}, \overline{tag}) = (c^*, tag^*)$ then : return $\perp$
10 : $\text{cnt} := \text{cnt} + 1, \text{flag} \leftarrow \$ \{0, 1, 2\}$	22 : $\text{cnt} := \text{cnt} + 1, k' := \text{Decaps}(sk, \bar{c})$
11 : $k' := \text{Decaps}(sk, \bar{c}) \quad // \quad G_{11}^B$	23 : if $k' = k^*$ then
12 : if $\text{flag} = 0$ then	if $\overline{tag} \neq tag^*$ then : return $\perp$
if $\overline{tag} \neq tag^*$ then : return $\perp$	else : return test
else : return test	24 : $\hat{k} := \text{S.Ext}_f(\overline{tag}) \quad // \quad G_{10}^B$
13 : $\hat{k} := \text{S.Ext}_f(\overline{tag})$	25 : if $k' = \perp$ or $H'(k') \neq \overline{tag}$ then
14 : $tag' := H'(k') \quad // \quad G_{11}^B$	return $\perp$
15 : if $\hat{k} = \perp$ or $\text{flag} = 1$ then : return $\perp$	26 : $\hat{k} := \text{S.Ext}_f(\overline{tag}) \quad // \quad G_8^B - G_9^B$
16 : if $G(\hat{k}) \in \{k_0, k_1\}$ then : return test	27 : if $\hat{k} = \perp$ then
17 : return $k := G(\hat{k})$	$\text{bad}' := \text{true}; \perp \quad // \quad G_9^B - G_{10}^B$
	28 : if $k' \neq \hat{k} \neq \perp$ then
	$\text{bad} := \text{true}; \perp \quad // \quad G_7^B - G_{10}^B$
	29 : if $G(k') \in \{k_0, k_1\}$ then
	return test
	30 : return $k := G(k')$

Figure 25: Games G_5^B to G_{13}^B for the proof of Theorem H.2.

Game G_7^B . In this game, we assume the event **bad** does not occur, where **bad** denotes the event that for a query $(\bar{c}, \overline{tag})$, the decapsulation oracle does not return $\perp$ (i.e., $H'(k') = \overline{tag}$), but $\hat{k} \neq \perp$ and $\hat{k} \neq k'$. By Property 8 in Definition H.1, we have $\Pr[\text{bad}] \leq \epsilon_1$, where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$. It follows that

$$|\Pr[Z_7] - \Pr[Z_6]| \leq \epsilon_1.$$

In Fig. 25, we check whether **bad** occurs in the decapsulation oracle (i.e., line 28) for simplicity, but this is technically inaccurate. This issue will be addressed in the next game.

Game G_8^B . This game moves the invocation of S.Ext_f into the decapsulation oracle, just after the verification of $H'(k')$. This move needs to swap S.Ext_f and S.RO at most $(q_{H'} + 1)$ times. Thus, by Property 4 of Definition H.1, we have

$$|\Pr[Z_8] - \Pr[Z_7]| \leq (q_{H'} + 1)\epsilon_2,$$

where $\epsilon_2 := 8\sqrt{2/|H'|}$.

Game G_9^B . In this game, we assume the event **bad'** does not occur, where **bad'** denotes the event that, for the query $(\bar{c}, \overline{tag})$, the decapsulation oracle does not output $\perp$ (i.e., $H'(k') = \overline{tag}$), but

$\hat{k} = \perp$. By Property 7 of Definition H.1, we have $\Pr[\text{bad}'] \leq \epsilon_3$, where $\epsilon_3 := 2/|H'|$. Thus, we have

$$|\Pr[Z_9] - \Pr[Z_8]| \leq \epsilon_3 .$$

Game G_{10}^B . This game moves the invocation of S.Ext_f just before the verification of $H'(k')$. By Property 4 of Definition H.1, we have

$$|\Pr[Z_{10}] - \Pr[Z_9]| \leq \epsilon_2 .$$

Game G_{11}^B . In this game, the decapsulation oracle no longer computes the return value based on k' . Instead, it first randomly samples $\text{flag} \leftarrow_{\$} \{0, 1, 2\}$, and then determines the return value based on the values of flag and $\hat{k}$, following the same procedure as in G_{11}^A in the proof of Lemma H.1. Similar to the analysis of G_{11}^A , we have

$$\Pr[Z_{10}] \leq 3 \Pr[Z_{11}] + \epsilon_3 .$$

Game G_{12}^B . In this game, the computations used to obtain k' and tag' in the decapsulation oracle (i.e., lines 11 and 14 in Fig. 25) are removed. Similar to the analysis of G_{12}^A in the proof of Lemma H.1, this modification does not affect $\Pr[Z_{11}]$. Thus, we have

$$\Pr[Z_{12}] = \Pr[Z_{11}] .$$

Game G_{13}^B . In this game, k^* is reset to a new value sampled from $\mathcal{K}_{\text{kem}}$ at the beginning, i.e., line 2 in Fig. 25. Since k^* is independent of H_{12} , O_{Dec} , and all other public values, by Lemma 3.3, we have

$$\Pr[Z_{13}] \leq (q_{H_{12}})^2 / |\mathcal{K}_{\text{kem}}| ,$$

where $q_{H_{12}}$ is the total number of queries to H_{12} in $A^{H_{12}, O_{\text{Dec}}}(z)$, and $q_{H_{12}} \leq q_G + q_{H'} + 2$ as analyzed in the proof of Lemma H.1.

Note that in both games G_{12}^B and G_{13}^B , all the oracles can be simulated without sk . Now, we claim that, if B can detect the modification from G_{12}^B to G_{13}^B , we can construct an adversary $\mathcal{B}'$ to break the IND-CPA security of KEM. Specially, upon receiving $(\text{pk}^*, c^*, k_{b'})$ from the IND-CPA challenger, $\mathcal{B}'$ sets $k^* := k_{b'}$, lets $z := (\text{pk}^*, (c^*, \text{tag}^*), k_0, k_1, b)$ (where $\text{tag}^* \leftarrow_{\$} \text{Range}(H')$, $k_0, k_1 \leftarrow_{\$} \text{Range}(G)$, and $b \leftarrow_{\$} \{0, 1\}$), and runs $B^{H_{12}, |G_{12}|, O_{\text{Dec}}}(z)$ (where the oracles are simulated by $\mathcal{B}'$ exactly as in G_{12}^B without sk^*). Once $B^{H_{12}, |G_{12}|, O_{\text{Dec}}}(z)$ outputs $\hat{k}$, $\mathcal{B}'$ outputs $\hat{b}' := [k^* = \hat{k}]$.

We can see that, if $b' = 0$, then $k^* = k_{b'}$ is the real key, and $\mathcal{B}'$ behaves as the challenger in G_{12}^B ; if $b' = 1$, then $k^* = k_{b'}$ is random, and $\mathcal{B}'$ behaves as the challenger in G_{13}^B . Thus, we can obtain $\Pr[\hat{b}' = 1 | b' = 0] = \Pr[Z_{12}]$, and $\Pr[\hat{b}' = 1 | b' = 1] = \Pr[Z_{13}]$. Now, we have

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') &\geq \left| \Pr[b' = \hat{b}'] - \frac{1}{2} \right| \\ &= \left| \Pr[b' = \hat{b}' \wedge b' = 0] + \Pr[b' = \hat{b}' \wedge b' = 1] - \frac{1}{2} \right| \\ &= \frac{1}{2} \left| \Pr[\hat{b}' = 0 | b' = 0] + \Pr[\hat{b}' = 1 | b' = 1] - 1 \right| \\ &= \frac{1}{2} \left| \Pr[\hat{b}' = 1 | b' = 1] - \Pr[\hat{b}' = 1 | b' = 0] \right| \\ &= \frac{1}{2} |\Pr[Z_{13}] - \Pr[Z_{12}]| . \end{aligned}$$

Combining all the results, we can obtain

$$|\Pr[W_4] - \Pr[W_3]| \leq 2\sqrt{6\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{B}') + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3 + \frac{3(q_G + q_{H'} + 2)^2}{|\mathcal{K}_{\text{kem}}|}},$$

where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$, q_G (or $q_{H'}$) is the bounded number of the queries made by $\mathcal{A}$ to the G (or H') random oracle, and $|H'|$ is the output space size of H' . $\square$

I Supporting Material: OW-CPA Secure Rigid DKEM tightly implies IND-1RCCA Secure KEM

This section analyzes the IND-1RCCA security of the *intermediate* KEM, where the underlying KEM is built from a rigid deterministic PKE (called *rigid DKEM* or *rigid D-OW-CPA KEM* in [81]). Specifically, we will analyze the IND-1RCCA security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$ in the ROM and the QROM, where $\text{KEM}_{\text{sim}}^r := T_{\text{sim}}[\text{PKE}]$ is derived via a simple transformation T_{sim} (Fig. 26) from rigid deterministic PKE PKE, and $\mathcal{K}_{\text{kem}}$ (resp. $\mathcal{C}_{\text{kem}}$) of $\text{KEM}_{\text{sim}}^r$ corresponds exactly to $\mathcal{M}_{\text{pke}}$ (resp. $\mathcal{C}_{\text{pke}}$) of PKE. We will recall some relevant PKE notions.

Gen(1^λ)	Encaps(pk)	Decaps(sk, c)
1 : (pk, sk) $\leftarrow$ Gen _{pke} (1^λ)	1 : $k := m \leftarrow \$ \mathcal{M}_{\text{pke}}$	1 : $m' := \text{Dec}_{\text{pke}}(\text{sk}, c)$
2 : return (pk, sk)	2 : $c \leftarrow \text{Enc}_{\text{pke}}(\text{pk}, m)$	2 : if $m' = \perp$ then : return $\perp$
	3 : return (k, c)	3 : return $k' := m'$

Figure 26: $\text{KEM}_{\text{sim}} := T_{\text{sim}}[\text{PKE}]$ [81]. $\mathcal{M}_{\text{pke}}$ is the message space of $\text{PKE} = (\text{Gen}_{\text{pke}}, \text{Enc}_{\text{pke}}, \text{Dec}_{\text{pke}})$. As shown by [81], if PKE is OW-CPA secure, then KEM_{sim} is also OW-CPA secure.

I.1 Background on Public-Key Encryption

Definition I.1 (Public-Key Encryption). The public-key encryption (PKE) scheme consists of three PPT algorithms with security parameter λ , a message space $\mathcal{M}$, and a ciphertext space $\mathcal{C}$: (1) The key generation algorithm Gen is a probabilistic algorithm that takes 1^λ as input and outputs a public/private key pair (pk, sk). (2) The encryption algorithm Enc is a probabilistic algorithm that takes pk and a message $m \in \mathcal{M}$ as input and outputs a ciphertext $c \in \mathcal{C}$. (3) The decryption algorithm Dec is a deterministic algorithm that takes sk and $c \in \mathcal{C}$ as input and outputs either $m \in \mathcal{M}$ or a special value $\perp \notin \mathcal{M}$. We say that $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ is defined over $(\mathcal{M}, \mathcal{C})$.

The correctness requirement of a PKE scheme is that for all $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, and for all $c \leftarrow \text{Enc}(\text{pk}, m)$, we have $\text{Dec}(\text{sk}, c) = m$. A PKE scheme is considered deterministic if Enc is a deterministic algorithm.

Definition I.2 (δ -correctness of PKE [34]). A PKE scheme is δ -correct if

$$\mathbb{E}_{(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)} \left[\max_{m \in \mathcal{M}} \Pr[\text{Dec}(\text{sk}, c) \neq m : c \leftarrow \text{Enc}(\text{pk}, m)] \right] \leq \delta.$$

We say a PKE scheme is perfectly correct, if $\delta = 0$.

Note that in the case where PKE is deterministic, for any $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$, the value of $\max_{m \in \mathcal{M}} \Pr[\text{Dec}(sk, c) \neq m : c \leftarrow \text{Enc}(pk, m)]$ is either equal to 1 or 0. Furthermore, if $\max_{m \in \mathcal{M}} \Pr[\text{Dec}(sk, c) \neq m : c \leftarrow \text{Enc}(pk, m)] = 0$, it implies that the mapping $\text{Enc}(pk, \cdot) : \mathcal{M} \rightarrow \mathcal{C}$ is injective¹⁵. We call the key pair (pk, sk) that makes $\max_{m \in \mathcal{M}} \Pr[\text{Dec}(sk, c) \neq m : c \leftarrow \text{Enc}(pk, m)] = 1$ as “bad” key pair.

Definition I.3 (Rigidity [8, Definition 6.1], adapted). We say a deterministic PKE scheme is rigid if

$$\Pr[\text{Enc}(pk, m) = c : (pk, sk) \leftarrow \text{Gen}(1^\lambda), c \leftarrow \mathcal{C}, m := \text{Dec}(sk, c) \in \mathcal{M}] = 1$$

when the PKE is correct.

Definition I.4 (OW-CPA Security of PKE). The OW-CPA security of a PKE scheme $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ is defined through an attack game between a challenger and an adversary $\mathcal{A}$, as shown in Fig. 27. The adversary’s advantage is defined as $\text{Adv}_{\text{PKE}}^{\text{OW-CPA}}(\mathcal{A}) := \Pr[m^* = \hat{m}]$. If

OW-CPA Game:
1 : $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$
2 : $m^* \leftarrow \mathcal{M}$
3 : $c^* \leftarrow \text{Enc}(pk, m^*)$
4 : $\hat{m} \leftarrow \mathcal{A}(pk, c^*)$
5 : return $[m^* = \hat{m}]$

Figure 27: Security games for OW-CPA security of $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$.

this advantage is negligible for all PPT adversaries, we say that PKE is OW-CPA secure. The message m^* and the ciphertext c^* generated by the challenger are referred to as the challenge message and challenge ciphertext, respectively.

I.2 Security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$ in the ROM

The security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$ in the ROM is captured in Theorem I.1, where $\text{KEM}_{\text{sim}}^r$ is derived from a rigid deterministic PKE PKE.

Theorem I.1 (Security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$ in the ROM). *Assume that the key-derivation function G is modeled as a random oracle. For any PPT adversary $\mathcal{A}$ that attacks the IND-1RCCA security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$, where $\text{KEM}_{\text{sim}}^r := T_{\text{sim}}[\text{PKE}]$ and PKE is rigid deterministic with δ -correctness, there exists a PPT adversary $\mathcal{B}$ that attacks the OW-CPA security of $\text{KEM}_{\text{sim}}^r$, such that*

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 3\text{Adv}_{\text{KEM}_{\text{sim}}^r}^{\text{OW-CPA}}(\mathcal{B}) + \delta.$$

Theorem I.1. Fig. 28 shows the simulation of the challenger for the adversary $\mathcal{A}$ in game G_j , where $j = 0, \dots, 5$. In each game, b is a bit randomly sampled by the challenger, and $\hat{b}$ is the bit output by $\mathcal{A}$. We use W_j to denote the event that $b = \hat{b}$ in game G_j .

Game G_0 . In this game, the challenger explicitly initializes an empty associative array $\text{Map} : \mathcal{K}_{\text{kem}} \rightarrow \text{Range}(G)$ to implement the random oracle G , where $\mathcal{K}_{\text{kem}}$ is the key space of $\text{KEM}_{\text{sim}}^r$.

¹⁵ $\max_{m \in \mathcal{M}} \Pr[\text{Dec}(sk, c) \neq m : c \leftarrow \text{Enc}(pk, m)] = 0$ means $\text{Dec}(sk, \text{Enc}(pk, m)) = m$ for all $m \in \mathcal{M}$. If the mapping $\text{Enc}(pk, \cdot)$ is not injective, it means that $\exists m_0 \neq m_1, \text{ s.t. } \text{Enc}(pk, m_0) = \text{Enc}(pk, m_1)$. Thus, $m_0 = \text{Dec}(sk, \text{Enc}(pk, m_0)) = \text{Dec}(sk, \text{Enc}(pk, m_1)) = m_1$. This is contradictory to the assumption that $m_0 \neq m_1$.

Games G_0 to G_5 :	Decapsulation Oracle $O_{\text{Dec}}(c \neq c^*)$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	16 : if $\text{cnt} \neq 0$ then : return $\perp$
2 : if BAD then : return $\perp$ $\parallel G_1 - G_5$	17 : $\text{cnt} := \text{cnt} + 1$
3 : $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$	18 : $\text{List}^{O_{\text{Dec}}}.append(c)$ $\parallel G_4 - G_5$
4 : $k_0, k_1 \leftarrow \$ \text{Range}(G), R \leftarrow \$ \text{Range}(G)$	19 : $k' := \text{Decaps}(\text{sk}, c)$ $\parallel G_0 - G_4$
5 : initialize an empty associative array $\text{Map} : \mathcal{K}_{\text{kem}} \rightarrow \text{Range}(G)$	20 : if $k' = \perp$ then $\parallel G_0 - G_2$
6 : $\text{Map}[k^*] := k_0$ $\parallel G_0 - G_1$	21 : if $\text{flag} = 0$ then $\parallel G_3 - G_5$
7 : $b \leftarrow \$ \{0, 1\}, \text{cnt} := 0$	22 : return $\perp$
8 : $\text{flag} \leftarrow \$ \{0, 1, 2\}$ $\parallel G_3 - G_5$	23 : if $G(k') \in \{k_0, k_1\}$ then $\parallel G_0 - G_2$
9 : $\text{List}^{O_{\text{Dec}}} := \perp$ $\parallel G_4 - G_5$	24 : if $\text{flag} = 1$ then $\parallel G_3 - G_5$
10 : $\hat{b} \leftarrow \mathcal{A}^{G, O_{\text{Dec}}}(\text{pk}, c^*, k_b)$	25 : return test
11 : return $[b = \hat{b}]$	26 : return $k := G(k')$ $\parallel G_0 - G_3$
Random Oracle $G(k)$:	27 : if $\exists k' \in \text{Domain}(\text{Map})$ s.t. $\text{Enc}_{\text{pke}}(\text{pk}, k') = c$ then return $k := \text{Map}[k']$ return $k := R$ $\parallel G_4 - G_5$
12 : if $k \notin \text{Domain}(\text{Map})$ then	
13 : $\text{Map}[k] \leftarrow \$ \text{Range}(G)$	
14 : if $(\text{Enc}_{\text{pke}}(\text{pk}, k) \in \text{List}^{O_{\text{Dec}}})$ and $\text{flag} = 2$ then $\text{Map}[k] := R$ $\parallel G_4 - G_5$	
15 : return $\text{Map}[k]$	

Figure 28: Games G_0 to G_5 for the proof of Theorem I.1.

Note that the challenger samples $k_0 \leftarrow \$ \text{Range}(G)$ at the beginning of the game, and then stores it in $\text{Map}[k^*]$. It is equivalent to setting the random oracle $G(k^*) := k_0$. There is an additional $R \leftarrow \$ \text{Range}(G)$, but it is not used. We can see that from the view of $\mathcal{A}$, the behavior of the challenger is consistent with that in the IND-1RCCA game of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$. Therefore, we have

$$|\Pr[W_0] - 1/2| = \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) .$$

Game G_1 . This game is the same as game G_0 , except that we assume that the event BAD does not occur here. The event BAD denotes that the key pair (pk, sk) is “bad”, i.e., there exist $(k, c) \leftarrow \text{Encaps}(\text{pk})$ such that $\text{Decaps}(\text{sk}, c) \neq k$ under this key pair. Since $\text{KEM}_{\text{sim}}^r := T_{\text{sim}}[\text{PKE}]$ and PKE is deterministic with δ -correctness, according to Definition I.2, the probability of BAD occurring is bounded by δ . Thus, we have

$$|\Pr[W_1] - \Pr[W_0]| \leq \delta .$$

Game G_2 . This game is the same as G_1 , except that the step assigning k_0 to $\text{Map}[k^*]$ is deleted. Let Z_j be the event that $\mathcal{A}$ queries the random oracle G at k^* in game G_j . It is not hard to see that games G_2 and G_1 proceed identically, unless Z_2 occurs. By the difference lemma (Lemma 3.4), we have

$$|\Pr[W_2] - \Pr[W_1]| \leq \Pr[Z_2] .$$

Since both k_0 and k_1 are randomly sampled from $\text{Range}(G)$, and are independent of all the oracles and other public values, b is independent of $\mathcal{A}$'s view. Thus, we have $\Pr[W_2] = 1/2$.

Game G_3 . In this game, we modify the initialization phase and the decapsulation oracle in game G_2 as follows, trying to guess whether $k' = \perp$, $k' \neq \perp \wedge G(k') \in \{k_0, k_1\}$, or $k' \neq \perp \wedge G(k') \notin \{k_0, k_1\}$, where $k' := \text{Decaps}(\text{sk}, c)$. In the initialization phase, the challenger samples a random value $\text{flag} \leftarrow \{0, 1, 2\}$. In the decapsulation oracle, we replace the condition $k' = \perp$ by $\text{flag} = 0$, and the condition $G(k') \in \{k_0, k_1\}$ by $\text{flag} = 1$. We can see that if $k' = \perp$ when $\text{flag} = 0$, if $k' \neq \perp \wedge G(k') \in \{k_0, k_1\}$ when $\text{flag} = 1$, or if $k' \neq \perp \wedge G(k') \notin \{k_0, k_1\}$ when $\text{flag} = 2$, games G_3 and G_2 proceed identically¹⁶. Thus, we can obtain

$$\begin{aligned} \Pr[Z_2] &= \Pr[Z_2 \wedge k' = \perp] + \Pr[Z_2 \wedge k' \neq \perp \wedge G(k') \in \{k_0, k_1\}] \\ &\quad + \Pr[Z_2 \wedge k' \neq \perp \wedge G(k') \notin \{k_0, k_1\}] \\ &= \Pr[Z_3 \wedge k' = \perp | \text{flag} = 0] + \Pr[Z_3 \wedge k' \neq \perp \wedge G(k') \in \{k_0, k_1\} | \text{flag} = 1] \\ &\quad + \Pr[Z_3 \wedge k' \neq \perp \wedge G(k') \notin \{k_0, k_1\} | \text{flag} = 2] . \end{aligned}$$

Game G_4 . In this game, we make the following modifications to game G_3 so that the challenger can answer the decapsulation oracle without sk :

- In the initialization phase: the challenger initializes an empty list $\text{List}^{O_{\text{Dec}}}$ to store the decapsulation oracle query issued by $\mathcal{A}$.
- In the decapsulation oracle $O_{\text{Dec}}(c)$:
 - when $\text{flag} = 0$ (or $\text{flag} = 1$): it still returns $\perp$ (or test).
 - when $\text{flag} = 2$: if there exists $k' \in \text{Domain}(\text{Map})$ such that $\text{Enc}_{\text{pke}}(\text{pk}, k') = c$, it returns $\text{Map}[k']$; otherwise, it returns R .
- In the random oracle $G(k)$: for the new query $k \notin \text{Domain}(\text{Map})$, if $\text{flag} = 2$ and $\text{Enc}_{\text{pke}}(\text{pk}, k) \in \text{List}^{O_{\text{Dec}}}$, set $\text{Map}[k]$ to R .

In the case where $\text{flag} = 0$ (or $\text{flag} = 1$), the decapsulation oracle returns $\perp$ (or test) and the random oracle is not modified, which is consistent with the behavior of the challenger in game G_3 . In the case where $\text{flag} = 2$, recall that we have assumed that BAD would not occur since game G_1 , and PKE is rigid deterministic. It implies that for the decapsulation oracle query $c \in \mathcal{C}$ issued by $\mathcal{A}$, where $\text{Decaps}(\text{pk}, c) = k' \neq \perp$, k' is the only value satisfying $\text{Enc}_{\text{pke}}(\text{pk}, k') = c$. Therefore, in the decapsulation oracle $O_{\text{Dec}}(c)$, when $\text{flag} = 2$, if $\mathcal{A}$ has queried the random oracle on k' such that $\text{Enc}_{\text{pke}}(\text{pk}, k') = c$, it will return $G(k')$ as the challenger does in game G_3 ; otherwise, it returns a random R , but in the subsequent corresponding random oracle query $G(k')$, it will use the same R as response, which is also the same as the behavior of the challenger in game G_3 in the same case, since R is randomly chosen by the challenger and is independent from the view of $\mathcal{A}$. Therefore,

$$\begin{aligned} \Pr[Z_4 \wedge k' = \perp | \text{flag} = 0] &= \Pr[Z_3 \wedge k' = \perp | \text{flag} = 0] , \\ \Pr[Z_4 \wedge k' \neq \perp \wedge G(k') \in \{k_0, k_1\} | \text{flag} = 1] &= \Pr[Z_3 \wedge k' \neq \perp \wedge G(k') \in \{k_0, k_1\} | \text{flag} = 1] , \\ \Pr[Z_4 \wedge k' \neq \perp \wedge G(k') \notin \{k_0, k_1\} | \text{flag} = 2] &= \Pr[Z_3 \wedge k' \neq \perp \wedge G(k') \notin \{k_0, k_1\} | \text{flag} = 2] . \end{aligned}$$

¹⁶One may note that if $k' = \perp$ but $\text{flag} = 2$, we need to extend the domain of the random oracle to make $G(\perp)$ well-defined, but since this case is not related with the relation between games G_3 and G_2 , so we omit this for brevity.

Thus, we can obtain

$$\begin{aligned}
\Pr[Z_4] &\geq \frac{1}{3} \left(\Pr[Z_4 \wedge k' = \perp \mid \text{flag} = 0] \right. \\
&\quad + \Pr[Z_4 \wedge k' \neq \perp \wedge G(k') \in \{k_0, k_1\} \mid \text{flag} = 1] \\
&\quad \left. + \Pr[Z_4 \wedge k' \neq \perp \wedge G(k') \notin \{k_0, k_1\} \mid \text{flag} = 2] \right) \\
&= \frac{1}{3} \left(\Pr[Z_3 \wedge k' = \perp \mid \text{flag} = 0] \right. \\
&\quad + \Pr[Z_3 \wedge k' \neq \perp \wedge G(k') \in \{k_0, k_1\} \mid \text{flag} = 1] \\
&\quad \left. + \Pr[Z_3 \wedge k' \neq \perp \wedge G(k') \notin \{k_0, k_1\} \mid \text{flag} = 2] \right) \\
&= \frac{1}{3} \Pr[Z_2] .
\end{aligned}$$

Game G_5 . Since the response of the decapsulation oracle in game G_4 no longer relies on $k' := \text{Decaps}(\text{sk}, c)$, we can remove it without affecting $\Pr[Z_4]$. Thus, we have

$$\Pr[Z_5] = \Pr[Z_4] .$$

At this point, sk and k^* are not used in any oracles. Thus, when Z_5 occurs, we can construct a new adversary $\mathcal{B}$ to break the OW-CPA security of $\text{KEM}_{\text{sim}}^r$. In particular, when $\mathcal{B}$ receives (pk, c^*) from his challenger in the OW-CPA game of $\text{KEM}_{\text{sim}}^r$, he can sample a random $k \leftarrow \$ \text{Range}(G)$ and send (pk^*, c^*, k) to $\mathcal{A}$. Then $\mathcal{B}$ can simulate the random oracle and the decapsulation oracle as described in game G_5 to answer $\mathcal{A}$'s oracle queries. At the end of the game, when Z_5 occurs, $\mathcal{B}$ can search k^* in $\text{Domain}(\text{Map})$ such that $\text{Enc}_{\text{pke}}(\text{pk}^*, k^*) = c^*$ and output it. Recall that we have assumed BAD would not occur, which implies such k^* is indeed the solution of the OW-CPA game of $\text{KEM}_{\text{sim}}^r$. Thus,

$$\Pr[Z_5] \leq \text{Adv}_{\text{KEM}_{\text{sim}}^r}^{\text{OW-CPA}}(\mathcal{B}) .$$

To summarize, we have

$$\text{Adv}_{\text{KEM}_{\text{tag}}^r}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 3\text{Adv}_{\text{KEM}_{\text{sim}}^r}^{\text{OW-CPA}}(\mathcal{B}) + \delta .$$

□

I.3 Security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$ in the QROM

We prove that the IND-1RCCA security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$ can be tightly reduced to the OW-CPA security of $\text{KEM}_{\text{sim}}^r$ in the QROM, except for the unavoidable quadratic loss [56], if $\text{KEM}_{\text{sim}}^r := T_{\text{sim}}[\text{PKE}]$ and PKE is rigid deterministic.

Theorem I.2 (Security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$ in the QROM). *Assume the key-derivation function G is modeled as a quantum-accessible random oracle. For any adversary $\mathcal{A}$ against the IND-1RCCA security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$, where $\text{KEM}_{\text{sim}}^r := T_{\text{sim}}[\text{PKE}]$ and PKE is rigid deterministic with δ -correctness, there exists an adversary $\mathcal{B}$ that attacks the OW-CPA security of $\text{KEM}_{\text{sim}}^r$, such that*

$$\text{Adv}_{\text{KEM}_{\text{tag}}^r}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 2\sqrt{3\text{Adv}_{\text{KEM}_{\text{sim}}^r}^{\text{OW-CPA}}(\mathcal{B})} + \delta .$$

Theorem I.2. In Fig. 29, we describe the simulation of the challenger for the adversary $\mathcal{A}$ in game G_j , where $j = 0, \dots, 3$. In each game, the random bit sampled by the challenger is denoted

Games G_0 to G_3 :	Random Oracle $G(k)$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	10 : if $\text{Enc}_{\text{pke}}(\text{pk}, k) = c^*$ then
2 : if BAD then : return $\perp$ $\parallel G_1 - G_3$	return k_0 $\parallel G_3$
3 : $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$	11 : return $G(k)$
4 : $G \leftarrow \$ \Omega_G$	Decapsulation Oracle $O_{\text{Dec}}(c \neq c^*)$:
5 : $k_0 \leftarrow \$ \text{Range}(G)$, $k_1 \leftarrow \$ \text{Range}(G)$	12 : if cnt $\neq 0$ then : return $\perp$
6 : $k_0 := G(k^*)$ $\parallel G_0 - G_1$	13 : cnt := cnt + 1
7 : $b \leftarrow \$ \{0, 1\}$, cnt := 0	14 : $k' := \text{Decaps}(\text{sk}, c)$
8 : $\hat{b} \leftarrow \mathcal{A}^{[G], O_{\text{Dec}}}(\text{pk}, c^*, k_b)$	15 : if $k' = \perp$ then : return $\perp$
9 : return $[b = \hat{b}]$	16 : if $G(k') \in \{k_0, k_1\}$ then
	return test
	17 : return $k := G(k')$

Figure 29: Games G_0 to G_3 for the proof of Theorem 1.2.

by b , while the bit output by $\mathcal{A}$ is denoted by $\hat{b}$. We define W_j be the event that $b = \hat{b}$ in game G_j .

Game G_0 . At the beginning of the game, the challenger randomly samples a function $G \leftarrow \$ \Omega_G$ to simulate the random oracle G , and randomly samples $k_0 \leftarrow \$ \text{Range}(G)$, but k_0 is reset to $G(k^*)$ immediately. We can see that the behavior of the challenger is consistent with that in the IND-1RCCA game of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\text{sim}}^r, G]$. Therefore,

$$|\Pr[W_0] - 1/2| = \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-1RCCA}}(\mathcal{A}) .$$

Game G_1 . This game assumes that the event BAD does not occur, where BAD denotes that the key pair (pk, sk) is “bad”, i.e., there exist $(k, c) \leftarrow \text{Encaps}(\text{pk})$ such that $\text{Decaps}(\text{sk}, c) \neq k$ under the pair (pk, sk) . Since $\text{KEM}_{\text{sim}}^r := T_{\text{sim}}[\text{PKE}]$ and PKE is deterministic with δ -correctness, according to Definition 1.2, $\Pr[\text{BAD}]$ is not greater than δ . Therefore,

$$|\Pr[W_1] - \Pr[W_0]| \leq \delta .$$

Game G_2 . In this game, we remove the step that sets $k_0 := G(k^*)$, i.e., line 6 in Fig. 29. Since both k_0 and k_1 are independent of all the oracles and other public values, b is random and independent of $\mathcal{A}$'s view. Thus, we have

$$\Pr[W_2] = 1/2 .$$

Game G_3 . This game is the same as game G_2 , except that we make the following modification to the random oracle. When the random oracle query is on the k such that $\text{Enc}_{\text{pke}}(\text{pk}, k) = c^*$, it responds with k_0 instead of $G(k)$. Since we have assumed BAD would not occur, k^* is the only value in $\mathcal{M}_{\text{pke}}$ such that $\text{Enc}_{\text{pke}}(\text{pk}, k^*) = c^*$. Therefore, this is equivalent to saying that we use a new function $G' \in \Omega_G$ to simulate the random oracle, where $G(k) = G'(k)$ for $k \neq k^*$, but $G(k^*)$ is random and independent of $G'(k^*)$, and $G'(k^*) = k_0$. We can see that the behavior of the challenger is consistent with that in the game G_1 . Therefore,

$$\Pr[W_3] = \Pr[W_1] .$$

Note that $\mathcal{A}$ is not allowed to make the decapsulation query on c^* , so the decapsulation oracle in game G_3 is consistent with that in game G_2 . Now, we can use two oracle

Games G_4 to G_7 :	Decapsulation Oracle $O_{\text{Dec}}(c \neq c^*)$:
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	13 : if $\text{cnt} \neq 0$ then : return $\perp$
2 : if BAD then : return $\perp$	14 : $\text{cnt} := \text{cnt} + 1$
3 : $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$	15 : $k' := \text{Decaps}(\text{sk}, c)$ $\parallel G_4 - G_6$
4 : $G \leftarrow \$ \Omega_G, b \leftarrow \$ \{0, 1\}$	16 : if $k' = \perp$ then $\parallel G_4 - G_5$
5 : $k_0, k_1 \leftarrow \$ \text{Range}(G), \text{cnt} := 0$	17 : if $\text{flag} = 0$ then $\parallel G_6 - G_7$
6 : $\text{flag} \leftarrow \$ \{0, 1, 2\}$ $\parallel G_6 - G_7$	18 : return $\perp$
7 : $\hat{k} \leftarrow B^{(G), (G'), O_{\text{Dec}}}(\text{pk}, c^*, k_0, k_1, b)$	19 : if $G(k') \in \{k_0, k_1\}$ then $\parallel G_4 - G_5$
8 : return $[k^* = \hat{k}]$	20 : if $\text{flag} = 1$ then $\parallel G_6 - G_7$
Random Oracle $G'(k)$:	21 : return test
9 : if $\text{Enc}_{\text{pke}}(\text{pk}, k) = c^*$ then	22 : return $k := G(k')$ $\parallel G_4$
10 : return k_0	23 : return $k := R(c)$ $\parallel G_5 - G_7$
11 : return $G(k)$ $\parallel G_4$	Random Oracle $G(k)$:
12 : return $R(\text{Enc}_{\text{pke}}(\text{pk}, k))$ $\parallel G_5 - G_7$	24 : return $G(k)$ $\parallel G_4$
	25 : return $R(\text{Enc}_{\text{pke}}(\text{pk}, k))$ $\parallel G_5 - G_7$

Figure 30: Games G_4 to G_7 for the proof of Theorem 1.2.

algorithms $A^{(G), O_{\text{Dec}}}(z)$ and $A^{(G'), O_{\text{Dec}}}(z)$ to simulate games G_2 and G_3 , respectively, where $z := (\text{pk}, c^*, k_0, k_1, b)$, $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$, $(k^*, c^*) \leftarrow \text{Encaps}(\text{pk})$, $k_0, k_1 \leftarrow \$ \text{Range}(G)$, and $b \leftarrow \$ \{0, 1\}$. Upon receiving z , $A^{(G), O_{\text{Dec}}}(z)$ (or $A^{(G'), O_{\text{Dec}}}(z)$) simulates O_{Dec} as described in game G_2 (or G_3), runs $\hat{b} \leftarrow \mathcal{A}^{(G), O_{\text{Dec}}}(\text{pk}, c^*, k_b)$, and outputs $[b = \hat{b}]$, where the quantum random oracle G is simulated by G (or G'). It is not hard to see

$$\Pr[1 \leftarrow A^{(G), O_{\text{Dec}}}(z)] = \Pr[W_2] \quad \text{and} \quad \Pr[1 \leftarrow A^{(G'), O_{\text{Dec}}}(z)] = \Pr[W_3] .$$

Since G and G' only differs at k^* , from Lemma 3.2, we can get that there exists an oracle algorithm $B^{(G), (G'), O_{\text{Dec}}}(z)$ such that

$$\begin{aligned} & \left| \Pr[1 \leftarrow A^{(G), O_{\text{Dec}}}(z)] - \Pr[1 \leftarrow A^{(G'), O_{\text{Dec}}}(z)] \right| \\ & \leq 2\sqrt{\Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{(G), (G'), O_{\text{Dec}}}(z)]} . \end{aligned}$$

The game played between the challenger and the oracle algorithm B in game G_j is defined in Fig. 30, where $j = 4, \dots, 7$. In each game, k^* is computed by the challenger, while $\hat{k}$ is output by the oracle algorithm. We also define Z_j to be the event that $k^* = \hat{k}$ in game G_j .

Game G_4 . The challenger is described in Fig. 29. Note that we still assume that the event BAD does not occur. The challenger selects a random function $G \leftarrow \$ \Omega_G$ to simulate the random oracle G and G' : a query k to G returns $G(k)$; a query k to $G'(k)$ returns k_0 if $\text{Enc}_{\text{pke}}(\text{pk}, k) = c^*$, and $G(k)$ otherwise. This is equivalent to having direct access to G and G' . The oracle algorithm B is also not allowed to make the decapsulation oracle query on $c = c^*$, so when $k' := \text{Decaps}(\text{sk}, c) \neq \perp$, k' is not equal to k^* , where $G(k')$ is always equal to $G'(k')$. We can see that

$$\Pr[Z_4] = \Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{(G), (G'), O_{\text{Dec}}}(z)] = \Pr[k^* = \hat{k} : \hat{k} \leftarrow B^{(G), (G'), O_{\text{Dec}}}(z)] .$$

Game G_5 . In this game, we introduce a new random function $R : \mathcal{C}_{\text{pke}} \rightarrow \text{Range}(G)$ and replace the random function $G(\cdot)$ with $R(\text{Enc}_{\text{pke}}(\text{pk}, \cdot))$. Recall that we have assumed BAD does not

occur. It implies that the key pair (pk, sk) is not “bad”, which implies that in the underlying PKE, for all $m \in \mathcal{M}_{pke}$, $\text{Dec}_{pke}(sk, \text{Enc}_{pke}(pk, m)) = m$ always holds. Therefore, for any different $k, k' \in \mathcal{K}_{kem} = \mathcal{M}_{pke}$, $\text{Enc}_{pke}(pk, k) \neq \text{Enc}_{pke}(pk, k')$. This means that $R(\text{Enc}_{pke}(pk, k))$ is random and independent of $R(\text{Enc}_{pke}(pk, k'))$ for any different $k, k' \in \mathcal{K}_{kem}$, so $G(\cdot)$ can be simulated by $R(\text{Enc}_{pke}(pk, \cdot))$ perfectly. At this point, since PKE is rigid deterministic and BAD does not occur, for the decapsulation oracle query $c \neq c^*$, if $k' := \text{Decaps}(sk, c) \neq \perp$, then $\text{Enc}_{pke}(pk, k') = c$. Therefore, in this case, the decapsulation oracle can respond with $k := G(k') = R(\text{Enc}_{pke}(pk, k')) = R(c)$ directly. Thus, we have

$$\Pr[Z_5] = \Pr[Z_4] .$$

Game G_6 . This game is the same as game G_5 , except that the challenger additionally samples a random $\text{flag} \leftarrow \$ \{0, 1, 2\}$, and replaces the condition $k' = \perp$ in the decapsulation oracle with $\text{flag} = 0$, and $G(k') \in \{k_0, k_1\}$ with $\text{flag} = 1$. If $k' = \perp$ when $\text{flag} = 0$, if $k' \neq \perp \wedge G(k') \in \{k_0, k_1\}$ when $\text{flag} = 1$, or if $k' \neq \perp \wedge G(k') \notin \{k_0, k_1\}$ when $\text{flag} = 2$, games G_6 and G_5 proceed identically. Thus, we have

$$\begin{aligned} \Pr[Z_5] &= \Pr[Z_5 \wedge k' = \perp] + \Pr[Z_5 \wedge k' \neq \perp \wedge G(k') \in \{k_0, k_1\}] \\ &\quad + \Pr[Z_5 \wedge k' \neq \perp \wedge G(k') \notin \{k_0, k_1\}] \\ &= \Pr[Z_6 \wedge k' = \perp | \text{flag} = 0] + \Pr[Z_6 \wedge k' \neq \perp \wedge G(k') \in \{k_0, k_1\} | \text{flag} = 1] \\ &\quad + \Pr[Z_6 \wedge k' \neq \perp \wedge G(k') \notin \{k_0, k_1\} | \text{flag} = 2] . \end{aligned}$$

Game G_7 . We can observe that the behavior of the decapsulation oracle in game G_6 does not rely on $k' := \text{Decaps}(sk, c)$, thus we can delete this step in game G_7 without being detected by B . It means

$$\begin{aligned} \Pr[Z_7 \wedge k' = \perp | \text{flag} = 0] &= \Pr[Z_6 \wedge k' = \perp | \text{flag} = 0] , \\ \Pr[Z_7 \wedge k' \neq \perp \wedge G(k') \in \{k_0, k_1\} | \text{flag} = 1] &= \Pr[Z_6 \wedge k' \neq \perp \wedge G(k') \in \{k_0, k_1\} | \text{flag} = 1] , \\ \Pr[Z_7 \wedge k' \neq \perp \wedge G(k') \notin \{k_0, k_1\} | \text{flag} = 2] &= \Pr[Z_6 \wedge k' \neq \perp \wedge G(k') \notin \{k_0, k_1\} | \text{flag} = 2] . \end{aligned}$$

Thus, we can derive

$$\Pr[Z_7] \geq \frac{1}{3} \Pr[Z_5] .$$

We can note that the simulations of all the oracles and all the inputs of B in game G_7 do not require sk . Thus, if the event Z_7 occurs, we can construct an adversary $\mathcal{B}$ to attack the OW-CPA security of $\text{KEM}_{\text{sim}}^r$. In particular, upon receiving (pk^*, c^*) from $\mathcal{B}$'s challenger in the OW-CPA security game of $\text{KEM}_{\text{sim}}^r$, he can sample random $k_0, k_1 \leftarrow \$ \text{Range}(G)$, $b' \leftarrow \$ \{0, 1\}$, simulate the oracles in the same way described in game G_7 , run $\hat{k} \leftarrow B^{|G|, |G'|, O_{\text{Dec}}}(pk^*, c^*, k_0, k_1, b)$, and use $\hat{k}$ as his output. Thus,

$$\Pr[Z_7] \leq \text{Adv}_{\text{KEM}_{\text{sim}}^r}^{\text{OW-CPA}}(\mathcal{B}) .$$

To summarize, we have

$$\text{Adv}_{\text{KEM}_{\text{tag}}^{\text{IND-1RCCA}}}(\mathcal{A}) \leq 2\sqrt{3\text{Adv}_{\text{KEM}_{\text{sim}}^r}^{\text{OW-CPA}}(\mathcal{B})} + \delta .$$

□

J Supporting Material: Multi-Stage Security Model

This section provides a brief overview of the multi-stage model used by Dowling et al. [38] to prove the security of the TLS 1.3 handshake protocol. For detailed information and discussions, please refer to their work [38].

J.1 Syntax

This subsection briefly outlines concepts used to describe the properties of the protocol and the session in the multi-stage model. For each protocol, *protocol-specific* properties are represented as a vector $(M, \text{AUTH}, \text{FS}, \text{USE}, \text{REPLAY})$, where M denotes the number of stages (keys derived), AUTH indicates when keys become *unilaterally* (or *mutually*) *authenticated*, FS specifies when the keys achieve forward secrecy, USE denotes key usage (internal or external), and REPLAY identifies stages vulnerable to replay attacks.

The set of *identities* (or *users*) is denoted by $\mathcal{U}$, with each identity represented as $U \in \mathcal{U}$. Protocol sessions are uniquely identified by a *label* $\text{label} \in \mathcal{U} \times \mathcal{U} \times \mathbb{N}$, where $\text{label} = (U, V, n)$ indicates the n -th local session of identity U (the session *owner*) with V as the intended communication *partner*. In the public-key variant of the model, each identity U is associated with a certified *long-term* public key pk_U and secret key sk_U . In the pre-shared secret setting, a session instead holds an identifier $\text{pssid} \in \{0, 1\}^*$ for the pre-shared secret pss . Each session maintains a *session list* List_S to track *session-specific* properties:

- $\text{label} \in \mathcal{U} \times \mathcal{U} \times \mathbb{N}$: Unique session label.
- $\text{id} \in \mathcal{U}$: Identity of the session owner.
- $\text{pid} \in \mathcal{U} \cup \{*\}$: Identity of the communication partner (can be unknown initially).
- $\text{role} \in \{\text{initiator}, \text{responder}\}$: Session owner's role (e.g. client/server for TLS).
- $\text{auth} \in \text{AUTH}$: Authentication type vector indicating levels for each stage.
- $\text{pssid} \in \{0, 1\}^* \cup \{\perp\}$: Identifier for the pre-shared secret (initialized as unknown if needed).
- $\text{st}_{\text{exec}} \in \{\text{running}_i\}_{i \in \mathbb{N} \cup \{0\}} \cup \{\text{accepted}_i\}_{i \in \mathbb{N}} \cup \{\text{rejected}_i\}_{i \in \mathbb{N}}$: Execution state, indicating whether the session is in the process of running the i -th stage, has accepted, or rejected the i -th key.
- $\text{stage} \in \{0, \dots, M\}$: Current session stage, incremented upon acceptance or rejection of keys.
- $\text{sid} \in (\{0, 1\}^* \cup \{\perp\})^M$: Session identifier in each stage, set upon acceptance.
- $\text{cid} \in (\{0, 1\}^* \cup \{\perp\})^M$: Contributive identifier in each stage, set multiple times until acceptance.
- $\text{key} \in (\{0, 1\}^* \cup \{\perp\})^M$: Established session key in each stage, set upon acceptance.
- $\text{st}_{\text{key}} \in \{\text{fresh}, \text{revealed}\}^M$: State of the session key in each stage (fresh or revealed).
- $\text{tested} \in \{\text{true}, \text{false}\}^M$: Indicator showing if the key for stage i has been tested (true or false).
- $\text{corrupted} \in \{0, \dots, M, \infty\}$: Indicator of corruption stage, showing when **Corrupt** occurs (including before the session started).

We say that two distinct sessions, label and label' , are *partnered* in stage i if $\text{label.sid}_i = \text{label'.sid}_i \neq \perp$. Similarly, two distinct sessions are *contributive partner* if $\text{label.cid}_i = \text{label'.cid}_i \neq \perp$.

J.2 Adversary Model

The multi-stage model considers a PPT adversary $\mathcal{A}$ that can control communication among all parties. This adversary can create sessions, send and receive messages, reveal session keys, and corrupt long-term secrets. Additionally, $\mathcal{A}$ can issue test queries, which return either a real or random session key, requiring the adversary to distinguish between the two cases. Specifically, $\mathcal{A}$ interacts with the protocol through the following queries:

- **NewSecret**(U, V, pssid): It is only available in the pre-shared secret (sMSKE) variant. It generates a fresh pre-shared secret identified by pssid for parties U and V , ensuring the uniqueness of pssid .
- **NewSession**($U, V, \text{role}, \text{auth}[, \text{pssid}]$): It creates a new session with a unique label label for identity U , specifying its role, intended partner V , and authentication type. In the sMSKE variant, it may also reference the pre-shared secret identified by pssid . If label is corrupted, set $\text{label.corrupted} := 0$.
- **Send**(label, m): It sends a message m to the session identified by label . If the session exists, it processes the message on behalf of the session owner U and returns the response along with the updated execution state $\text{label.st}_{\text{exec}}$. If the session's state changes to accepted_i , execution is paused, allowing the adversary to resume it later. This query also facilitates key testing and corruption management. Upon key acceptance at stage i , if there exists a partnered session label' such that $\text{label'.tested}_i = \text{true}$, then label.tested_i is also set to true . If key_i is an internal key, we further set $\text{label.key}_i := \text{label'.key}_i$.
- **Reveal**(label, i): It reveals the session key label.key_i for stage i in the session identified by label . If the session does not exist or if the current stage is less than i , it returns $\perp$; otherwise, it marks the key as *revealed*.
- **Corrupt**(U) or **Corrupt**(U, V, pssid): It provides the adversary with the corresponding long-term secret, either the secret key for the public-key variant or the pre-shared secret for the pre-shared secret variant. It also updates the status of corrupted entities and reveals session keys based on the forward secrecy conditions.
- **Test**(label, i): It is used to check the session key for stage i associated with label . It relies on a random bit b , which the adversary $\mathcal{A}$ aims to guess. The session must be accepted and not previously tested; if stage i is internal and has a partnered session with $\text{label'.st}_{\text{exec}} \neq \text{accepted}_i$, a “lost” flag is set to true . Other flags are set depending on the level of authentication (see [38] for more details). If $b = 0$, the K is set to the real key label.key_i , and if $b = 1$, the key K is sampled at random. The oracle distinguishes between internal and external keys; internal keys should be replaced with K during testing, while external keys retain their original values. This ensures consistency in key usage and proper management of test results. Additionally, if a partnered session has accepted the key, it updates the test status and key accordingly. Return K .

J.3 Multi-Stage Security

We now describe the game that defines multi-stage security.

Definition J.1 (Multi-Stage Secure [38, Definition 4.2]). Let KE be a multi-stage key exchange protocol characterized by properties (M, AUTH, FS, USE, REPLAY). For any PPT adversary $\mathcal{A}$ playing following game $\text{Multi-Stage}_{\text{KE}}(\mathcal{A})$:

1. **Setup:** A random bit b is sampled, the lost flag is initialized to **false**, and long-term keys $(\text{pk}_U, \text{sk}_U)$ are generated for all $U \in \mathcal{U}$ in the public-key variant (pMSKE).
2. **Query:** Adversary $\mathcal{A}$ receives public keys (pMSKE) and has access to the oracles **NewSecret**, **NewSession**, **Send**, **Reveal**, **Corrupt**, and **Test**. These queries may set lost flag to **true**.
3. **Guess:** $\mathcal{A}$ stops and outputs a guess b' .
4. **Finalize:** If there are two sessions label and label' such that $\text{label.sid}_i = \text{label'.sid}_i$, $\text{label.st}_{\text{key},i} = \text{revealed}$, and $\text{label'.tested}_i = \text{true}$, then the challenger set lost flag to **true**. The game outputs 1 if $b = b'$ and the lost flag is **false**.

We defined $\text{Adv}_{\text{KE}}^{\text{multi-stage}}(\mathcal{A}) := |\Pr[1 \leftarrow \text{Multi-Stage}_{\text{KE}}(\mathcal{A})] - 1/2|$ to be the multi-stage advantage of $\mathcal{A}$ with respect to KE . We say KE is multi-stage secure if KE is match-secure (defined in [38, Section 4.3.1]) and, for PPT adversary $\mathcal{A}$, the advantage $\text{Adv}_{\text{KE}}^{\text{multi-stage}}(\mathcal{A})$ is negligible in the security parameter.

J.4 TLS 1.3 in the Multi-Stage Model

This subsection outlines the parameters of the TLS 1.3 full 1-RTT handshake relevant to our proof in the multi-stage model. A detailed description of the TLS 1.3 full 1-RTT handshake can be found in Fig. 1 of [38].

J.4.1 Protocol Properties

The protocol-specified properties (M, AUTH, FS, USE, REPLAY) of the TLS 1.3 full 1-RTT handshake are as follows:

- **M = 6:** The handshake consists of six stages, deriving in sequence the client and server handshake traffic keys tk_{chs} and tk_{shs} , the client and server application traffic secrets **CATS** and **SATS**, the exporter master secret **EMS**, and the resumption master secret **RMS**.
- **AUTH = $\{((3, m), (3, m), (3, m), (4, m), (5, m), (6, m)) \mid m \in \{6, \infty\}\}$:** The handshake traffic keys tk_{chs} and tk_{shs} are initially unauthenticated, but all keys become unilaterally authenticated after stage 3. With (optional) client authentication, all keys become mutually authenticated at stage $m = 6$; otherwise, they never reach mutual authentication, with $m = \infty$.
- **FS = 1:** All derived keys have forward secrecy for .
- **USE = (internal : $\{1, 2\}$, external : $\{3, 4, 5, 6\})$:** Traffic keys for stages 1 and 2 are used internally, while keys from stages 3 to 6 are used externally.
- **REPLAY = (nonreplayable : $\{1, 2, 3, 4, 5, 6\})$:** The keys of all stages are non-replayable.

J.4.2 Session and Contributive Identifiers

During the execution of the TLS 1.3 full 1-RTT handshake, session identifiers are set upon acceptance of each stage and consist of a label along with all handshake messages up to that point (entering the key derivation):

$$\begin{aligned} \text{sid}_1 &= (\text{"CHTS"}, \text{CH}, \text{CKS}, \text{SH}, \text{SKS}), \\ \text{sid}_2 &= (\text{"SHTS"}, \text{CH}, \text{CKS}, \text{SH}, \text{SKS}), \\ \text{sid}_3 &= (\text{"CATS"}, \text{CH}, \text{CKS}, \text{SH}, \text{SKS}, \text{EE}, \text{CR}^*, \text{SCRT}, \text{SCV}, \text{SF}), \\ \text{sid}_4 &= (\text{"SATS"}, \text{CH}, \text{CKS}, \text{SH}, \text{SKS}, \text{EE}, \text{CR}^*, \text{SCRT}, \text{SCV}, \text{SF}), \\ \text{sid}_5 &= (\text{"EMS"}, \text{CH}, \text{CKS}, \text{SH}, \text{SKS}, \text{EE}, \text{CR}^*, \text{SCRT}, \text{SCV}, \text{SF}), \\ \text{sid}_6 &= (\text{"RMS"}, \text{CH}, \text{CKS}, \text{SH}, \text{SKS}, \text{EE}, \text{CR}^*, \text{SCRT}, \text{SCV}, \text{SF}, \text{CCRT}^*, \text{CCV}^*, \text{CF}). \end{aligned}$$

The meanings of the shorthands for TLS 1.3 messages and derived keys/values can be found in [38, Table 1]. Messages marked with an asterisk (*) appear only in the mutual authentication mode. Note that the session identifier is defined over the *unencrypted* handshake messages.

For the contributive identifiers in stages 1 and 2, the client (or server) sets $\text{cid}_1 := (\text{"CHTS"}, \text{CH}, \text{CKS})$ and $\text{cid}_2 := (\text{"SHTS"}, \text{CH}, \text{CKS})$ upon sending (or receiving) CH and CKS. Later, after receiving (or sending) SH, SKS, they are extended to $\text{cid}_1 = (\text{"CHTS"}, \text{CH}, \text{CKS}, \text{SH}, \text{SKS})$ and $\text{cid}_2 = (\text{"SHTS"}, \text{CH}, \text{CKS}, \text{SH}, \text{SKS})$. For stages $i \in \{3, 4, 5, 6\}$, all other contributive identifiers are set as $\text{cid}_i = \text{sid}_i$ when the session identifier is set.

Since a client session only accepts the first stage key after receiving the SH message, a contributive partner of a tested client session will have the same $\text{cid}_1 = \text{sid}_1$, meaning the client and server exchanged identical messages in the first stage. On the other hand, a server session accepts the first stage key just after receiving the CH, CKS messages (and thus can be tested), ensuring only that the client and the server have the same client messages, but not necessarily that the server messages are the same.

K Supporting Material: Proof of Theorem 6.2

Theorem 6.2 (Security of the Full 1-RTT Handshake of TLS 1.3 with KEM). *Assume that HKDF.Ext, HKDF.Exp_j ($j = 0, 4, 5, 6$), and HKDF.TK shown in Fig. 2 are modeled as random oracles. Let the hash function H used to compute the hashed transcripts T_i be a random oracle. For any PPT multi-stage adversary $\mathcal{A}$, there exist PPT adversaries $\{\mathcal{B}_i\}_{i \in \{0, \dots, 6\}}$, such that*

$$\text{Adv}_{\text{TLS1.3-1RTT}}^{\text{multi-stage}}(\mathcal{A}) \leq 6t_s \cdot \left(\text{Adv}_H^{\text{CR}}(\mathcal{B}_1) + t_u \cdot \text{Adv}_{\text{Sig}}^{\text{euf-cma}}(\mathcal{B}_2) + t_s \cdot \left(\text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{B}_3) + 2 \cdot \right. \right.$$

$$\left. \left. \text{Adv}_{\text{HKDF.Exp}}^{\text{prf}}(\mathcal{B}_4) + \text{Adv}_{\text{HKDF.Ext}}^{\text{prf}}(\mathcal{B}_5) + \text{Adv}_{\text{HKDF.Exp}}^{\text{prf}}(\mathcal{B}_6) \right) \right), \text{ where } t_s \text{ (or } t_u) \text{ is the bounded number}$$

of sessions (or users), and Sig is the signature scheme used for the server authentication (not shown in Fig. 2).

As previously mentioned, our proof aligns completely with [52, Theorem 5] and [81, Theorem 6], except we replace the use of IND-1CCA-MAC and IND-1CCA-MAC* in their proofs with our IND-1CCA-1MAC. We first briefly analyze the rationale behind this substitution before proceeding with the detailed proof.

IND-1CCA-MAC is primarily used in Game $G_{B.2}$ (referred to as $\Gamma^{B.2}$ in [52]’s proof) to replace (CHTS, SHTS, dHS) with random secrets, providing a decapsulation oracle and a MAC

oracle for simulating a honest session without knowledge of its long-term secret. It is important to note that both oracles are only called once, with the decapsulation oracle invoked before the MAC oracle, and both queries are focused on the same SH message (i.e., the same (c, n)). From the structure of our IND-1CCA-1MAC, it can be observed that the decapsulation and the MAC oracles function identically to those provided by IND-1CCA-MAC, and IND-1CCA-1MAC is also used to distinguish between (CHTS, SHTS, dHS) and random secrets. Therefore, our IND-1CCA-1MAC can effectively replace IND-1CCA-MAC in the proof without compromising its integrity.

Theorem 6.2. The proof proceeds through a series of games. We emphasize that, apart from Game $G_{B,2}$, the other games are identical to the proofs provided by Huguenin-Dumittan and Vaudenay [52, Theorem 5], Zhou et al.'s [81, Theorem 6], and Dowling et al.'s [38, Theorem 5.2]. Thus, for these games, we offer only a brief overview, with further details available in their proofs.

Game G_0 . This is the original Multi-Stage game.

Game G_1 . We restrict $\mathcal{A}$ to making only one Test query, resulting in a loss factor of $6t_s$ in the security bound.

Game G_2 . If a collision occurs in the hash function H used to compute T_i in Fig. 2, the game aborts, introducing the error term $\text{Adv}_H^{\text{CR}}(\mathcal{B}_1)$. Our security analysis is divided into two disjoint cases: (A) The test session lacks an honest contributive partner in the first stage; (B) The test session has an honest contributive partner in the first stage. In case (A), one can show that the probability of $\mathcal{A}$ winning the game is upper bounded by $t_u \text{Adv}_{\text{Sig}}^{\text{euf-cma}}(\mathcal{B}_2)$, where Sig is the signature scheme used for server authentication (not shown in Fig. 2). Next, we focus on case (B).

Game $G_{B,0}$. This game is identical to game G_2 conditioned on the presence of an honest contributive partner in the first stage of the test session.

Game $G_{B,1}$. At the beginning of the game, we guess which session among all will be the contributive partner of the test session, leading to a loss factor of t_s .

Game $G_{B,2}$. This game is the only one that differs from the proof provided by Huguenin-Dumittan and Vaudenay [52]. In contrast to the work of Huguenin-Dumittan and Vaudenay, we use our IND-1CCA-1MAC game to analyze the impact of replacing (CHTS, SHTS, dHS) with random secrets on $\mathcal{A}$'s advantage, instead of employing the IND-1CCA-MAC game proposed by Huguenin-Dumittan and Vaudenay. We denote label_C as the client session that is either the test session or its contributive partner session, and label_S as the server session with similar definitions. Note that a session and its partner always have opposing roles. We represent (c, n_S) (or (pk, n_C)) as the SH (or CH) message sent by label_S (or label_C), where c is the ciphertext generated by the *intermediate* KEM, and n_S (or n_C) is the nonce for the server (or client) session.

To align the symbols in TLS 1.3 with those in our IND-1CCA-1MAC game, we express the key-derivation function $\text{HKDF.Exp}_j(\cdot, T_2)$ (where $j \in \{4, 5\}$) as $H_i(\cdot, H_T(c, n_S))$ (where $i \in \{1, 2\}$) with hash function H_T , while omitting other parameters irrelevant to the proof (such as public key and client nonce). Similarly, since T_0 and dES are constants, we express $\text{HKDF.Ext}(\text{dES}, \cdot)$ as $G(\cdot)$, $\text{HKDF.Exp}_0(\cdot, T_0)$ as $H_3(\cdot)$, and $\text{HKDF.Exp}_6(\cdot)$ as $H_4(\cdot)$. Finally, we represent HKDF.TK as H_D . H_i ($i = 1, 2, 3, 4, D$) and G are modeled as random oracles.

In this game, we implement the following two modifications:

1. We replace the derived (CHTS, SHTS, dHS) from label_S with random secrets.
2. If label_C receives (c, n_S) from the SH message, we replace the derived (CHTS, SHTS, dHS) for label_C with the *same* random values used in the first modification.

We will analyze the impact of the aforementioned modifications on $\mathcal{A}$'s advantage using our IND-1CCA-1MAC game in the following two different cases:

- **Case 1:** The client session label_C is the test session. Recall that label_C can only be tested after receiving the SH message. Since label_S is its contributive partner, this means that the SH received by label_C is consistent with what label_S sent. Consequently, we need to implement both modifications simultaneously in this game. The reduction is quite intuitive. When the adversary $\mathcal{B}_3$ in the IND-1CCA-1MAC game receives the instance $(\text{pk}^*, c^*, n^*, (\text{CHTS}_b, \text{SHTS}_b, \text{dHS}_b))$, it can simulate the test session as follows: First, $\mathcal{B}_3$ sends pk^* in the CH, and then uses n^* as the nonce for the contributive session label_S . Next, it sets $(\text{CHTS}_b, \text{SHTS}_b, \text{dHS}_b)$ as the secrets for label_S and sends c^* in the SH message. Finally, upon receiving the SH, label_C also sets its secret to $(\text{CHTS}_b, \text{SHTS}_b, \text{dHS}_b)$. Clearly, when $b = 0$ in the IND-1CCA-1MAC game, $(\text{CHTS}_0, \text{SHTS}_0, \text{dHS}_0)$ represents the real secret, allowing $\mathcal{B}_3$ to perfectly simulate game $G_{B.1}$; conversely, when $b = 1$, $(\text{CHTS}_1, \text{SHTS}_1, \text{dHS}_1)$ are random, and $\mathcal{B}_3$ simulates game $G_{B.2}$. Therefore, in case 1, we have:

$$|\Pr[1 \leftarrow G_{B.2}] - \Pr[1 \leftarrow G_{B.1}]| \leq \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{B}_3).$$

Note that in case 1, the simulation performed by $\mathcal{B}_3$ does not require the decapsulation and MAC oracles provided by the IND-1CCA-1MAC game.

- **Case 2:** The server session label_S is the test session. If the SH sent by label_S matches that received by the contributive session label_C , we can simulate the session using the adversary $\mathcal{B}_3$ in the same manner as in case 1. Since label_S can derive the keys of stages 1 and 2 just after receiving the CH, there may be a case where the SH sent by label_S differs from that received by label_C .

In this case, upon receiving the IND-1CCA-1MAC instance $(\text{pk}^*, c^*, n^*, (\text{CHTS}_b, \text{SHTS}_b, \text{dHS}_b))$, $\mathcal{B}_3$ sends pk^* in the CH and then includes (c^*, n^*) in the SH sent by label_S . After receiving the CH, label_S sets $(\text{CHTS}_b, \text{SHTS}_b, \text{dHS}_b)$ as its derived secrets. When label_C receives the tampered $\text{SH} = (c', n'_S) \neq (c^*, n^*)$, $\mathcal{B}_3$ queries the decapsulation oracle $O_{\text{Dec}}(c', n'_S)$ in the IND-1CCA-1MAC game to obtain the corresponding traffic keys (tk_C, tk_S) . This allows $\mathcal{B}_3$ to correctly simulate label_C and any queries until it receives the SF message.

After label_C receives the SF message including tag T_7 , $\mathcal{B}_3$ queries $O_{\text{MAC}}^{\text{Dec}}(c', n', \text{SF}, T_7)$. If the tag in SF is correct, $\mathcal{B}_3$ obtains HS and can derive all subsequent keys, correctly simulating label_C . If the tag in SF is incorrect, the MAC oracle returns $\perp$, allowing $\mathcal{B}_3$ to abort the session label_C , which also correctly simulates label_C . Thus, when $b = 0$ in the IND-1CCA-1MAC game, $\mathcal{B}_3$ simulates game $G_{B.1}$, while when $b = 1$, it simulates game $G_{B.2}$. Therefore, we have

$$|\Pr[1 \leftarrow G_{B.2}] - \Pr[1 \leftarrow G_{B.1}]| \leq \text{Adv}_{\text{KEM}}^{\text{IND-1CCA-1MAC}}(\mathcal{B}_3).$$

Game $G_{B.3}$. Note that, in game $G_{B.2}$, the secrets $(\text{CHTS}, \text{SHTS}, \text{dHS})$ are replaced with random secrets. Thus, if HKDF.TK is a secure pseudo-random function, we can substitute the relevant traffic keys (tk_C, tk_S) with random ones in this game.

Game $G_{B.4}$. If HKDF.Ext is a secure pseudo-random function, we can further replace the relevant master secret MS with a random one.

Game $G_{B.5}$. If HKDF.Exp is a secure pseudo-random function, we can further replace the remaining keys in the tested session (and in the partnered session if label_C is tested and has received the correct SH) with random ones. Consequently, all keys in the tested session are random and independent of the values in any non-partnered session. Thus, we can conclude that $\mathcal{A}$ cannot win the game where all tested keys are random. This completes the proof. $\square$

Games G_5 to G_6 :
1 : $(pk_1 pk_2, sk_1 sk_2) \leftarrow \text{Gen}_{ }(1^\lambda), (k_1^* k_2^*, c_1^* c_2^*) \leftarrow \text{Encaps}_{ }(pk_1 pk_2)$
2 : $k_1^* \leftarrow \$ \mathcal{K}_{\text{kem}_1} \quad // \quad G_6$
3 : $pk := pk_1 pk_2, c^* := c_1^* c_2^*, k^* := k_1^* k_2^*$
4 : initialize two empty associative arrays $Map : \mathcal{K}_{\text{kem}} \rightarrow \text{Range}(G), Map' : \mathcal{K}_{\text{kem}} \rightarrow \text{Range}(H')$
5 : $k_0, k_1 \leftarrow \$ \text{Range}(G), tag^* \leftarrow \$ \text{Range}(H'), b \leftarrow \$ \{0, 1\}, cnt := 0$
6 : $flag \leftarrow \$ \{0, 1, 2\}$
7 : $\hat{b} \leftarrow \mathcal{A}^{G, H', O_{\text{Dec}}}(\text{pk}, (c^*, tag^*), k_b)$
8 : return $[b = \hat{b}]$

Figure 31: Games G_5 to G_6 for the proof of Lemma 7.2 in the ROM setting.

L Supporting Material: Proof for Lemma 7.2

Lemma 7.2 (IND-1RCCA Security of $\text{KEM}_{tag} := T_{tag}[\text{KEM}_{||}, G, H']$). Assume G, H' in Fig. 7 are modeled as random oracles. For any PPT adversary $\mathcal{A}$ attacking the IND-1RCCA security of $\text{KEM}_{tag} := T_{tag}[\text{KEM}_1 || \text{KEM}_2, G, H']$, there exist a PPT adversary $\mathcal{B}'$ attacking the IND-CPA security of KEM_1 , such that

$$\text{Adv}_{\text{KEM}_{tag}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 6\text{Adv}_{\text{KEM}_1}^{\text{IND-CPA}}(\mathcal{B}') + \frac{(q_{H'}+1)^2+1}{|H'|} + \frac{3(q_G+q_{H'}+1)}{|\mathcal{K}_{\text{kem}_1}|},$$

where q_G (or $q_{H'}$) is the bounded number of $\mathcal{A}$'s queries to the random oracle G (or H'), $|H'|$ is the output space size of H' , and $\mathcal{K}_{\text{kem}_1}$ is the key space of KEM_1 .

When the random oracles G and H' are quantum-accessible, we have

$$\text{Adv}_{\text{KEM}_{tag}}^{\text{IND-1RCCA}}(\mathcal{A}) \leq 2\sqrt{6\text{Adv}_{\text{KEM}_1}^{\text{IND-CPA}}(\mathcal{B}') + \epsilon_1 + (q_{H'}+2)\epsilon_2 + 2\epsilon_3 + \frac{3(q_G+q_{H'}+2)^2}{|\mathcal{K}_{\text{kem}_1}|}},$$

where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$.

L.1 Security of $\text{KEM}_{tag} := T_{tag}[\text{KEM}_{||}, G, H']$ in the ROM

The proof of Lemma 7.2 in the ROM setting is obtained through a further analysis of game G_5 in the proof of Theorem 5.3 (see Appendix G).

For game G_5 in the proof of Theorem 5.3, when the underlying KEM is $\text{KEM} := \text{KEM}_{||} = \text{KEM}_1 || \text{KEM}_2$, the game can be further represented as illustrated in Fig. 31. In this representation, the random oracles G and H' , as well as the decapsulation oracle O_{Dec} accessed by the adversary $\mathcal{A}$, are simulated as specified for G_5 in Fig. 21.

Game G_6 . In this game, k_1^* is reset to a random key from $\mathcal{K}_{\text{kem}_1}$ (line 2 in Fig. 31). Since k_1^* is independent of all oracles and public values in G_6 , we have

$$\Pr[Z_6] \leq (q_G + q_{H'} + 1)/|\mathcal{K}_{\text{kem}_1}|,$$

where q_G and $q_{H'}$ are the upper bounds on the number of queries made by $\mathcal{A}$ to the random oracles G and H' , respectively, and $\mathcal{K}_{\text{kem}_1}$ is the key space of KEM_1 .

As the simulation of the decapsulation oracle in G_5 and G_6 does not require $sk := sk_1 || sk_2$, we can construct an adversary $\mathcal{B}'$ against the IND-CPA security of KEM_1 . Specifically, upon

Games G_{12}^B to G_{13}^B :
1 : $(\text{pk}_1 \parallel \text{pk}_2, \text{sk}_1 \parallel \text{sk}_2) \leftarrow \text{Gen}_{\parallel}(1^\lambda), (k_1^* \parallel k_2^*, c_1^* \parallel c_2^*) \leftarrow \text{Encaps}_{\parallel}(\text{pk}_1 \parallel \text{pk}_2)$
2 : $k_1^* \leftarrow \$ \mathcal{K}_{\text{kem}_1} \quad // \quad G_{13}^B$
3 : $\text{pk} := \text{pk}_1 \parallel \text{pk}_2, c^* := c_1^* \parallel c_2^*, k^* := k_1^* \parallel k_2^*$
4 : $k_0, k_1 \leftarrow \$ \text{Range}(G), \text{tag}^* \leftarrow \$ \text{Range}(H'), b \leftarrow \$ \{0, 1\}, \text{cnt} := 0$
5 : $\hat{k} \leftarrow B^{ H_{12}\rangle, G_{12}\rangle, O'_{\text{Dec}}}(\text{pk}, (c^*, \text{tag}^*), k_0, k_1, b)$
6 : return $[k^* = \hat{k}]$

Figure 32: Games G_{12}^B to G_{13}^B for the proof of Lemma 7.2 in the QROM setting.

receiving $(\text{pk}_1^*, c_1^*, k_{b'})$ from the IND-CPA challenger for KEM_1 , where b' is the challenger's random bit, $\mathcal{B}'$ sets $k_1^* := k_{b'}$, computes $(\text{pk}_2^*, \text{sk}_2^*) \leftarrow \text{Gen}_2(1^\lambda), (k_2^*, c_2^*) \leftarrow \text{Encaps}_2(\text{pk}_2^*)$, defines $\text{pk}^* := \text{pk}_1^* \parallel \text{pk}_2^*, c^* := c_1^* \parallel c_2^*$, and $k^* := k_1^* \parallel k_2^*$, samples $k \leftarrow \$ \text{Range}(G)$ and $\text{tag}^* \leftarrow \$ \text{Range}(H')$, and sends $(\text{pk}^*, (c^*, \text{tag}^*), k)$ to $\mathcal{A}$. The oracles are simulated by $\mathcal{B}'$ exactly as in G_5 . At the end of the game, $\mathcal{B}'$ checks whether $k_{b'} \in \text{Domain}(\text{Map}) \cup \text{Domain}(\text{Map}')$: if so, it outputs $\hat{b}' := 1$; otherwise, it outputs $\hat{b}' := 0$.

Note that if $b' = 0$, then $k_{b'}$ is the real key and $\mathcal{B}'$ behaves as in G_5 ; if $b' = 1$, then $k_{b'}$ is random and $\mathcal{B}'$ behaves as in G_6 . Therefore, $\Pr[\hat{b}' = 1 \mid b' = 0] = \Pr[Z_5]$, and $\Pr[\hat{b}' = 1 \mid b' = 1] = \Pr[Z_6]$. Following an analysis similar to that of G_6 in the proof of Theorem 5.3, we have

$$\text{Adv}_{\text{KEM}_1}^{\text{IND-CPA}}(\mathcal{B}') \geq \frac{1}{2} |\Pr[Z_6] - \Pr[Z_5]|.$$

Combined with the discussions of games G_0 to G_5 in the proof of Theorem 5.3, we conclude

$$\text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-IRCCA}}(\mathcal{A}) \leq 6\text{Adv}_{\text{KEM}_1}^{\text{IND-CPA}}(\mathcal{B}') + \frac{(q_{H'} + 1)^2 + 1}{|H'|} + \frac{3(q_G + q_{H'} + 1)}{|\mathcal{K}_{\text{kem}_1}|},$$

where q_G (or $q_{H'}$) is the bounded number of queries made by $\mathcal{A}$ to the random oracle G (or H'), and $|H'|$ denotes the output space size of H' .

This completes the proof of Lemma 7.2 in the ROM setting.

L.2 Security of $\text{KEM}_{\text{tag}} := T_{\text{tag}}[\text{KEM}_{\parallel}, G, H']$ in the QROM

The proof of Lemma 7.2 in the QROM setting is obtained through a further analysis of game G_{12}^B in Appendix H.3.

For the game G_{12}^B in Appendix H.3, when the underlying $\text{KEM} := \text{KEM}_{\parallel} = \text{KEM}_1 \parallel \text{KEM}_2$, the game G_{12}^B can be further represented as shown in Fig. 32. In this representation, the oracles H_{12} , G_{12} , and O'_{Dec} accessed by the oracle algorithm B are defined as in game G_{12}^B of Fig. 25.

Game G_{13}^B . In this game, k_1^* is reset to a random key sampled from $\mathcal{K}_{\text{kem}_1}$ (line 2 in Fig. 32). Since k_1^* is independent of H_{12} , O_{Dec} , and all other public values, by Lemma 3.3, we have

$$\Pr[Z_{13}] \leq (q_{H_{12}})^2 / |\mathcal{K}_{\text{kem}_1}|,$$

where $q_{H_{12}} = q_G + q_{H'} + 2$ is the total number of queries to H_{12} in $A^{|H_{12}\rangle, O_{\text{Dec}}}(z)$.

Since in both games G_{12}^B and G_{13}^B , all the oracles can be simulated without $\text{sk} := \text{sk}_1 \parallel \text{sk}_2$, we can construct an adversary $\mathcal{B}'$ to break the IND-CPA security of KEM_1 . Specially, upon receiving $(\text{pk}_1^*, c_1^*, k_{b'})$ from the IND-CPA challenger for KEM_1 , $\mathcal{B}'$ sets $k_1^* := k_{b'}$, computes $(\text{pk}_2^*, \text{sk}_2^*) \leftarrow \text{Gen}_2(1^\lambda)$ and $(k_2^*, c_2^*) \leftarrow \text{Encaps}_2(\text{pk}_2^*)$, sets $\text{pk}^* := \text{pk}_1^* \parallel \text{pk}_2^*, c^* := c_1^* \parallel c_2^*, k^* :=$

$k_1^* \| k_2^*$, lets $z := (\mathbf{pk}^*, (c^*, \text{tag}^*), k_0, k_1, b)$ (where $\text{tag}^* \leftarrow \$ \text{Range}(H')$, $k_0, k_1 \leftarrow \$ \text{Range}(G)$, and $b \leftarrow \$ \{0, 1\}$), and runs $B^{|H_{12}|, |G_{12}|, O_{\text{Dec}}}(z)$ (where the oracles are simulated by $\mathcal{B}'$ exactly as in G_{12}^B without $\mathbf{sk}^* := \mathbf{sk}_1^* \| \mathbf{sk}_2^*$). Once $B^{|H_{12}|, |G_{12}|, O_{\text{Dec}}}(z)$ outputs $\hat{k}$, $\mathcal{B}'$ outputs $\hat{b}' := [k^* = \hat{k}]$.

We can see that, if $b' = 0$, then $k_1^* = k_{b'}$ is the real key, and $\mathcal{B}'$ behaves as the challenger in G_{12}^B ; if $b' = 1$, then $k_1^* = k_{b'}$ is random, and $\mathcal{B}'$ behaves as the challenger in G_{13}^B . By the similar analysis as in G_{13}^B in Appendix H.3, we have

$$\text{Adv}_{\text{KEM}_1}^{\text{IND-CPA}}(\mathcal{B}') \geq \frac{1}{2} |\Pr[Z_{13}] - \Pr[Z_{12}]|.$$

Combined with the discussions in Appendices H.2 and H.3, we can obtain

$$\begin{aligned} \text{Adv}_{\text{KEM}_{\text{tag}}}^{\text{IND-IRCCA}}(\mathcal{A}) \\ \leq 2 \sqrt{6 \text{Adv}_{\text{KEM}_1}^{\text{IND-CPA}}(\mathcal{B}') + \epsilon_1 + (q_{H'} + 2)\epsilon_2 + 2\epsilon_3 + \frac{3(q_G + q_{H'} + 2)^2}{|\mathcal{K}_{\text{kem}_1}|}}, \end{aligned}$$

where $\epsilon_1 := \frac{40e^2(q_{H'}+2)^3+2}{|H'|}$, $\epsilon_2 := 8\sqrt{2/|H'|}$, $\epsilon_3 := 2/|H'|$, q_G (or $q_{H'}$ is the bounded numbers of the queries made by $\mathcal{A}$ to random oracle G (or H'), and $|H'|$ is the output space size of H' .

This completes the proof of Lemma 7.2 in the QROM setting.

M Supporting Material: Experimental Data

We performed experiments on the hybrid schemes (treated as KEMs) across two different processor architectures (with two distinct implementations): an Intel Core i7-1165G7 CPU (x86-64) and an Apple M4 chip (ARM).

M.1 Experimental Data on x86-64 Architecture

The experiments were conducted on an Ubuntu 24.04.3 LTS virtual machine with an Intel Core i7-1165G7 CPU and 16 GB RAM, using OpenSSL 3.6.0.

Table 2: Benchmark of hybrid schemes[†] (treated as KEMs) on x86-64 architecture. Here, $\mathbf{pk}_{\parallel}/c_{\parallel}/k_{\parallel}$ denote the hybrid public key/ciphertext/key, $\text{Encaps}_{\parallel}/\text{Decaps}_{\parallel}$ refer to the hybrid encapsulation/decapsulation algorithms, and H' denotes SHA3-512.

Hybrid Schemes	$\mathbf{pk}_{\parallel}$	$c_{\parallel}$	$\text{Encaps}_{\parallel}$	$\text{Decaps}_{\parallel}$	$H'(k_{\parallel})$
X25519MLKEM768	1216 B	1120 B	134.01 μs	75.78 μs	0.72 μs
X448MLKEM1024	1624 B	1624 B	584.40 μs	270.71 μs	1.24 μs
SecP256r1MLKEM768	1249 B	1153 B	149.47 μs	110.79 μs	0.72 μs
SecP384r1MLKEM1024	1665 B	1665 B	2766.3 μs	1317.8 μs	1.20 μs

[†] Based on OpenSSL 3.6.0 by removing the FO-like transformation [49] (cf. [7]).

M.2 Experimental Data on ARM Architecture

The experiments were conducted on a system running macOS Sequoia 15.6 with an Apple M4 chip and 16 GB of memory, using SIMD-optimized implementations (NEON on arm64 via liboqs 0.15.0 + oqsprovider 0.11.0 + OpenSSL 3.4.3).

Table 3: Benchmark of hybrid schemes[†] (treated as KEMs) on ARM architecture. Here, $\text{pk}_{\parallel}/c_{\parallel}/k_{\parallel}$ denote the hybrid public key/ciphertext/key, $\text{Encaps}_{\parallel}/\text{Decaps}_{\parallel}$ refer to the hybrid encapsulation/decapsulation algorithms, and H' denotes SHA3-512.

Hybrid Schemes	$\text{pk}_{\parallel}$	$c_{\parallel}$	$\text{Encaps}_{\parallel}$	$\text{Decaps}_{\parallel}$	$H'(k_{\parallel})$
X25519MLKEM768	1216 B	1120 B	43.20 μs	19.61 μs	0.22 μs
X448MLKEM1024	1624 B	1624 B	103.80 μs	72.48 μs	0.32 μs
SecP256r1MLKEM768	1249 B	1153 B	49.77 μs	34.26 μs	0.21 μs
SecP384r1MLKEM1024	1665 B	1665 B	745.95 μs	372.81 μs	0.33 μs

[†] Based on liboqs 0.15.0 + oqsprovider 0.11.0 + OpenSSL 3.4.3 by removing the FO-like transformation [49] (cf. [7]).